



US 20240178128A1

(19) **United States**(12) **Patent Application Publication**
CHEN et al.(10) **Pub. No.: US 2024/0178128 A1**(43) **Pub. Date: May 30, 2024**(54) **SEMICONDUCTOR DEVICE STRUCTURE
INCLUDING FORKSHEET TRANSISTORS
AND METHODS OF FORMING THE SAME***H01L 29/417* (2006.01)*H01L 29/66* (2006.01)*H01L 29/786* (2006.01)(71) Applicant: **TAIWAN SEMICONDUCTOR
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(2013.01)

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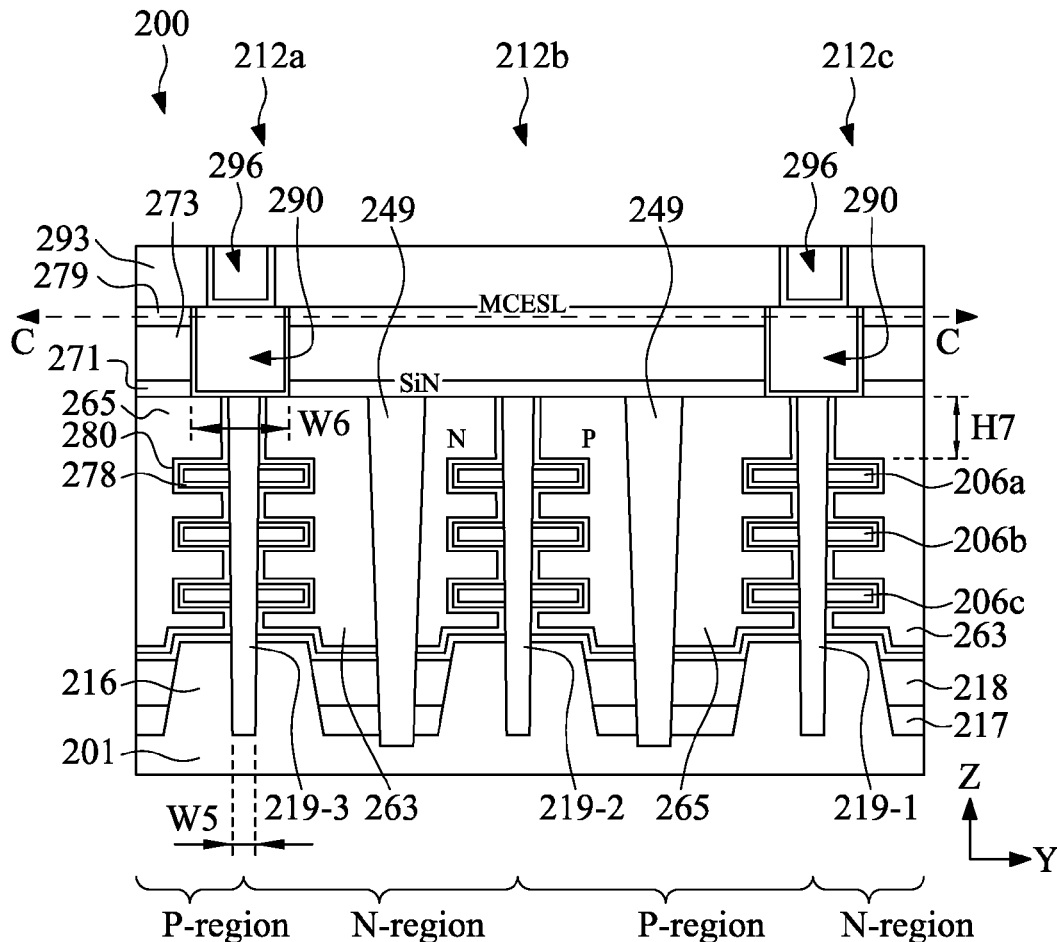
(57)

ABSTRACT

A semiconductor device structure includes a first dielectric wall, a plurality of first semiconductor layers vertically stacked and extending outwardly from a first side of the first dielectric wall, a plurality of second semiconductor layers vertically stacked and extending outwardly from a second side of the first dielectric wall. The structure also includes a first gate electrode layer surrounding at least three surfaces of each of the first semiconductor layers, the first gate electrode layer having a first conductivity type, and a second gate electrode layer surrounding at least three surfaces of each of the second semiconductor layers, the second gate electrode layer having a second conductivity type opposite the first conductivity type. The structure further includes a gate bridge contact disposed on the first dielectric wall, and a gate via contact disposed on the gate bridge contact.

(21) Appl. No.: **18/099,952**(22) Filed: **Jan. 22, 2023****Related U.S. Application Data**

(60) Provisional application No. 63/428,638, filed on Nov. 29, 2022.

Publication Classification(51) **Int. Cl.***H01L 23/522* (2006.01)*H01L 21/8234* (2006.01)*H01L 29/06* (2006.01)

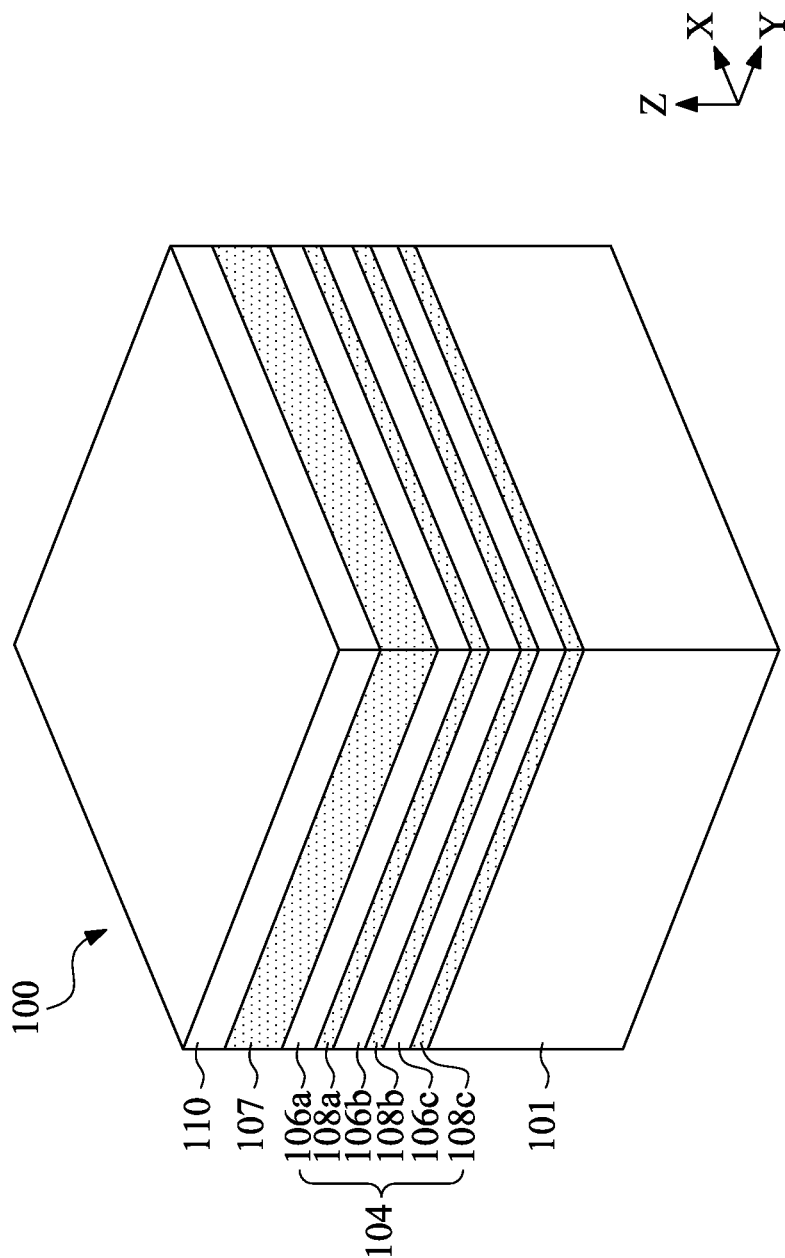


Fig. 1

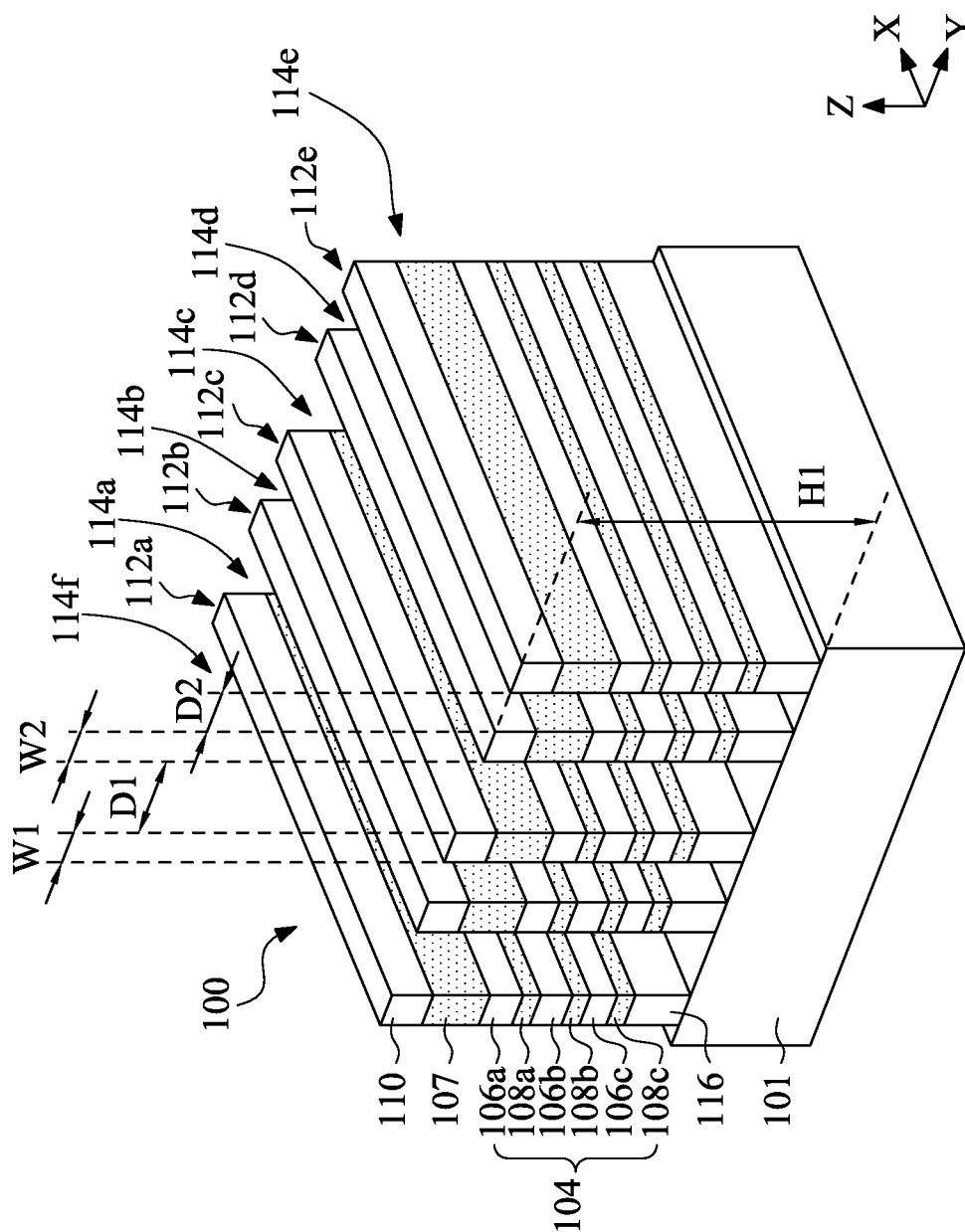


Fig. 2

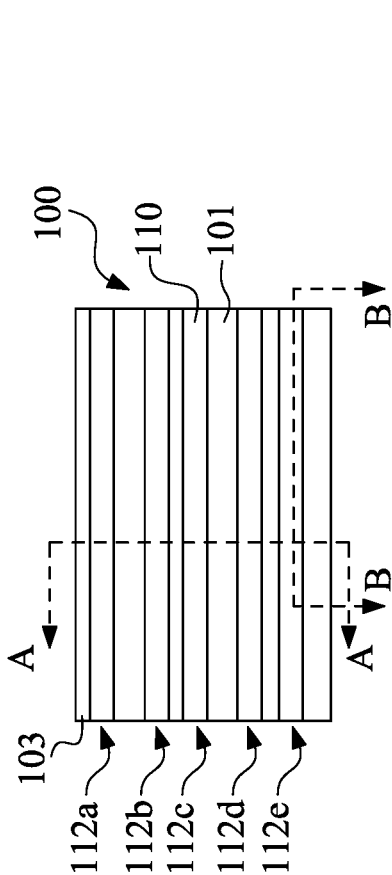


Fig. 3C

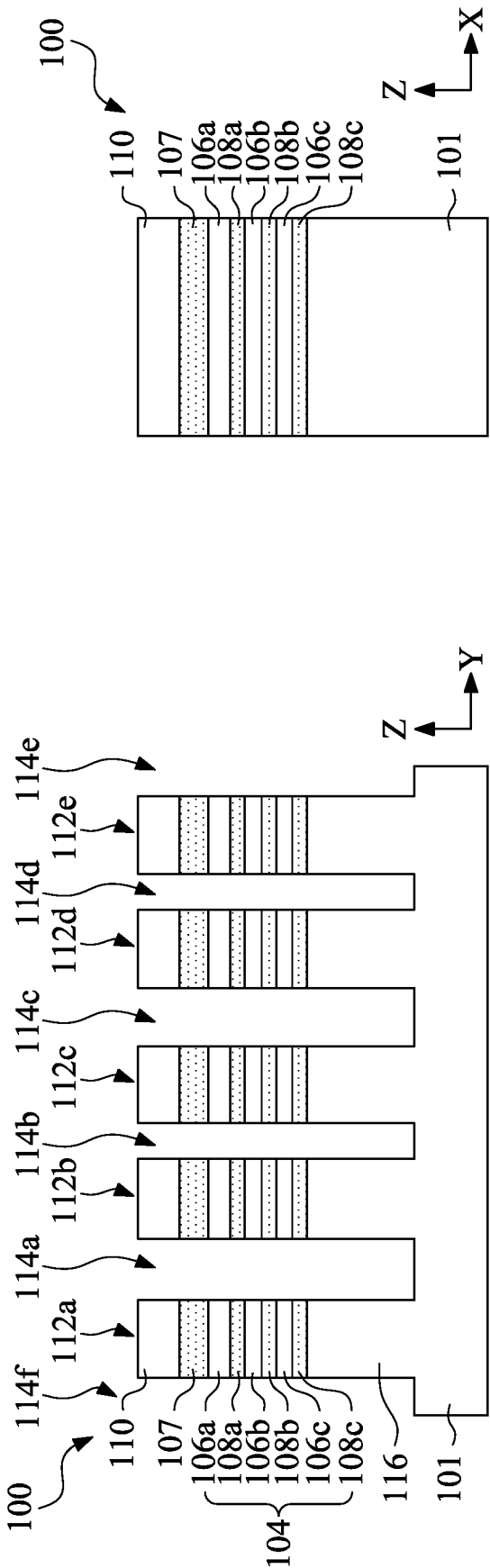


Fig. 3B

Fig. 3A

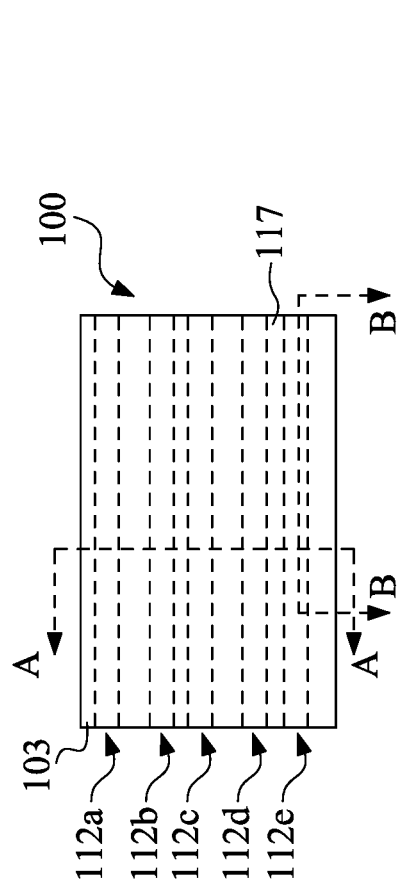


Fig. 4C

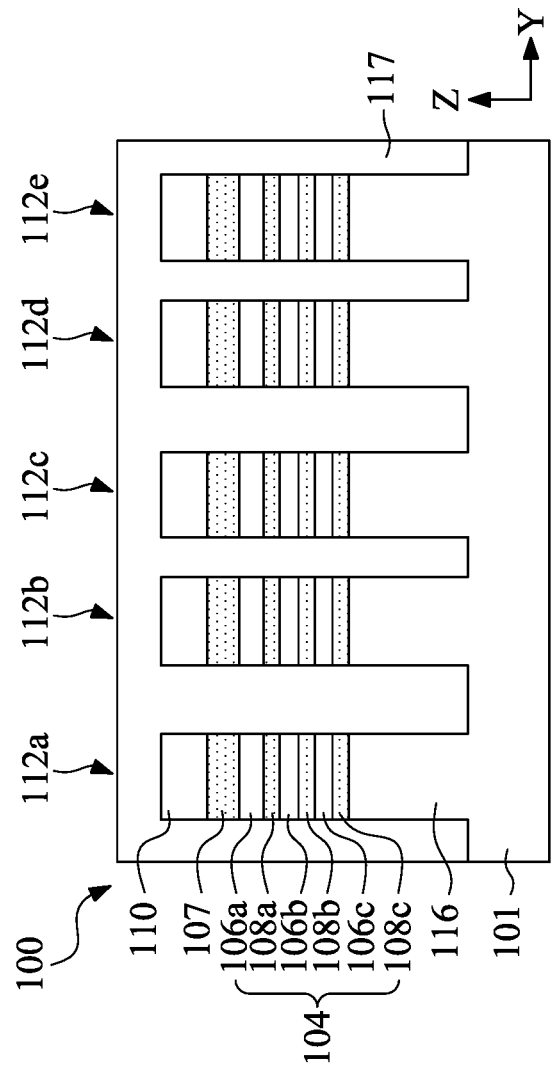


Fig. 4A

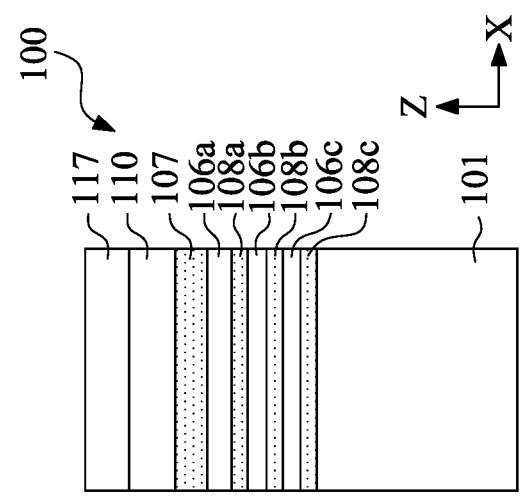


Fig. 4B

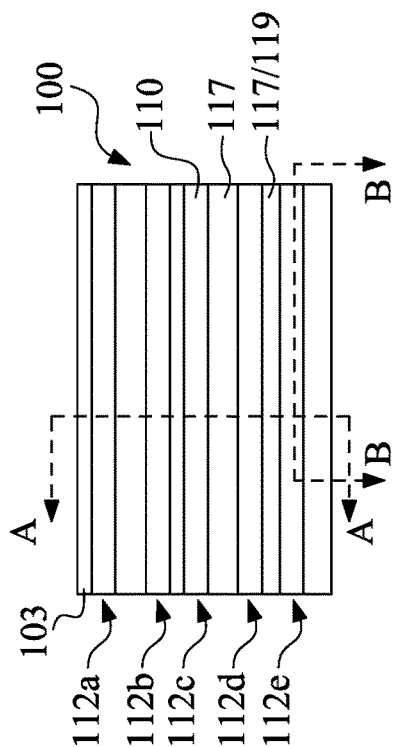


Fig. 5C

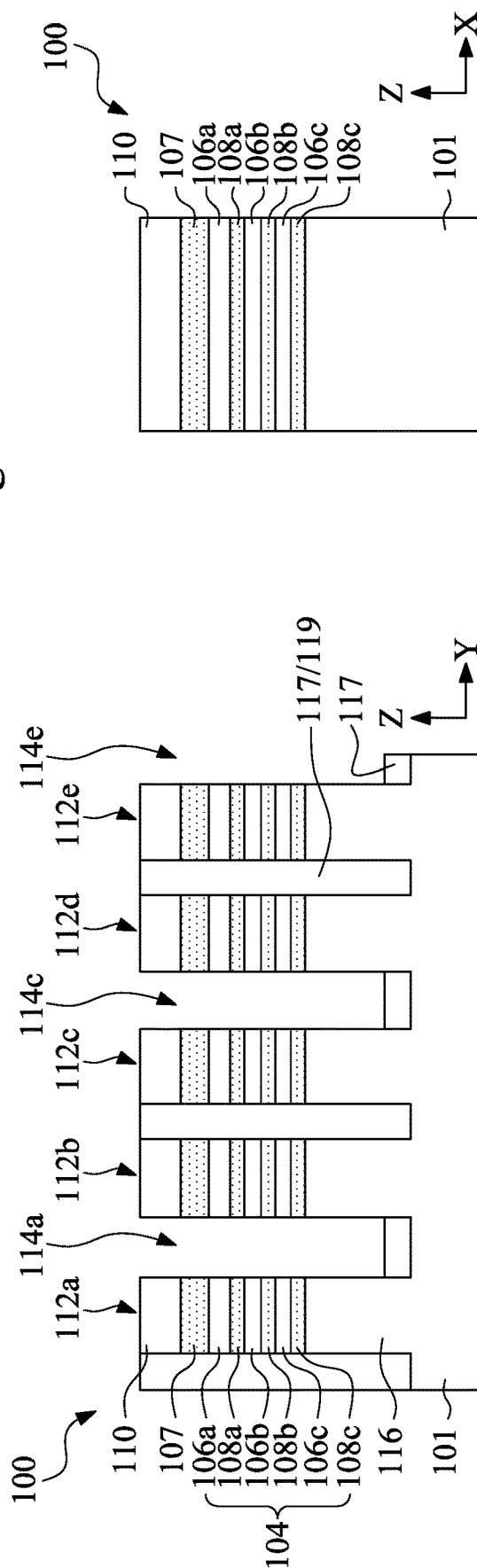


Fig. 5A

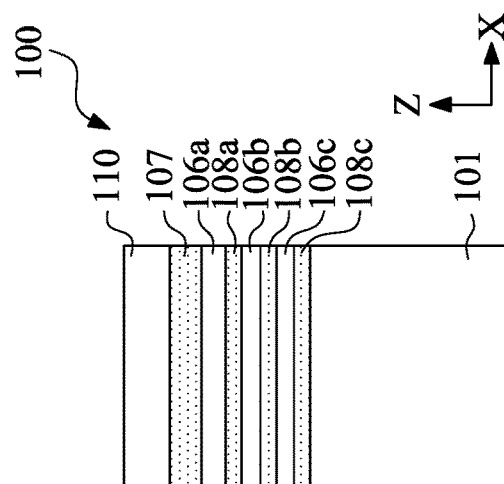


Fig. 5B

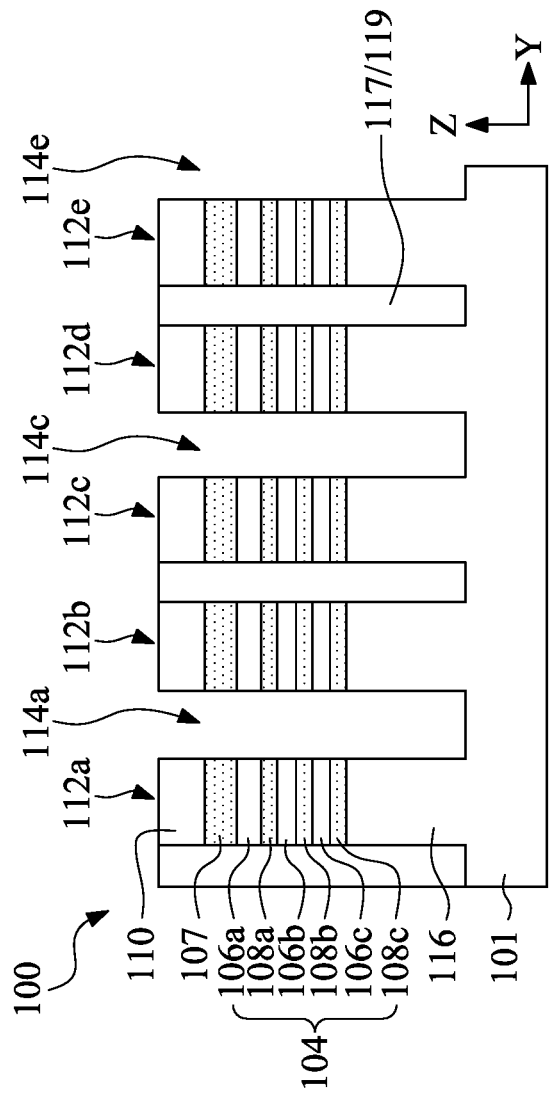


Fig. 5A-1

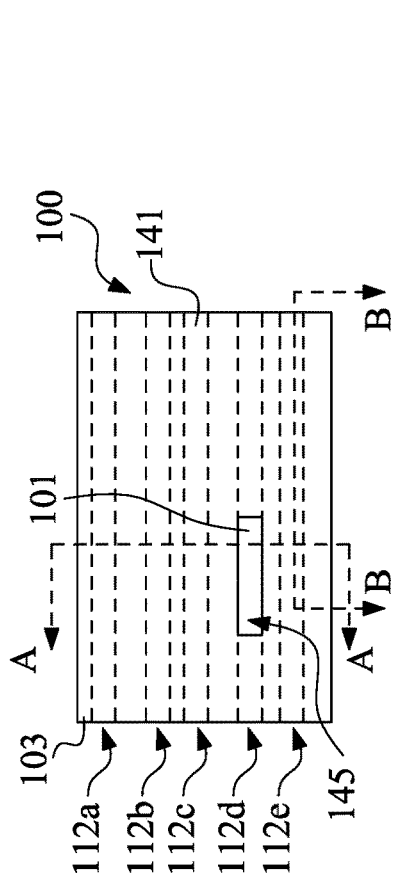


Fig. 6C

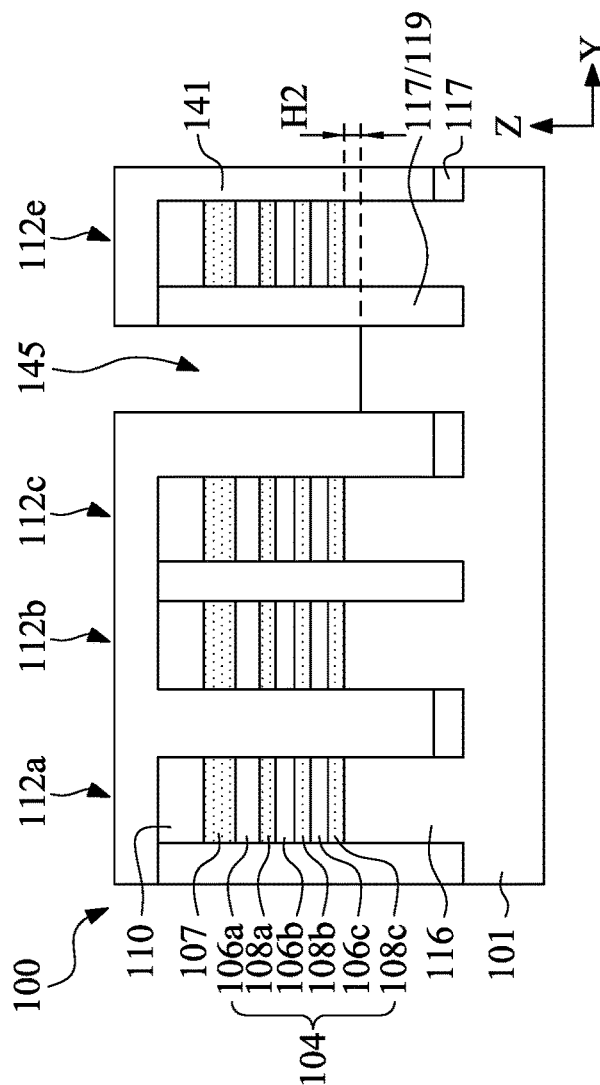


Fig. 6A

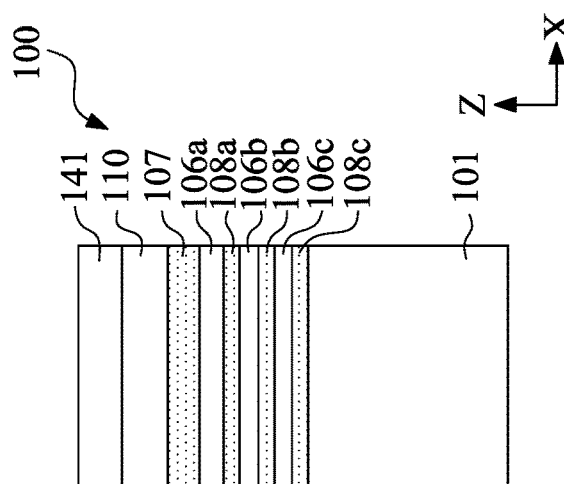


Fig. 6B

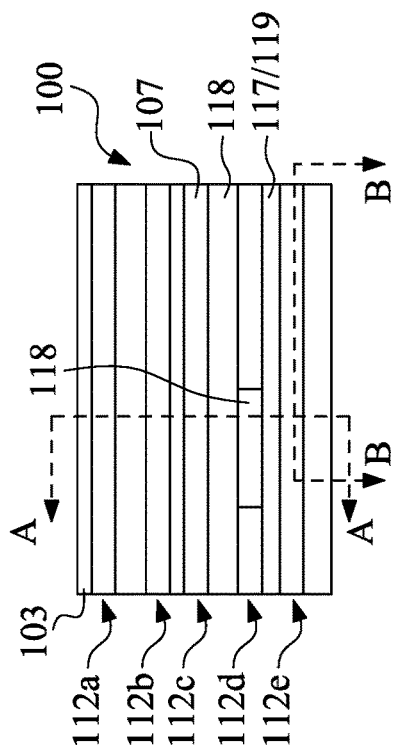


Fig. 7C

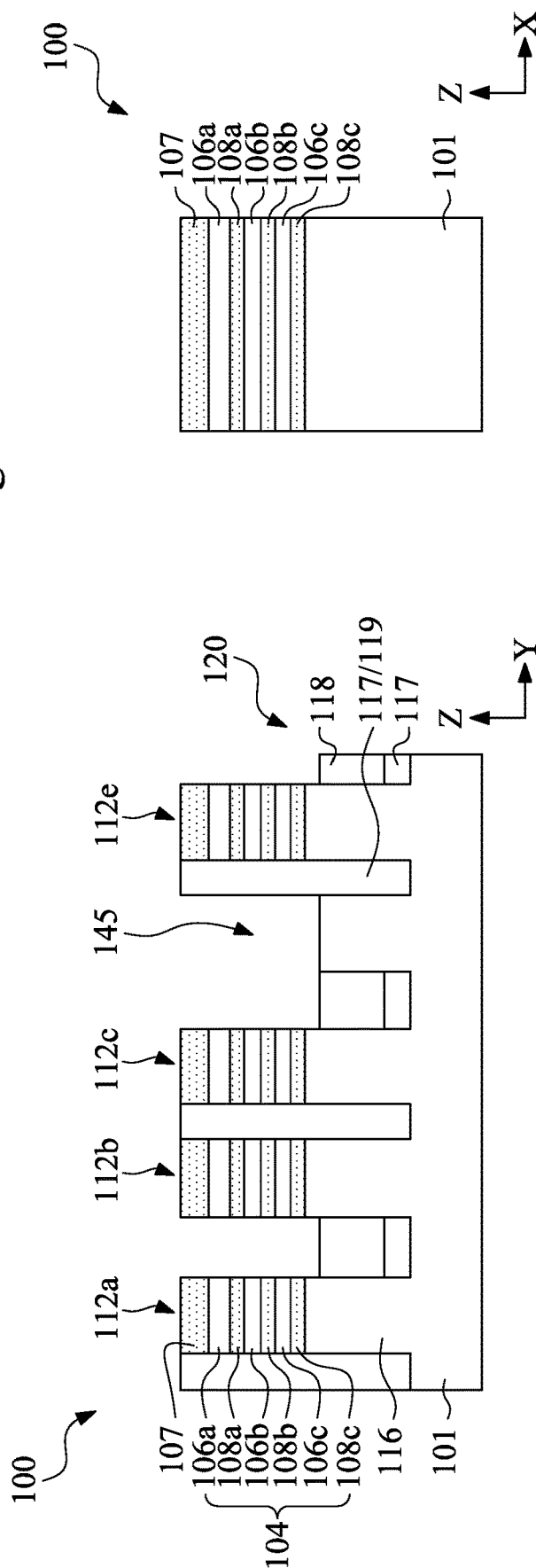


Fig. 7A

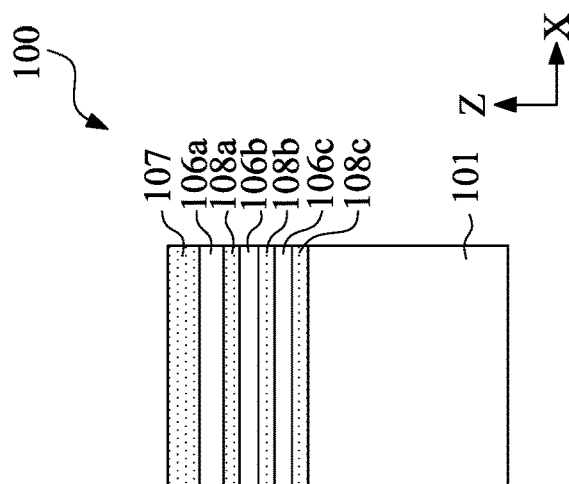


Fig. 7B

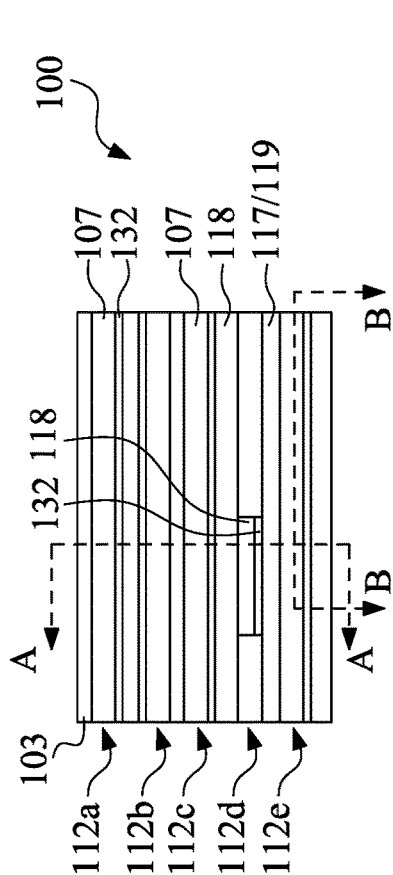


Fig. 8C

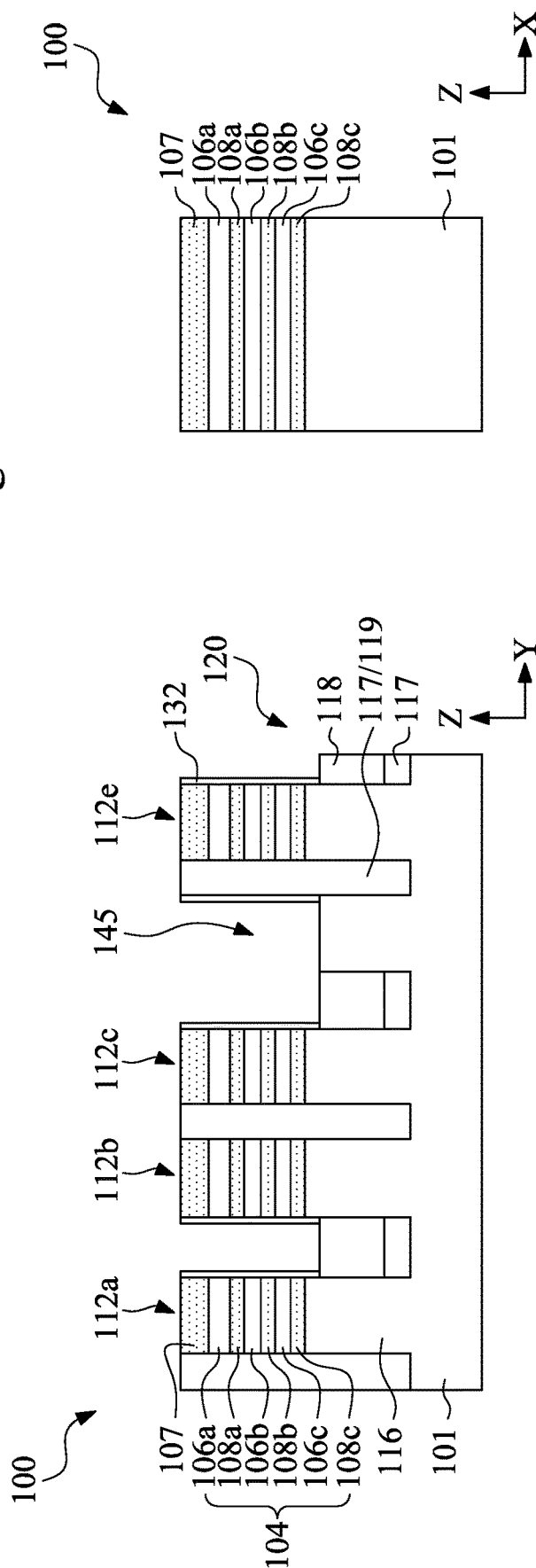


Fig. 8A

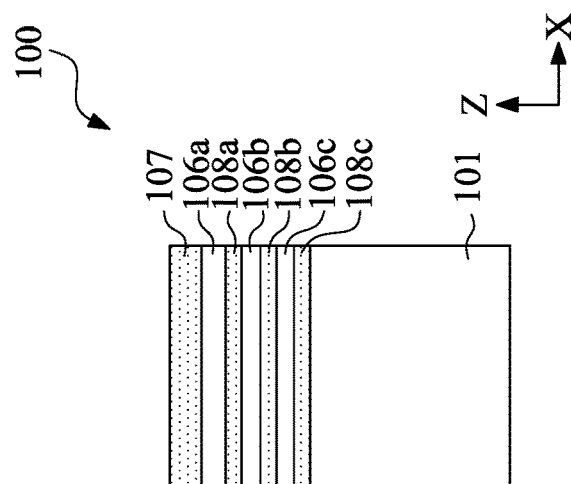


Fig. 8B

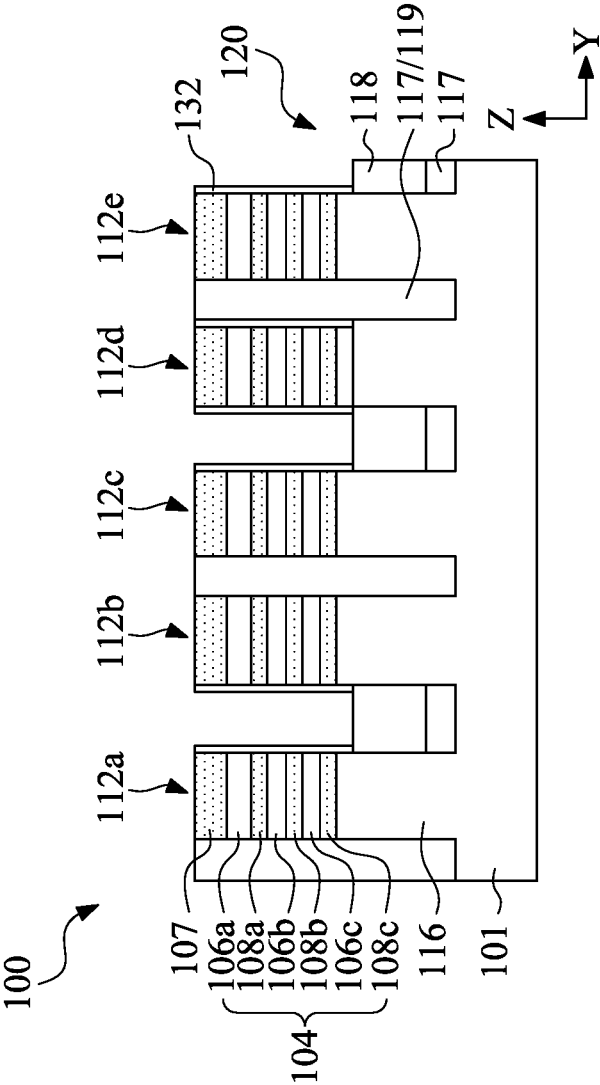


Fig. 8A-1

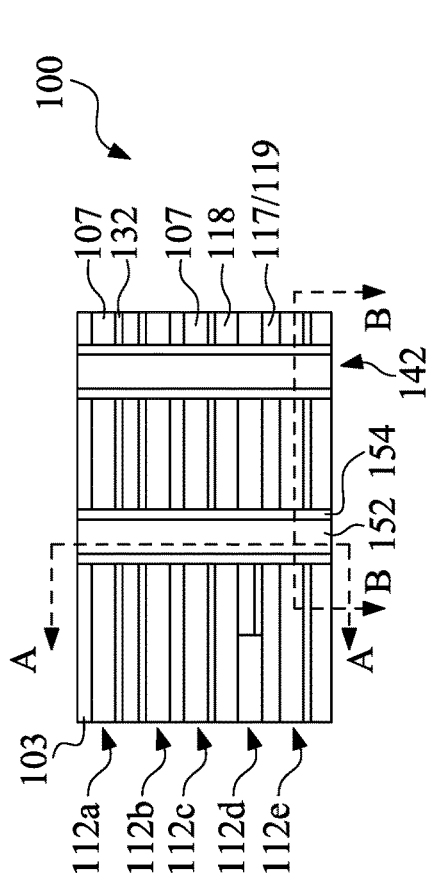


Fig. 9C

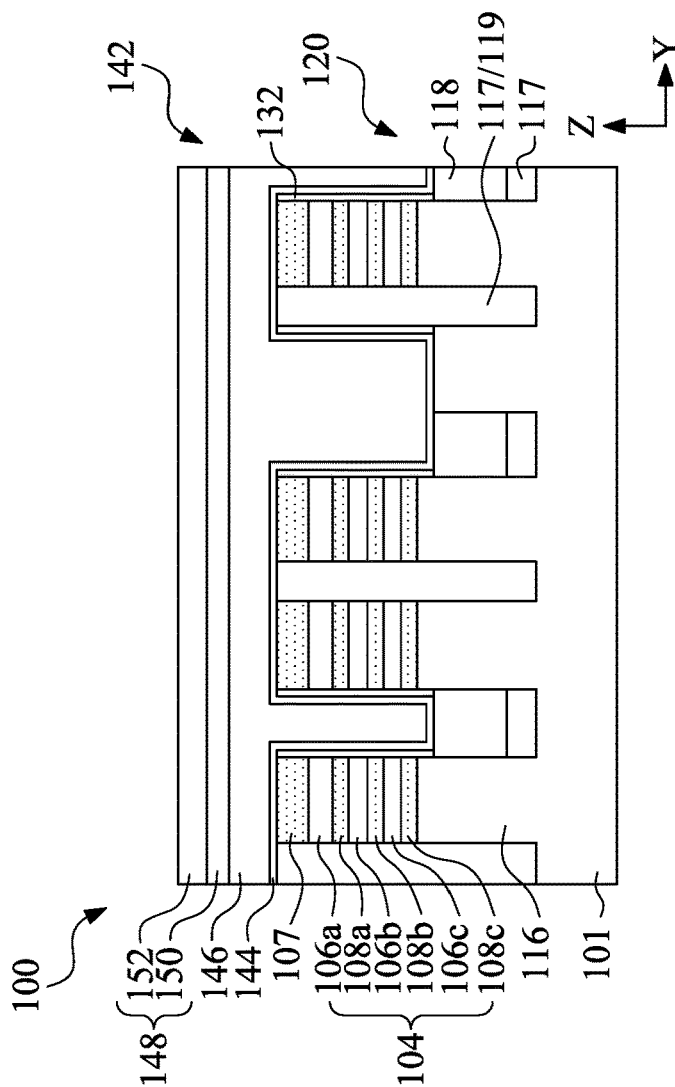


Fig. 9A

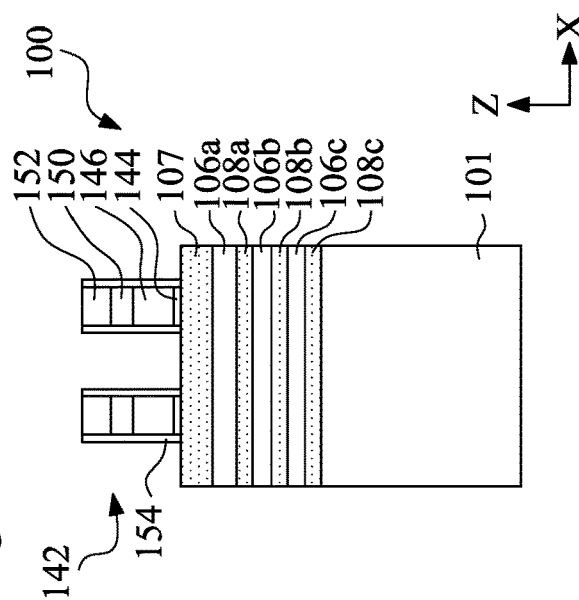


Fig. 9B

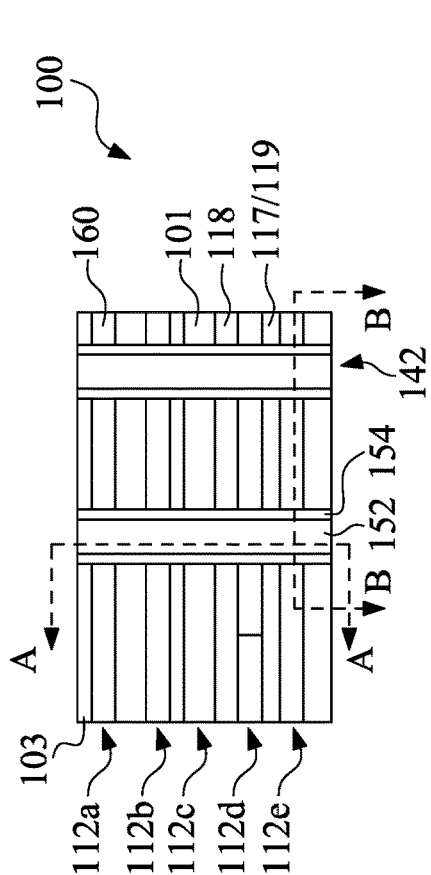


Fig. 12C

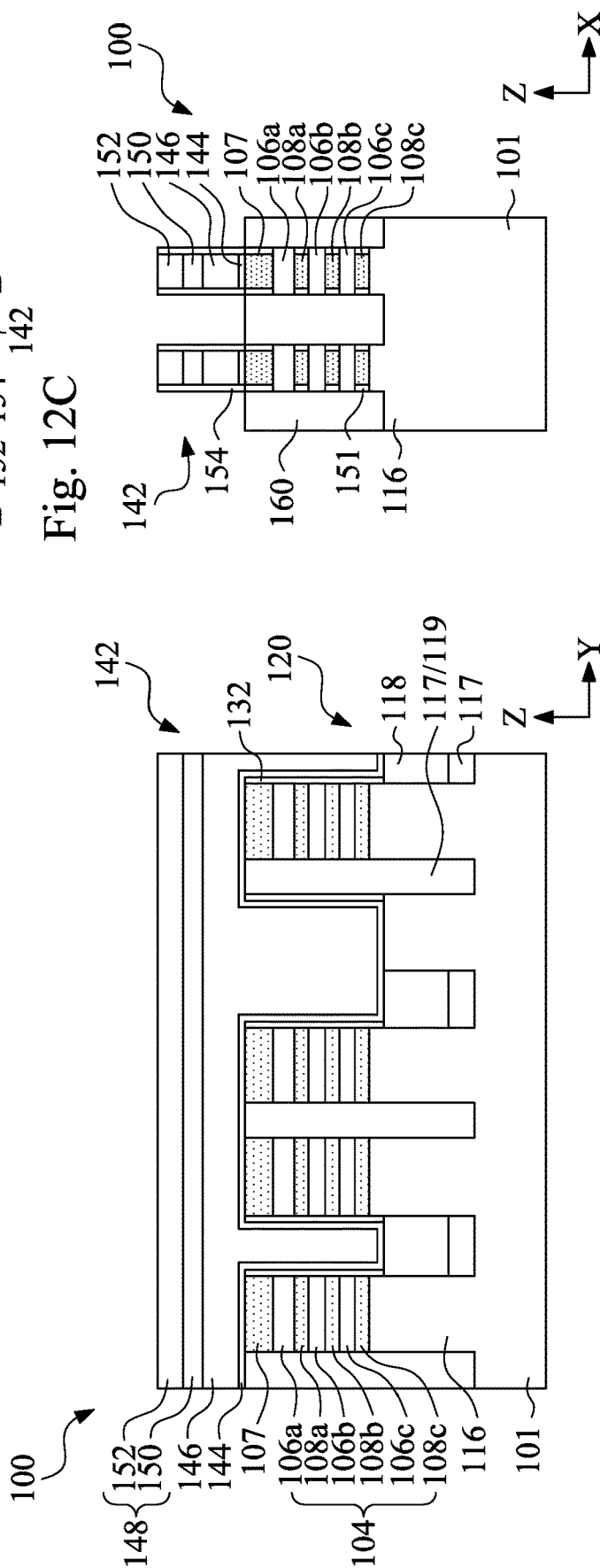


Fig. 12A

Fig. 12B

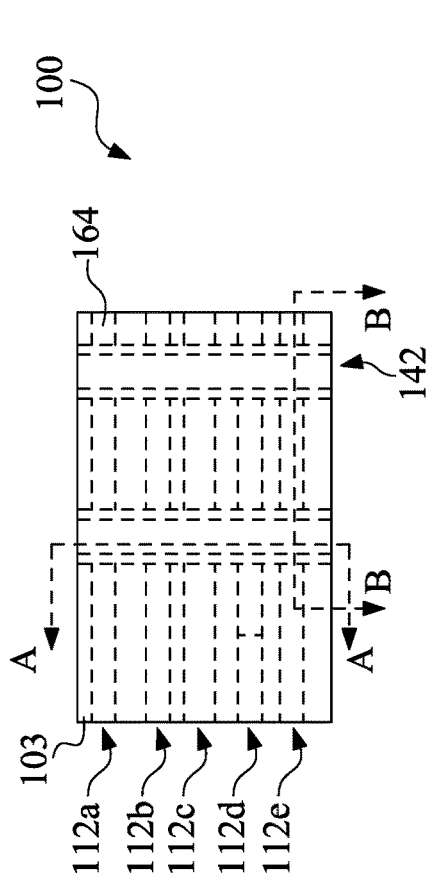


Fig. 13C

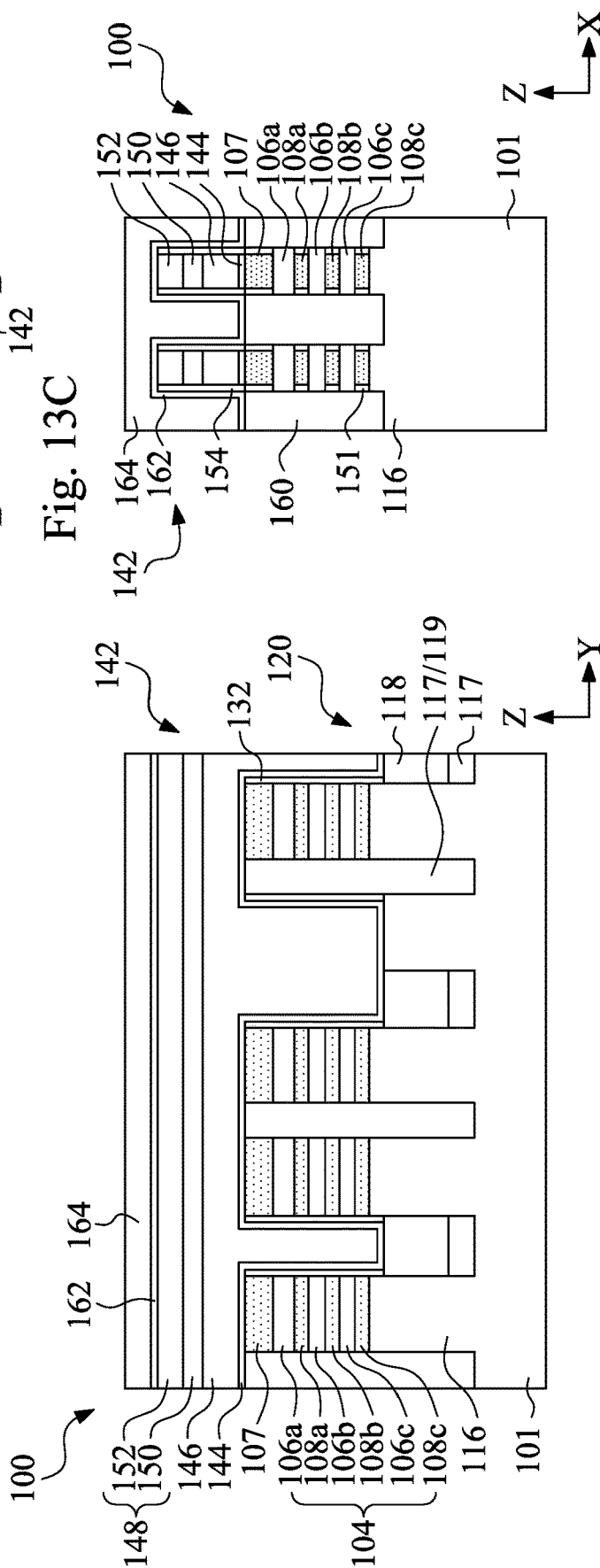


Fig. 13A

Fig. 13B

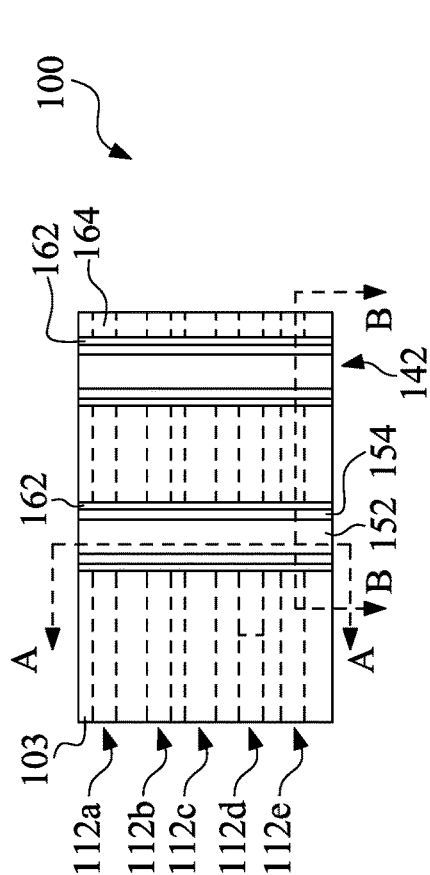


Fig. 14C

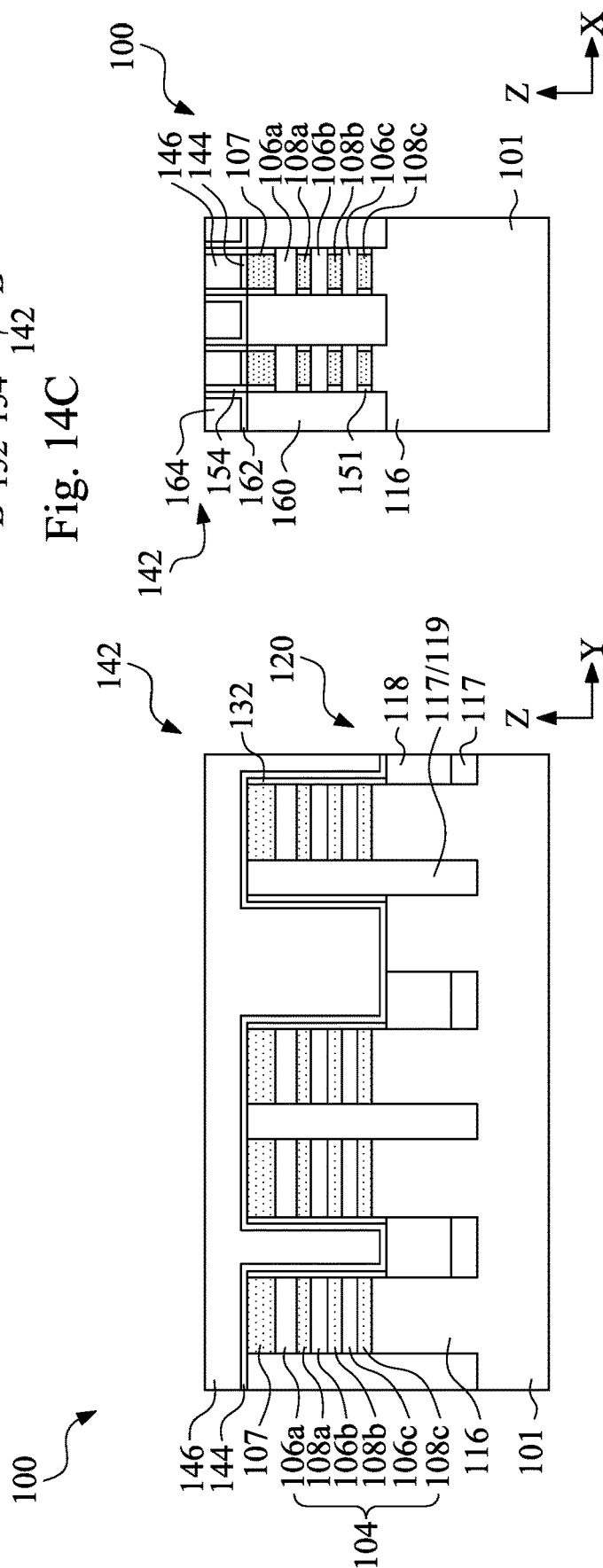


Fig. 14A

Fig. 14B

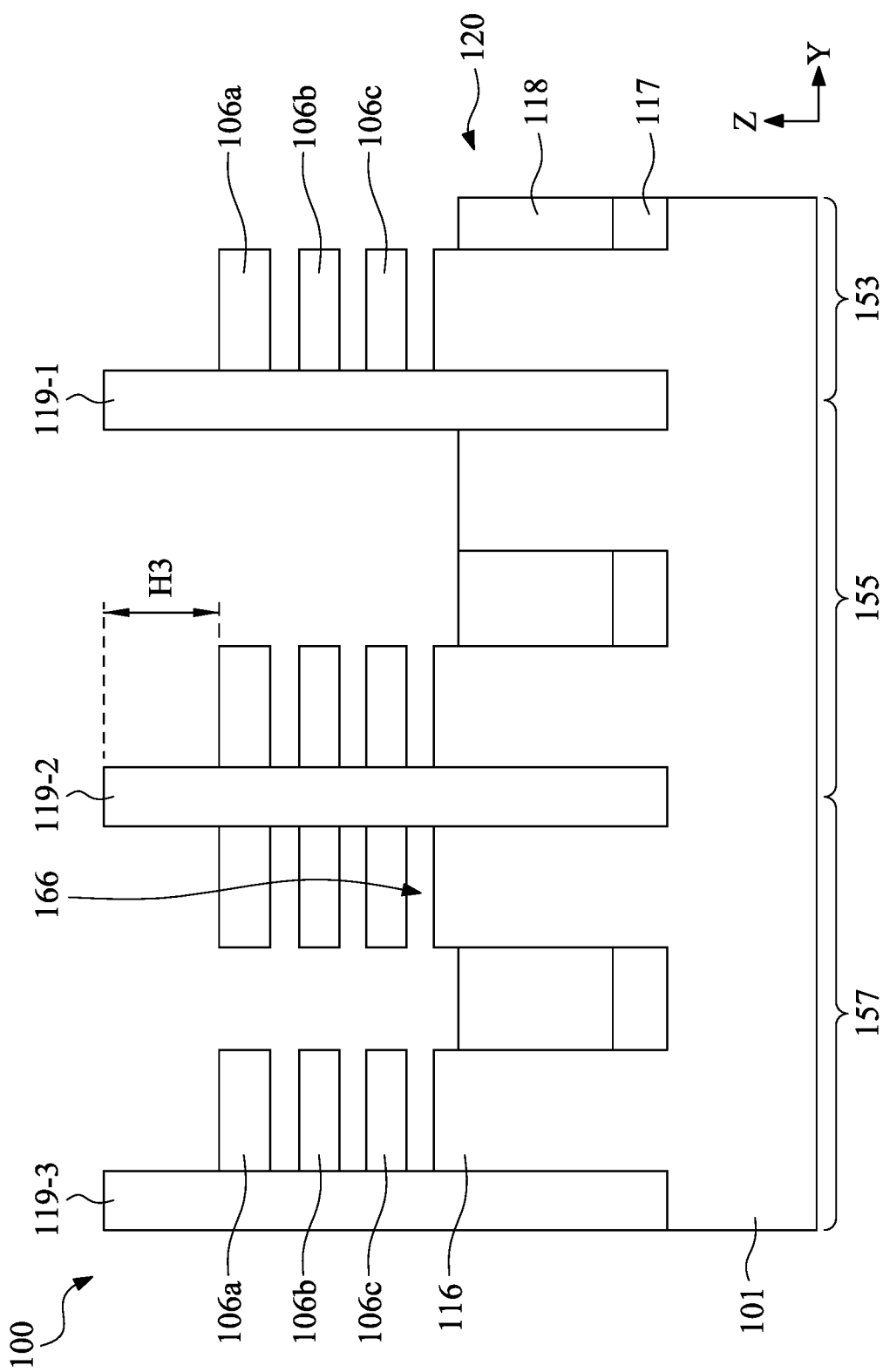


Fig. 15A

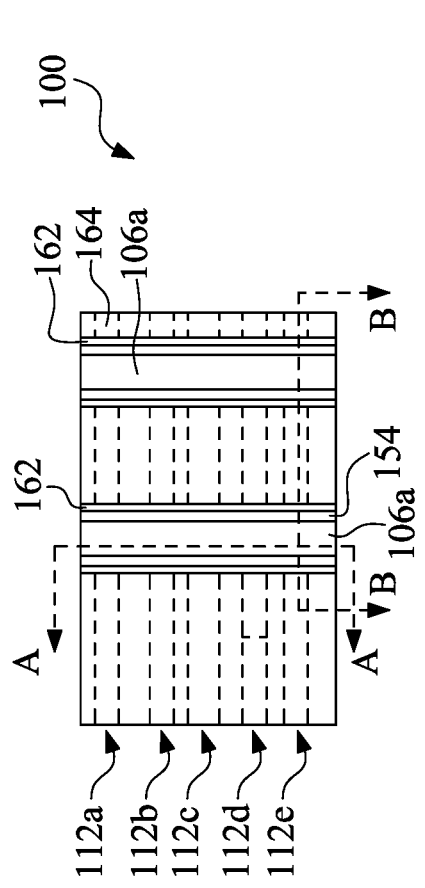


Fig. 15C

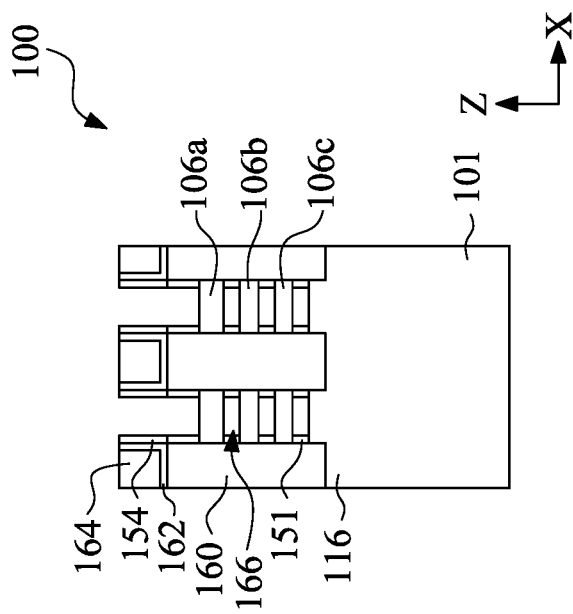


Fig. 15B

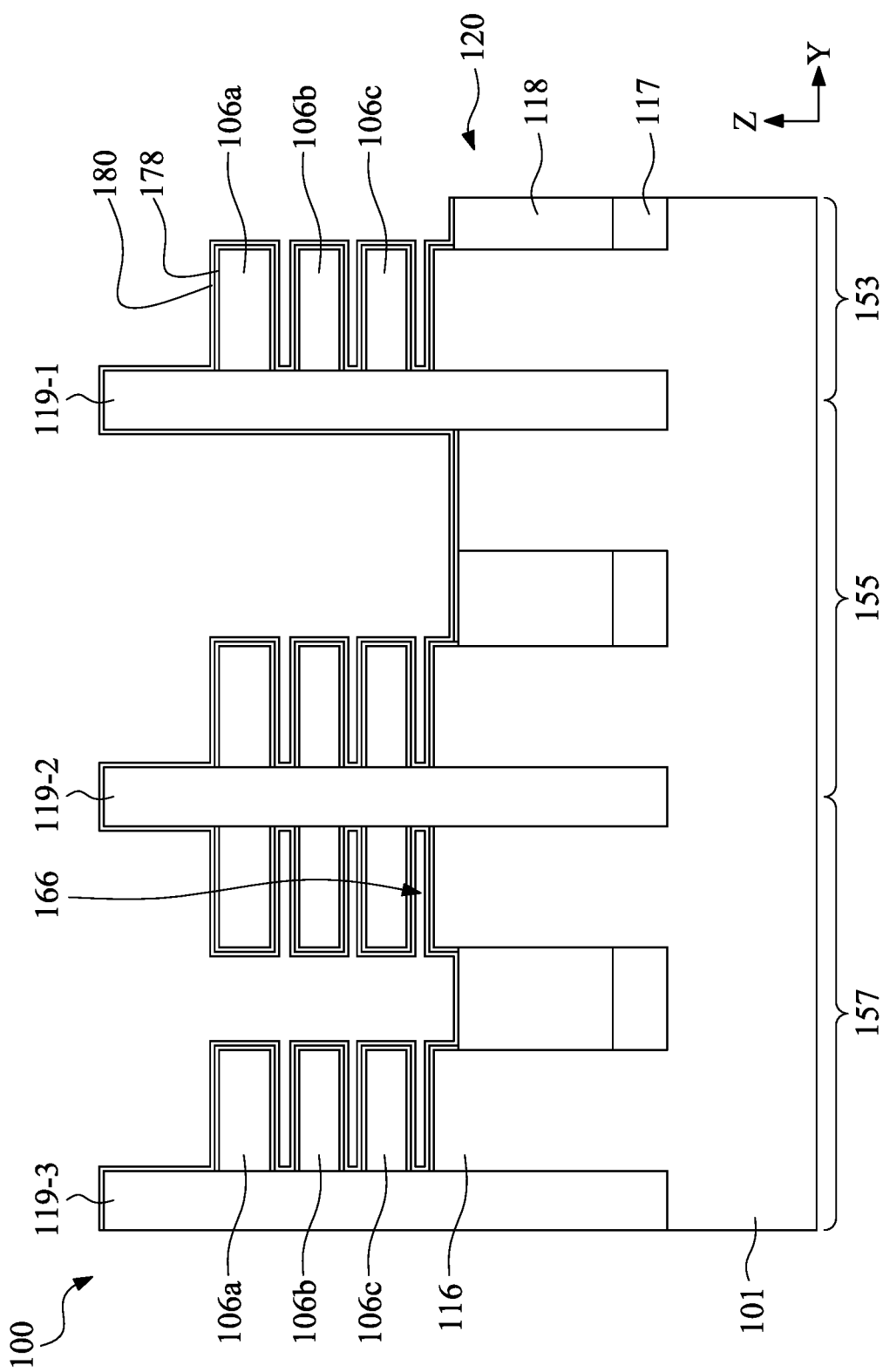


Fig. 16A

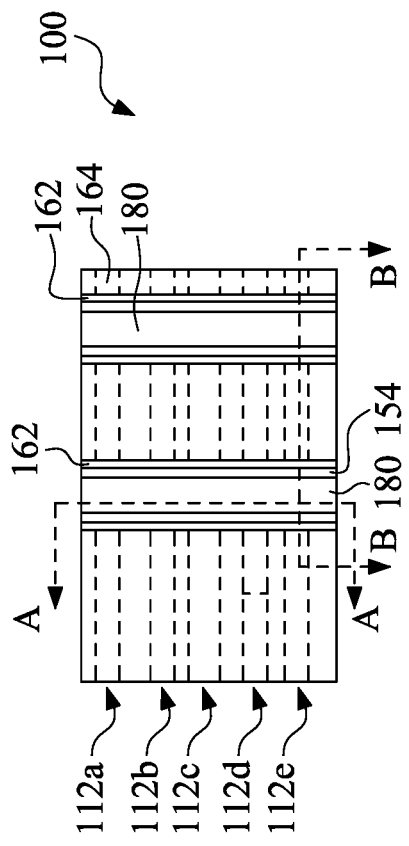


Fig. 16C

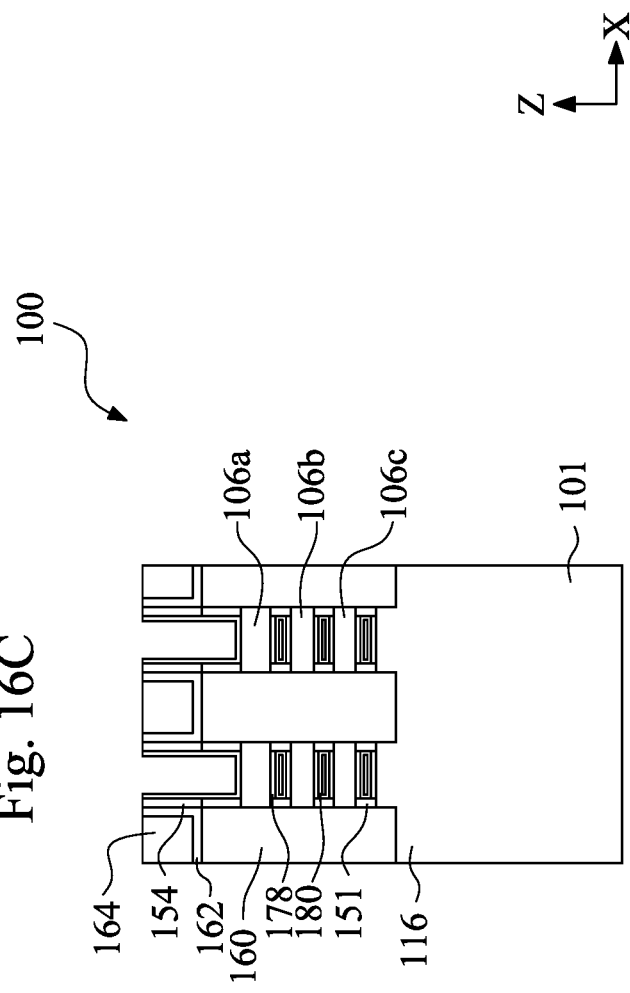


Fig. 16B

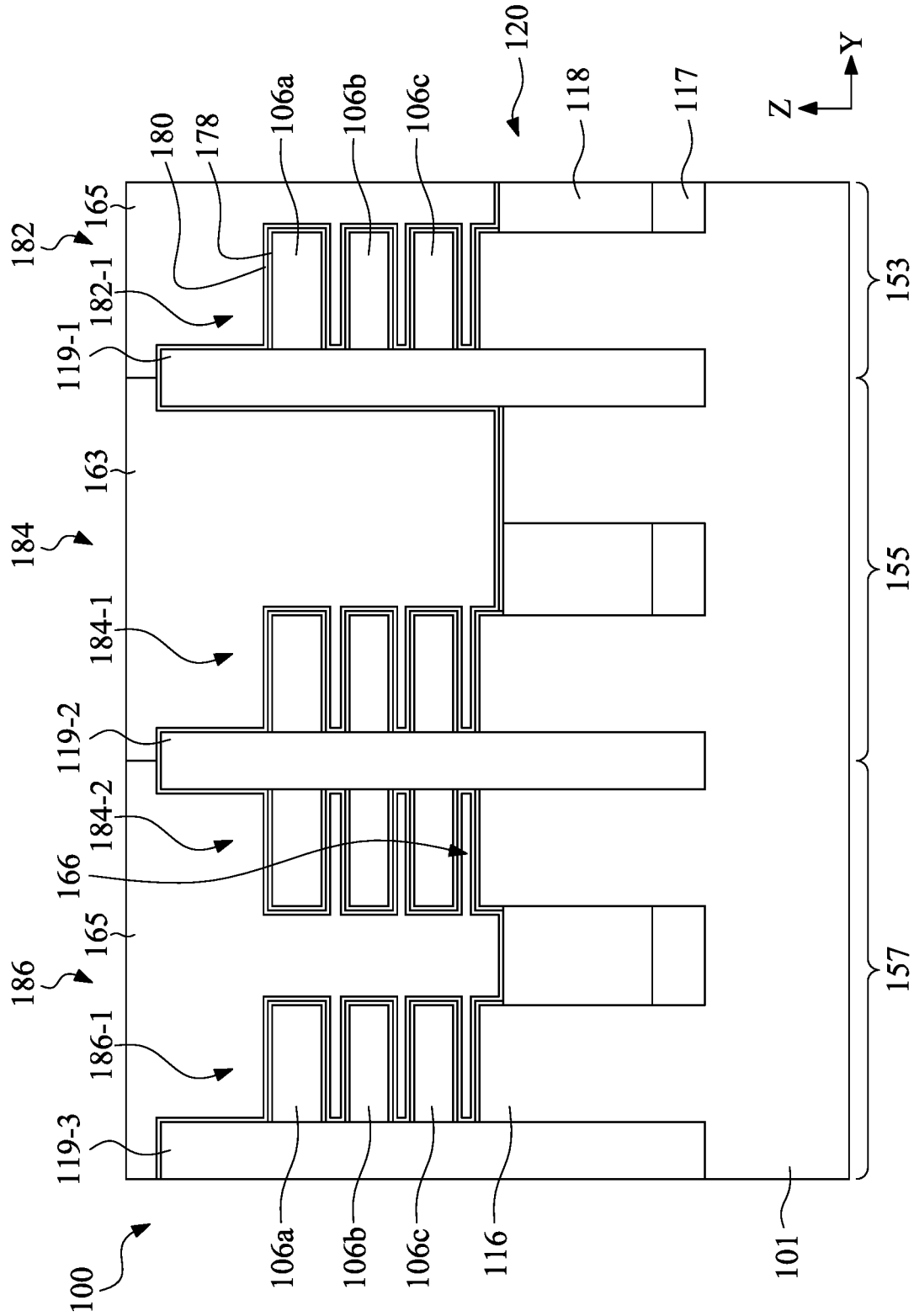


Fig. 17A

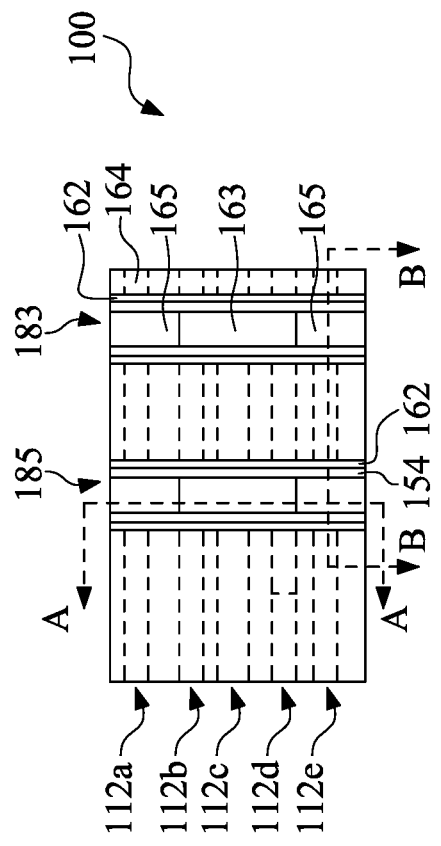


Fig. 17C

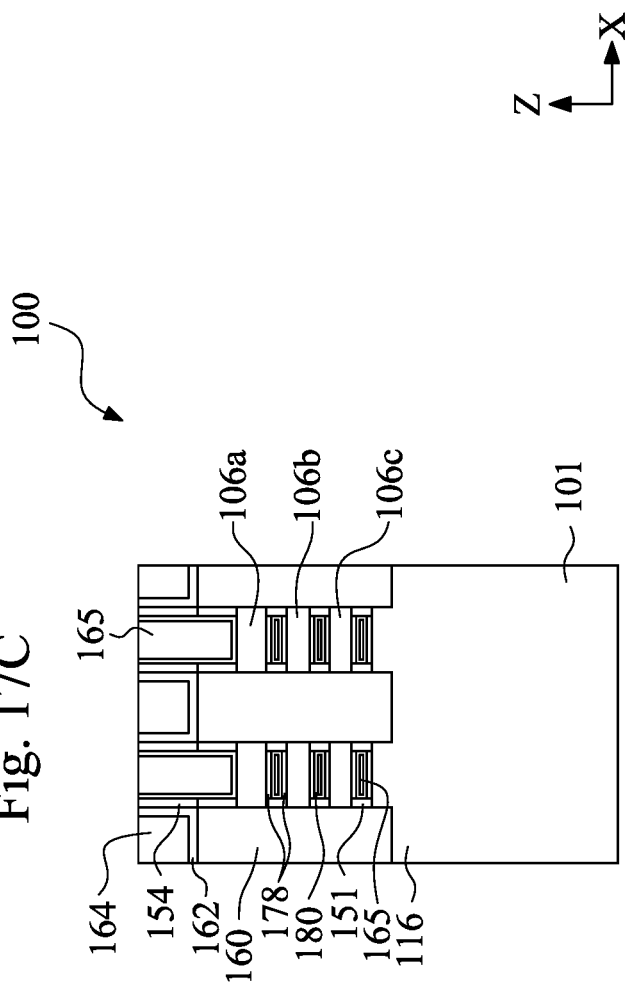


Fig. 17B

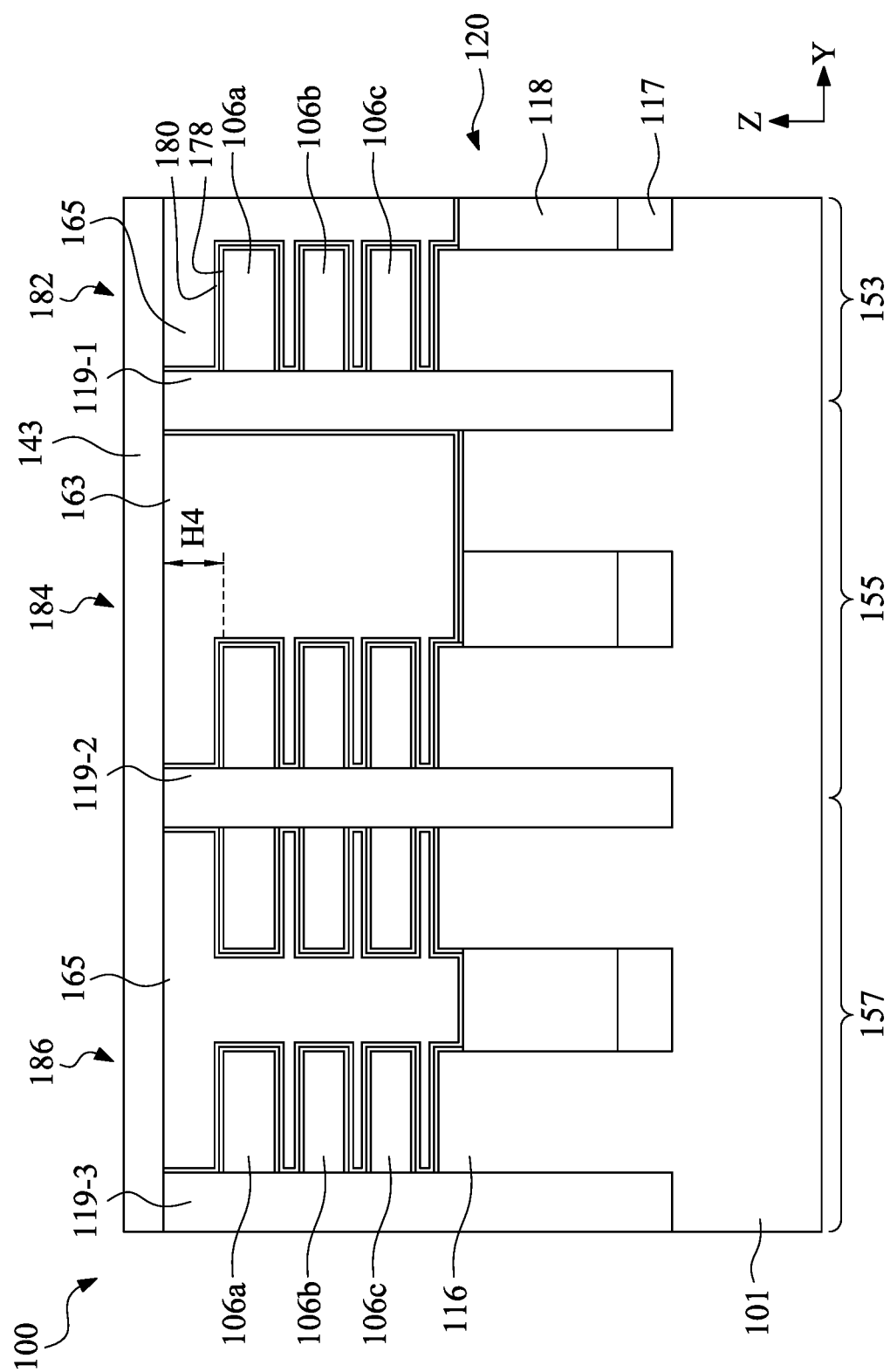


Fig. 18A

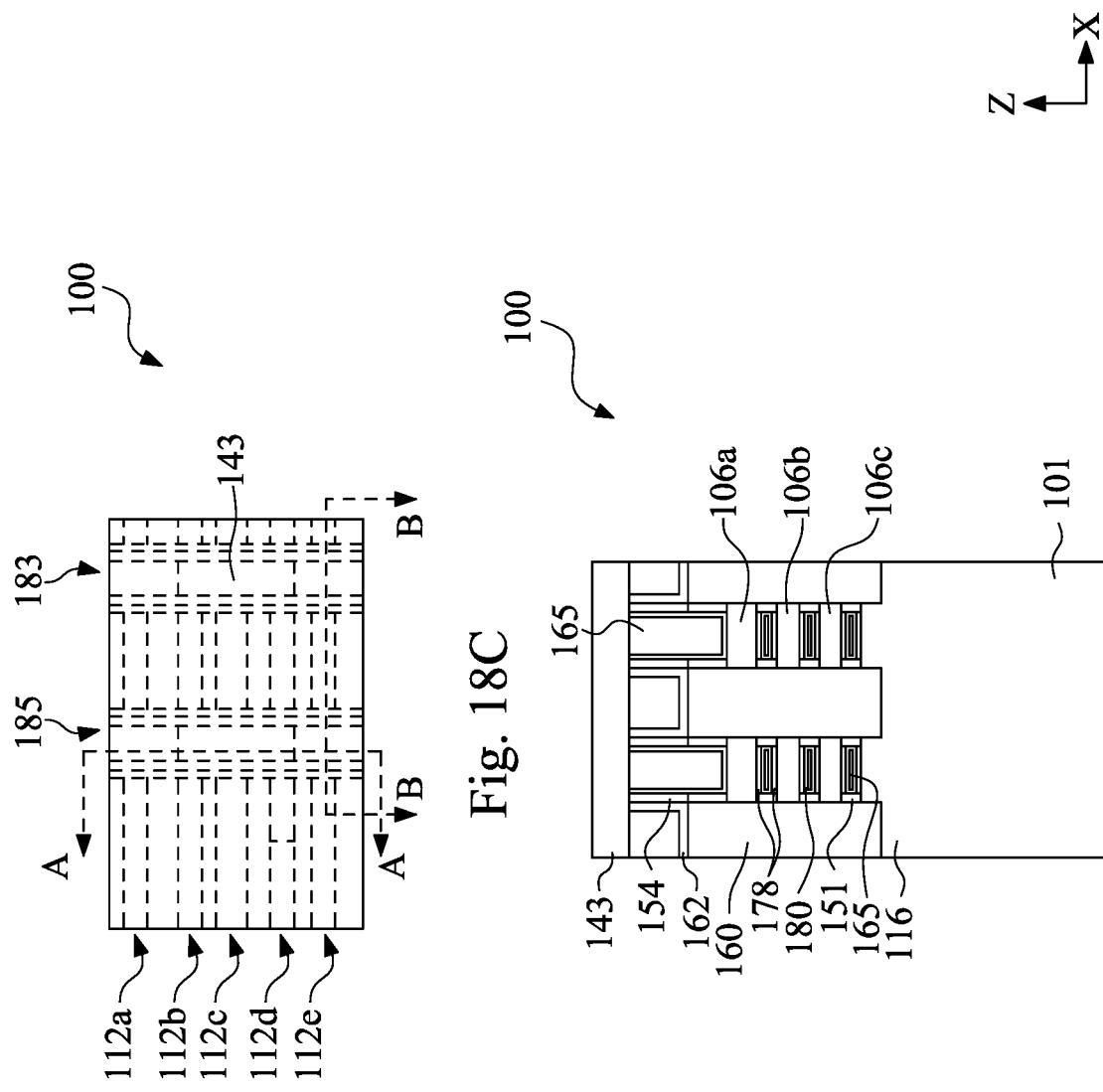


Fig. 18C

Fig. 18B

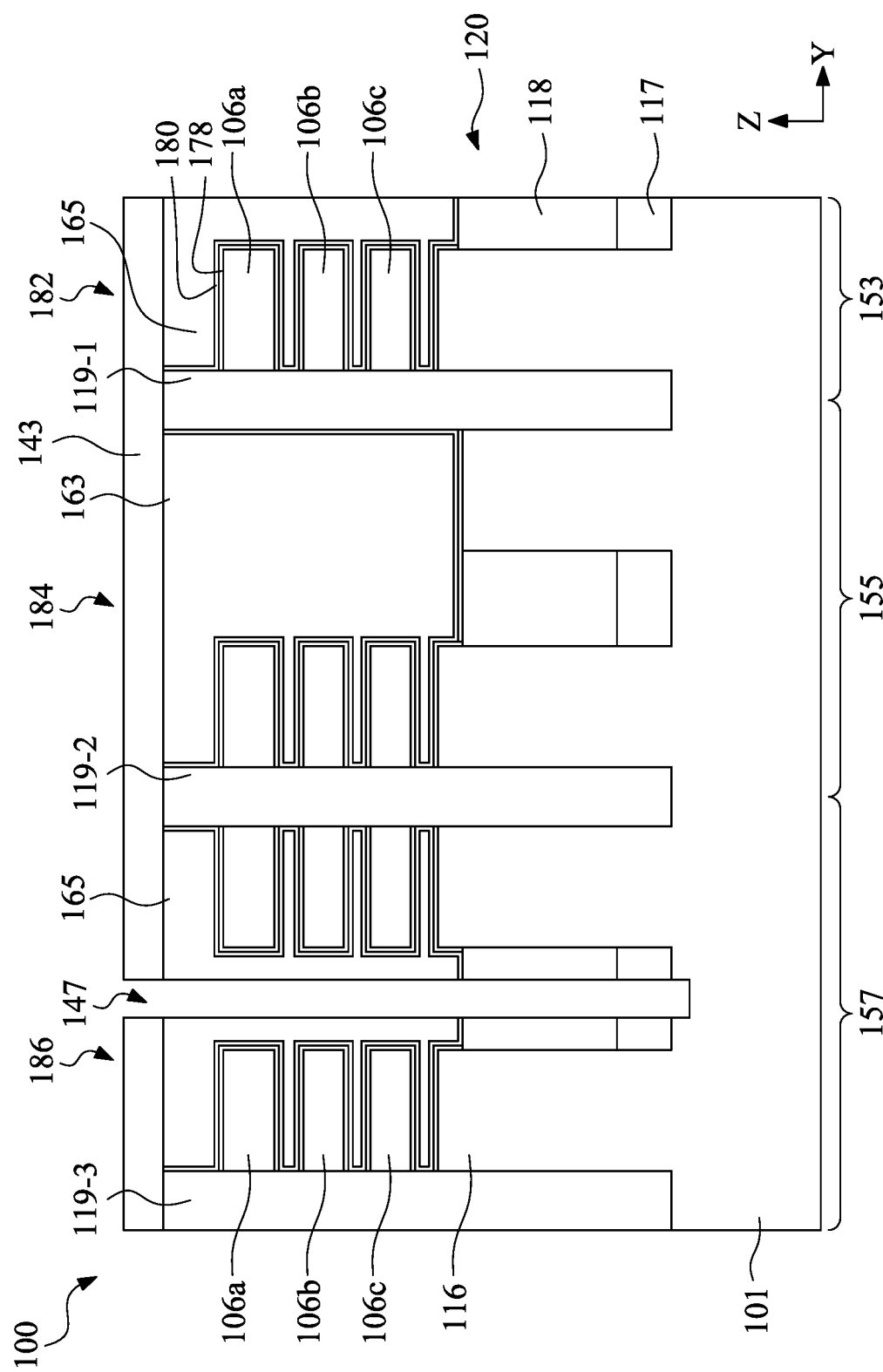


Fig. 19A

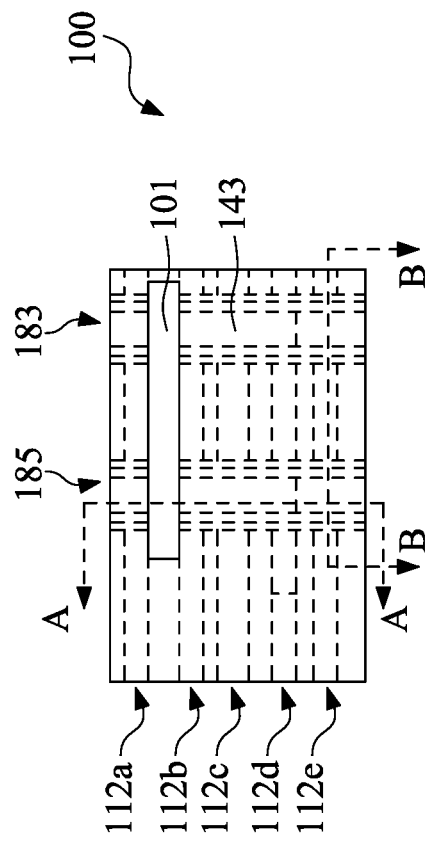


Fig. 19C

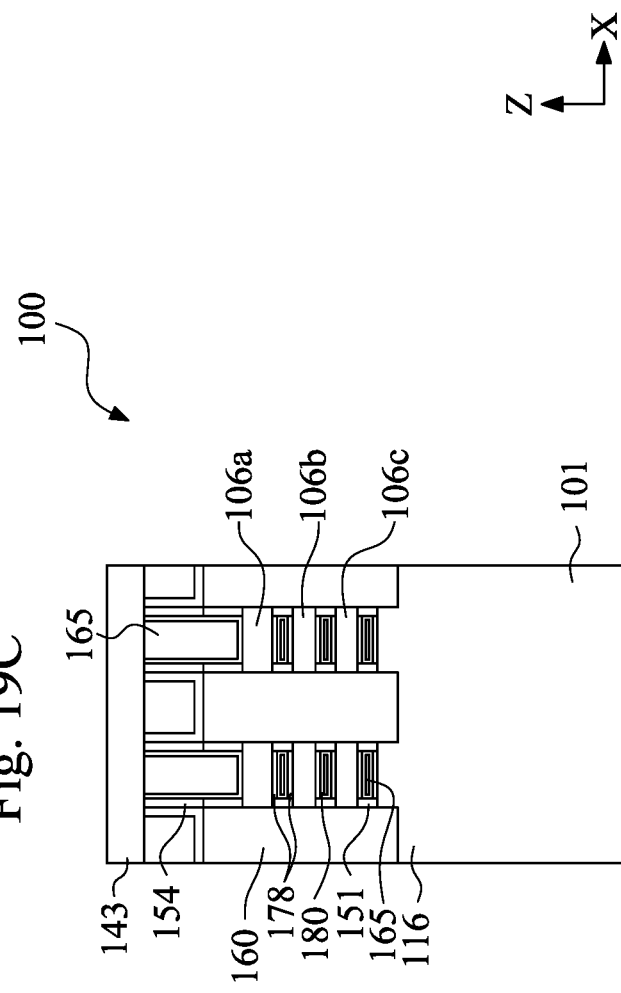


Fig. 19B

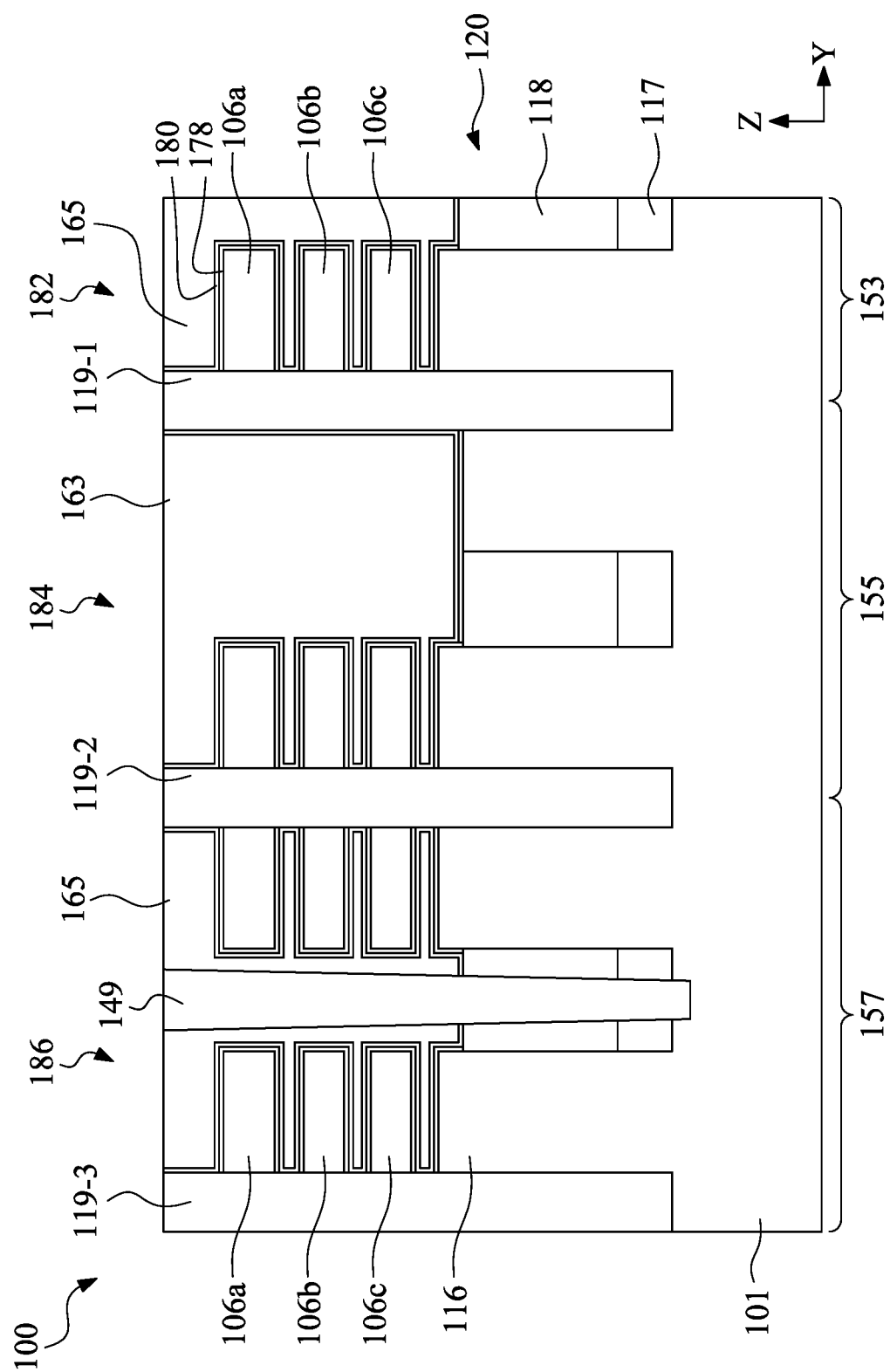


Fig. 20A

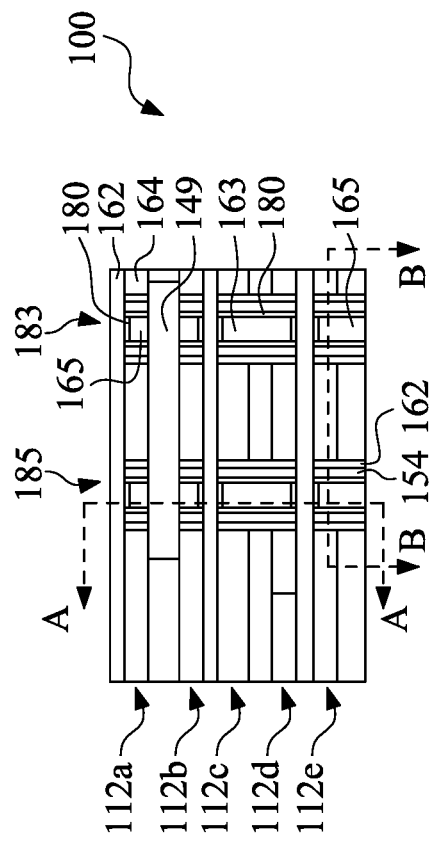


Fig. 20C

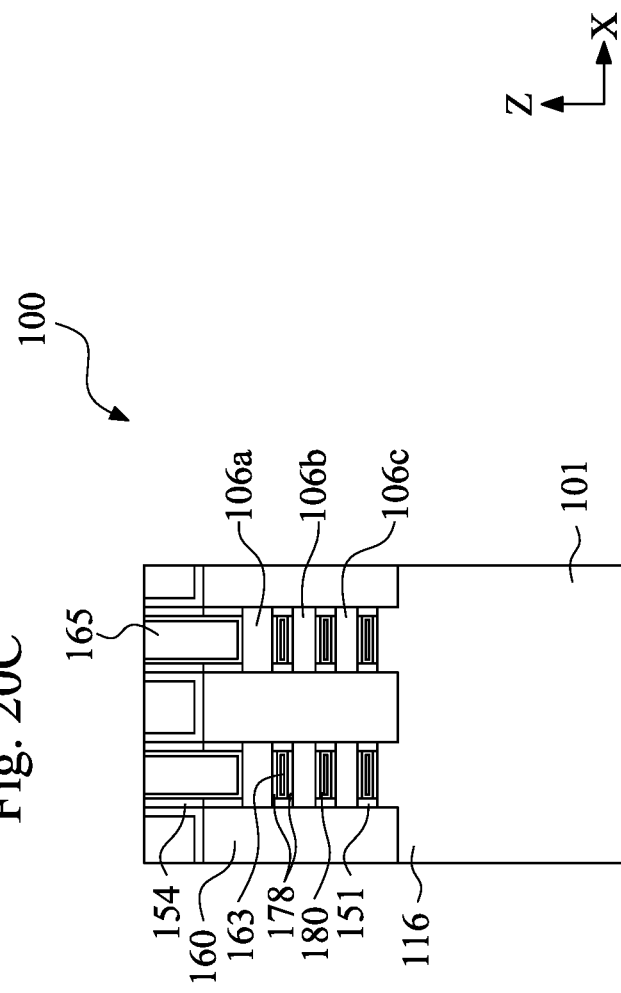


Fig. 20B

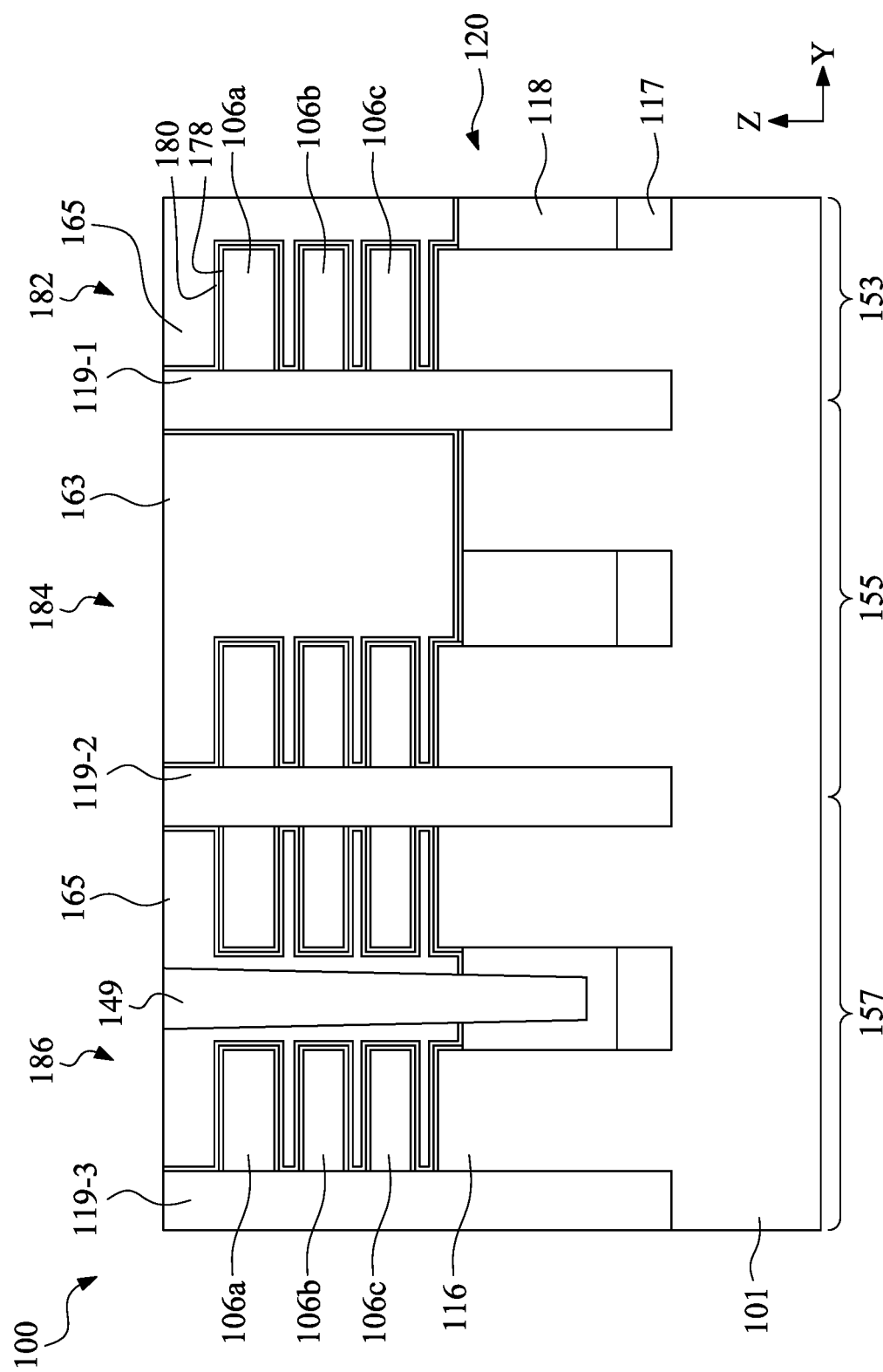


Fig. 20A-1

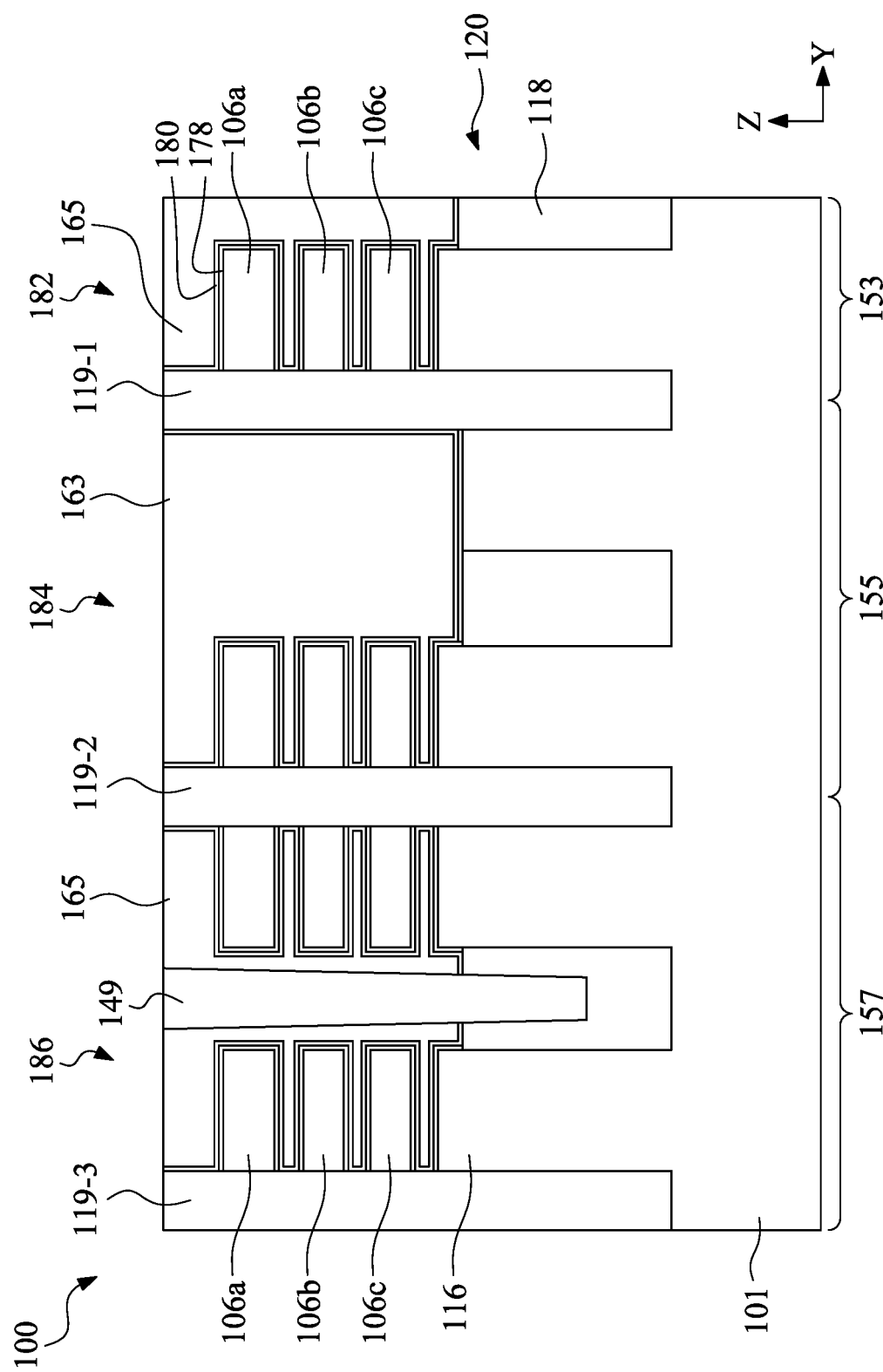


Fig. 20A-2

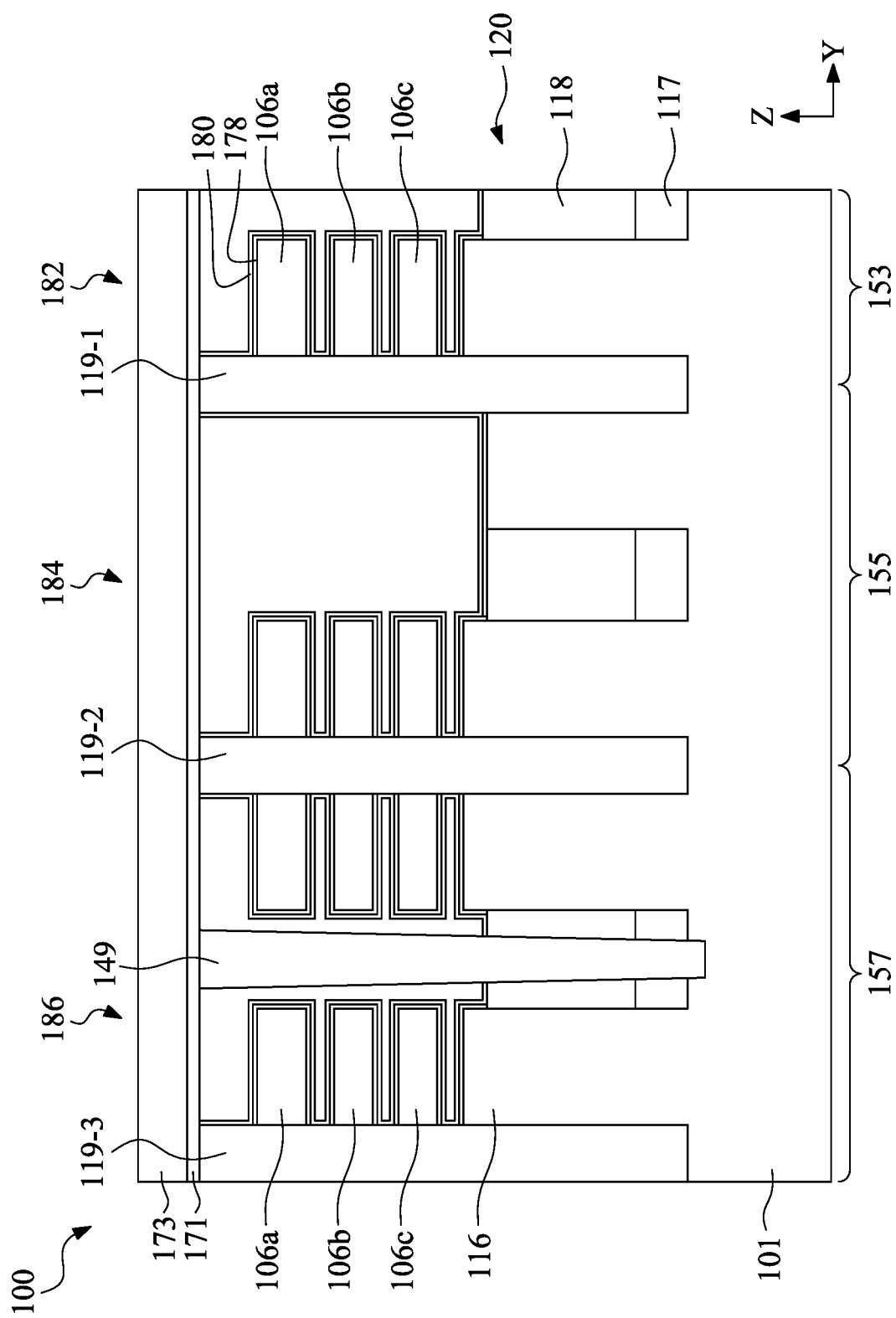


Fig. 21A

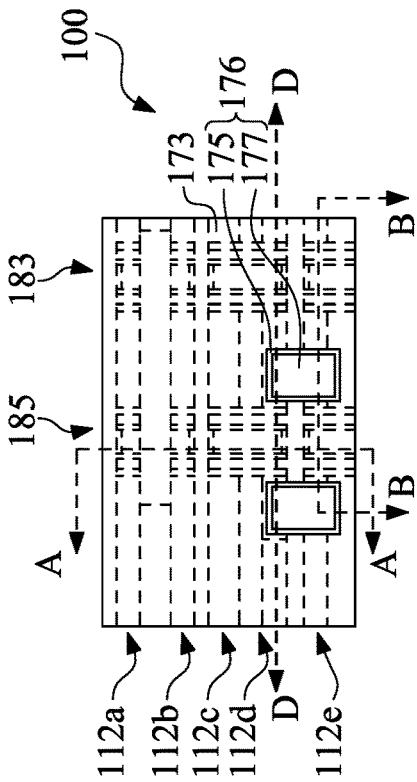


Fig. 21C

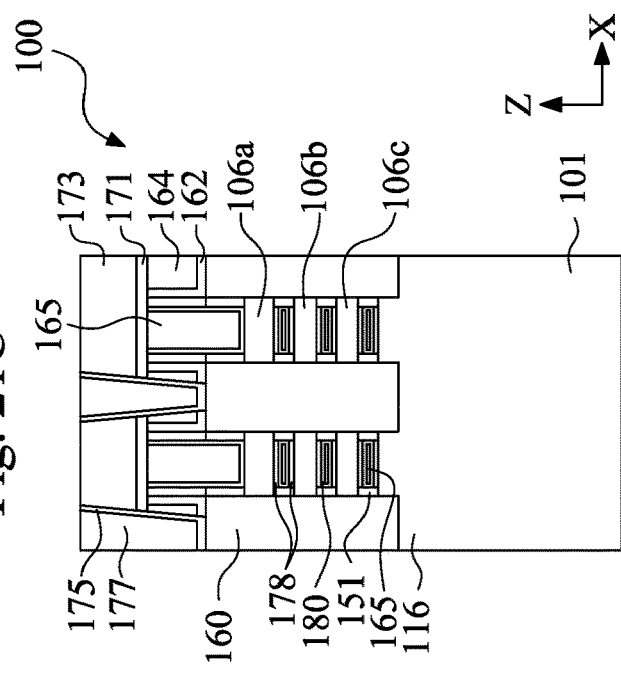


Fig. 21B

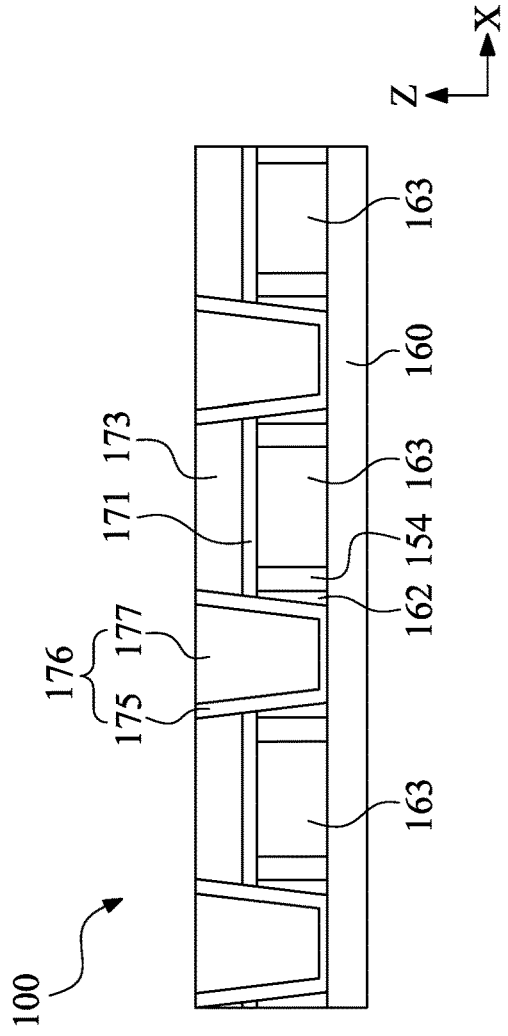


Fig. 21D

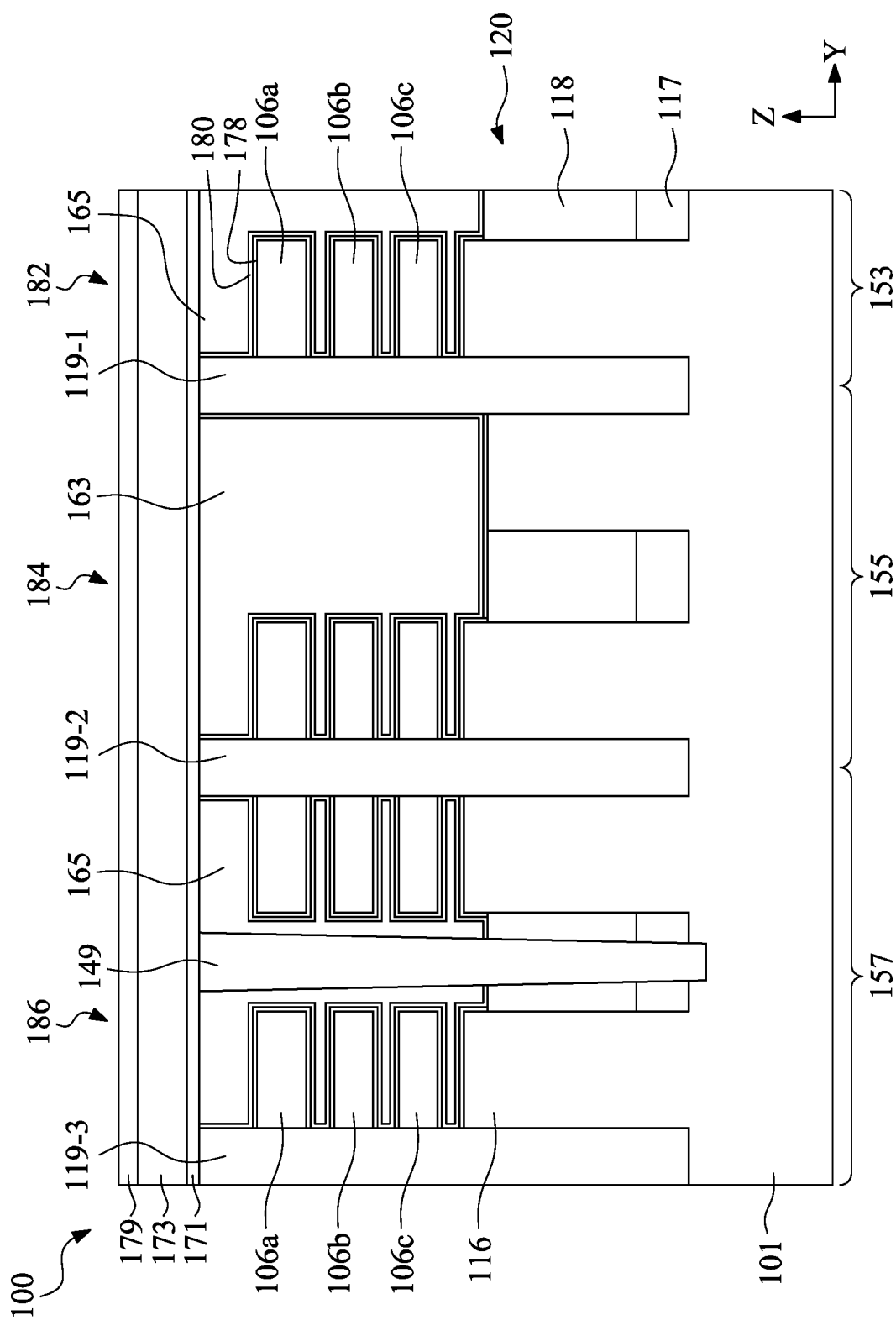


Fig. 22A

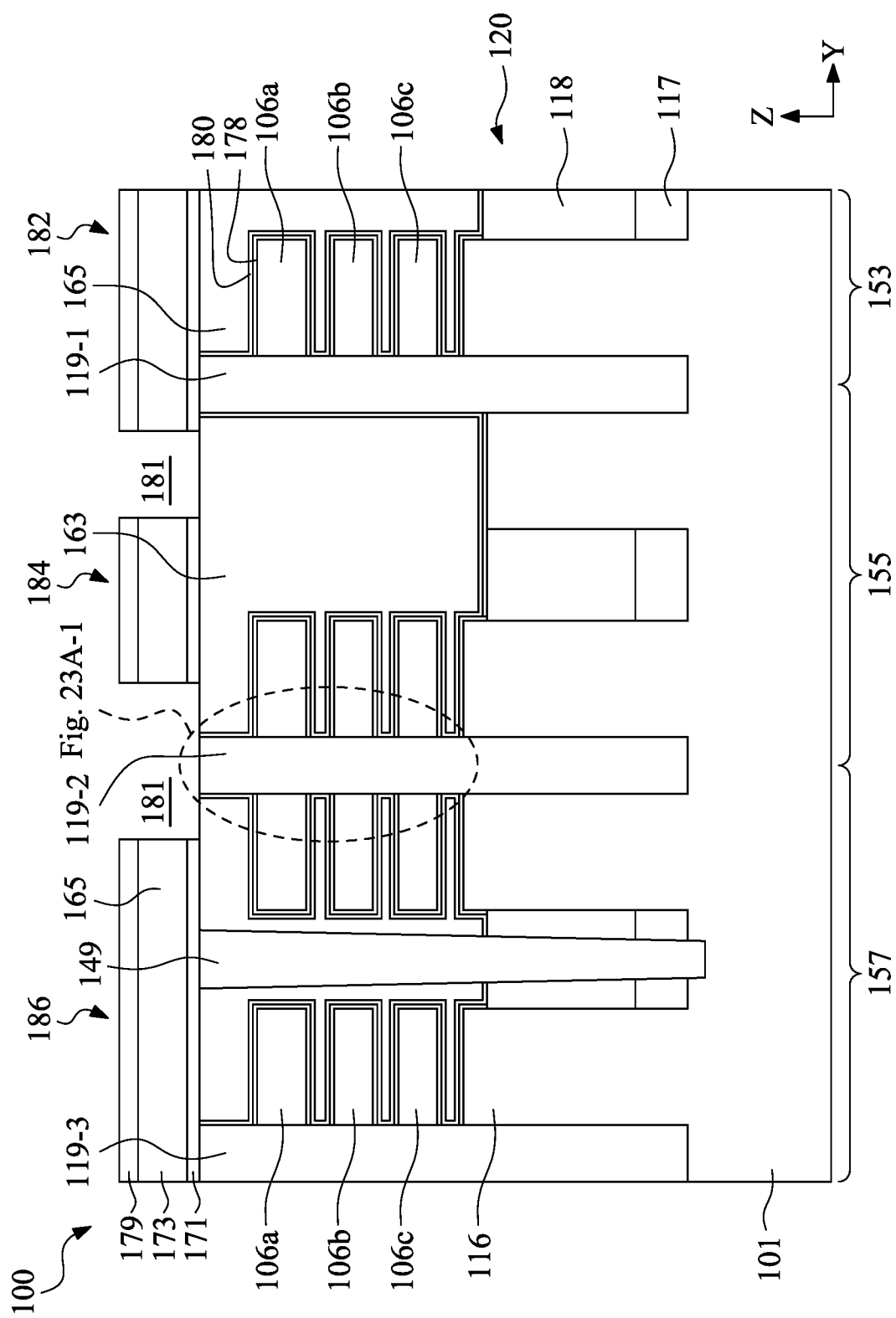


Fig. 23A

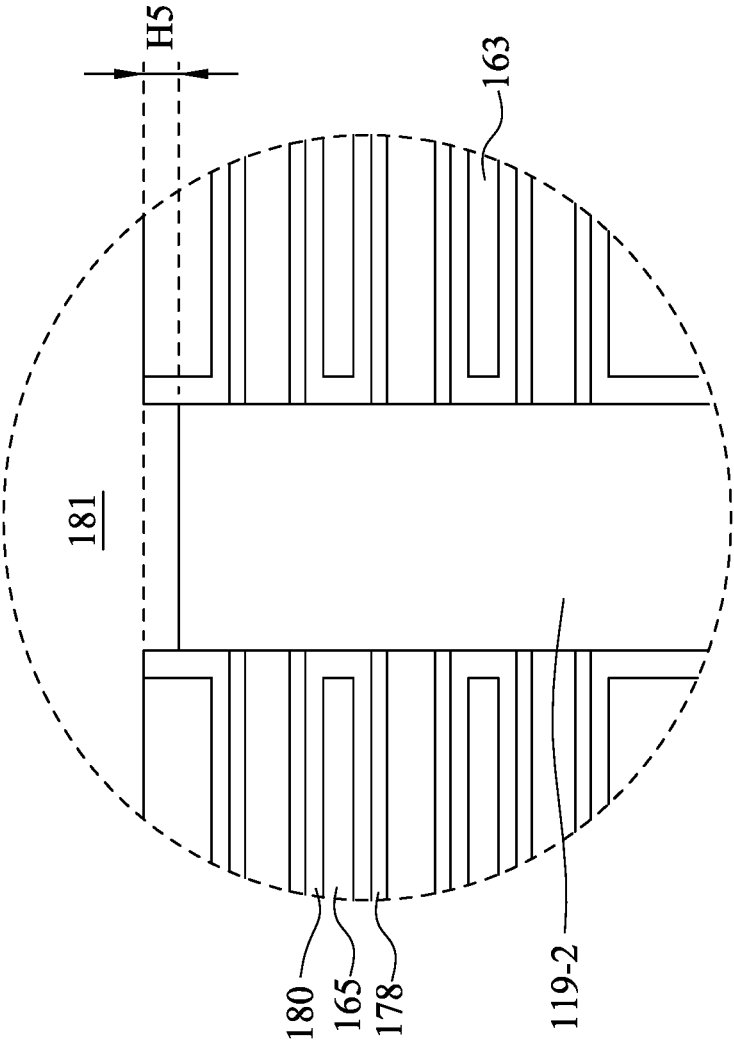


Fig. 23A-1

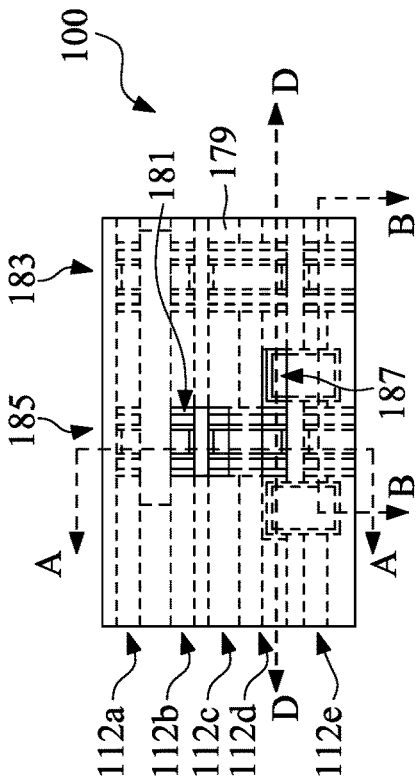


Fig. 23B

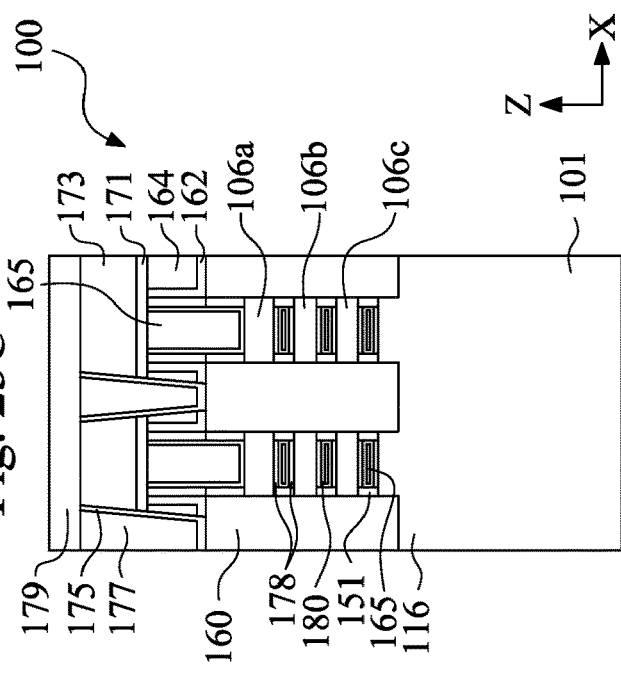
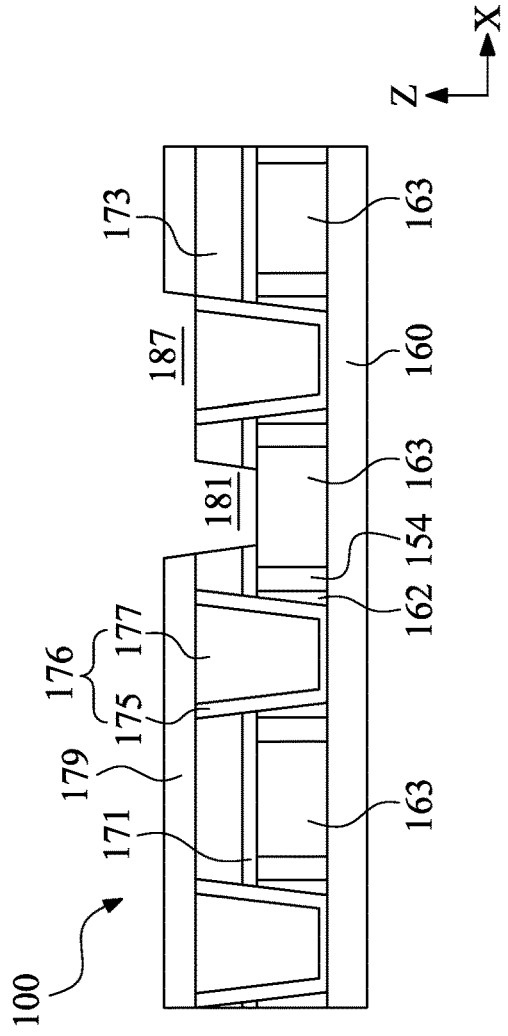


Fig. 23C



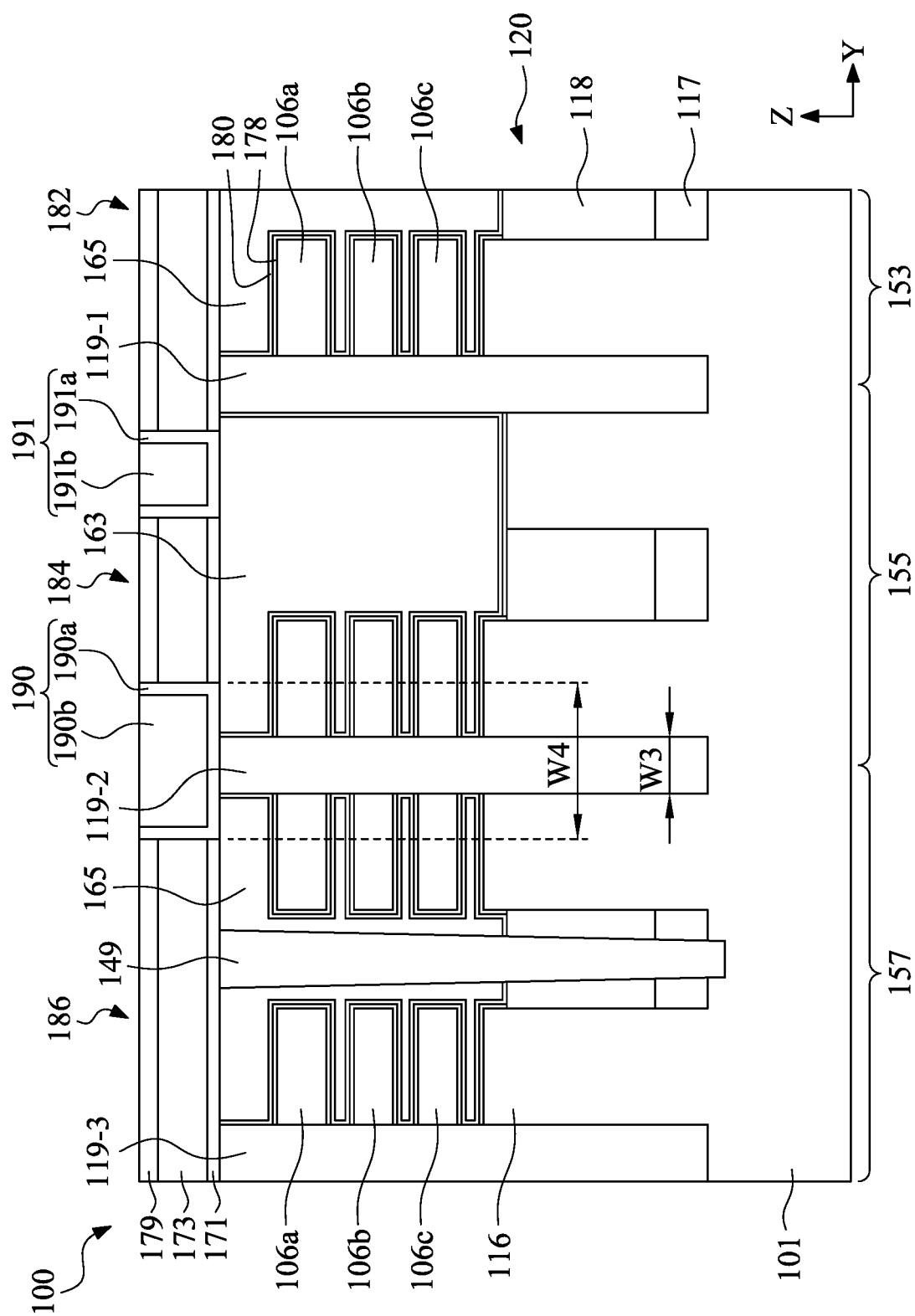


Fig. 24A

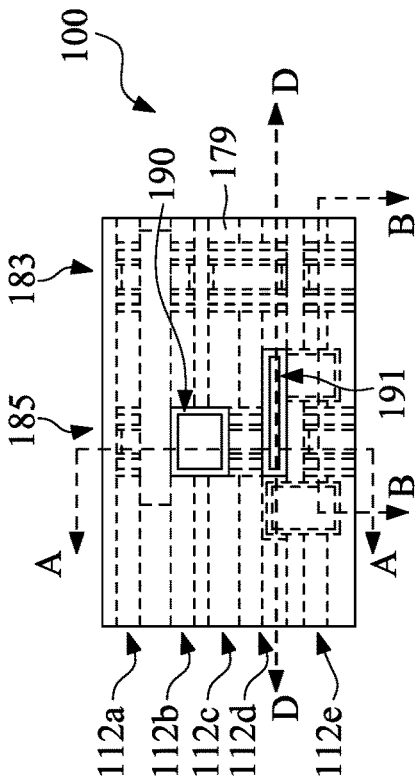


Fig. 24C

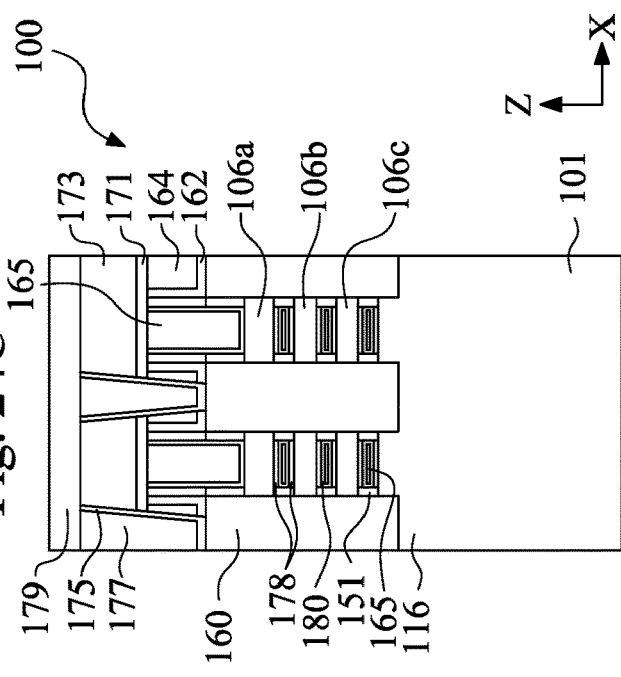


Fig. 24B

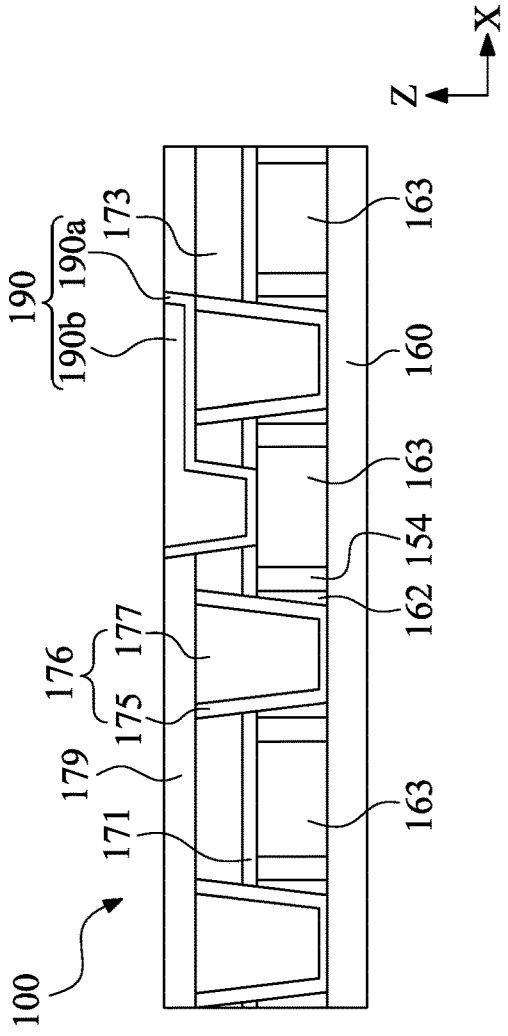


Fig. 24D

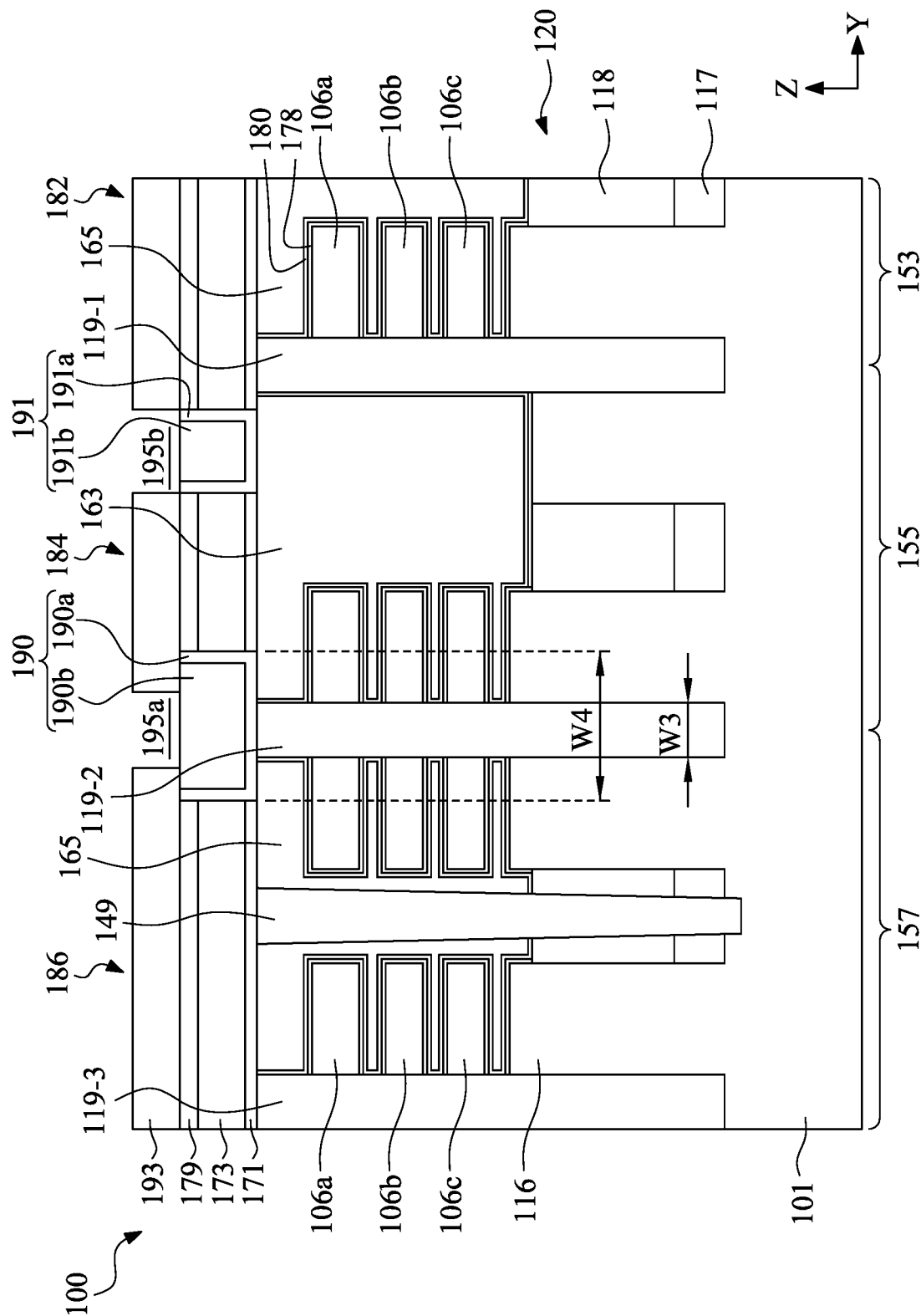


Fig. 25A

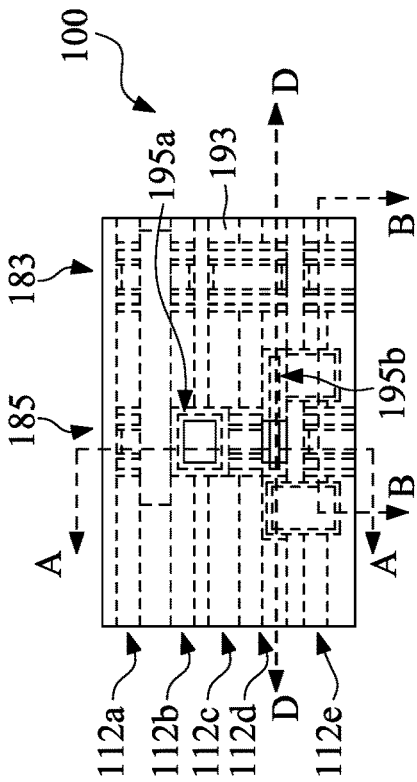


Fig. 25C

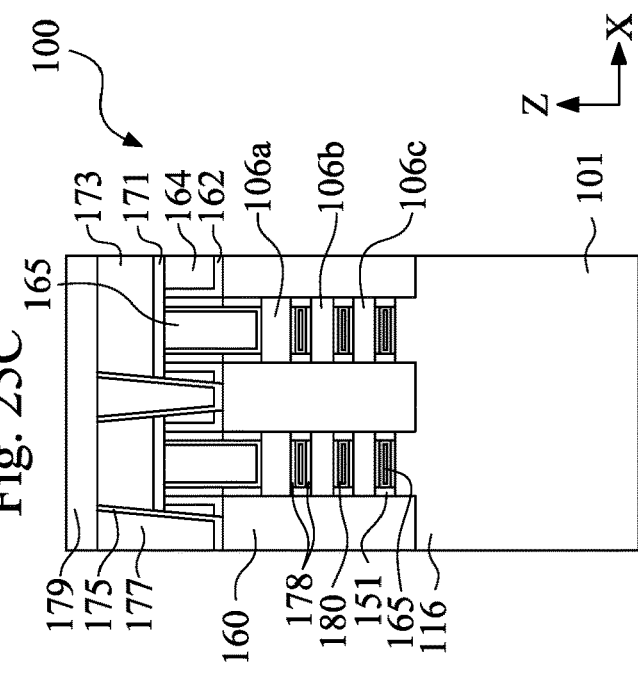


Fig. 25B

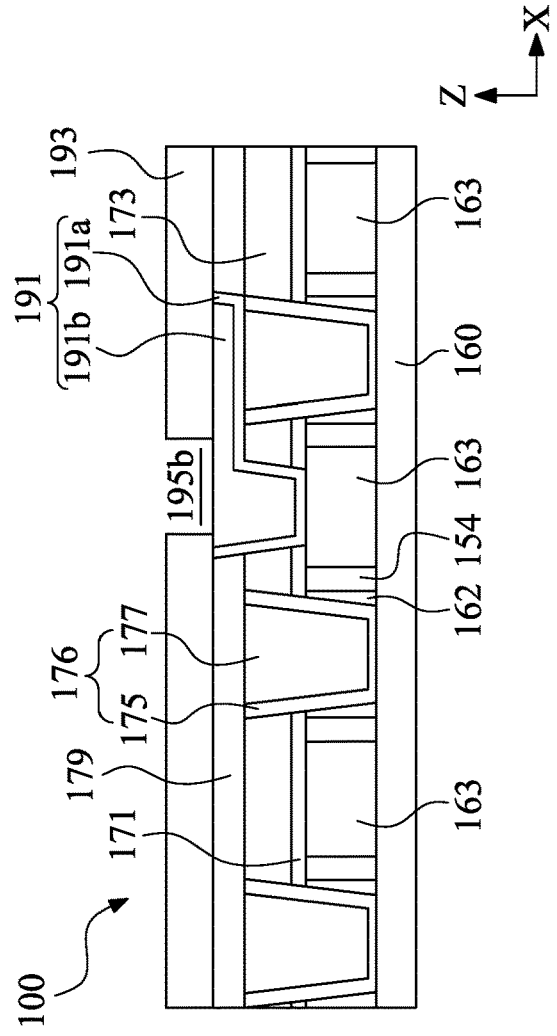


Fig. 25D

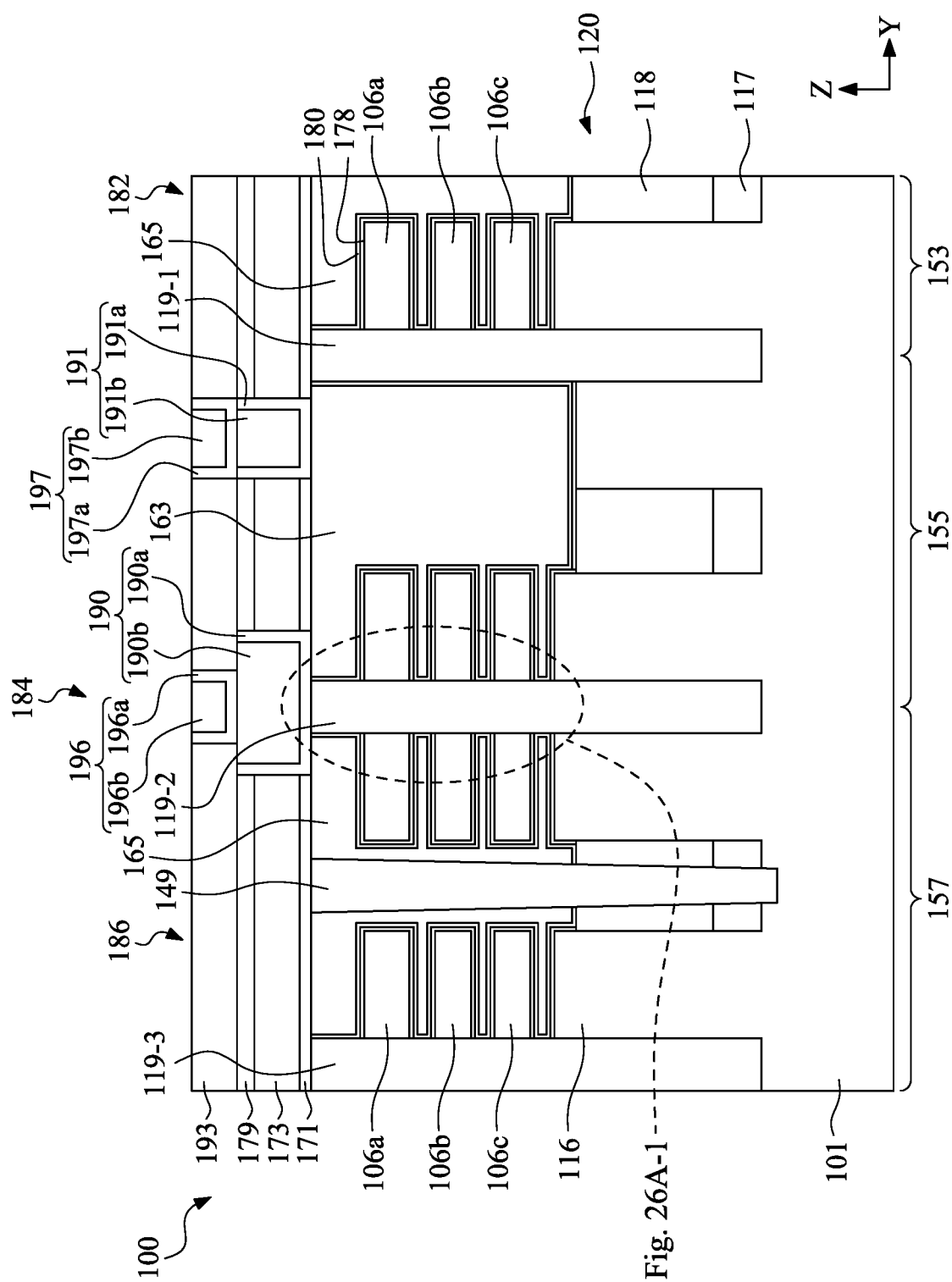


Fig. 26A

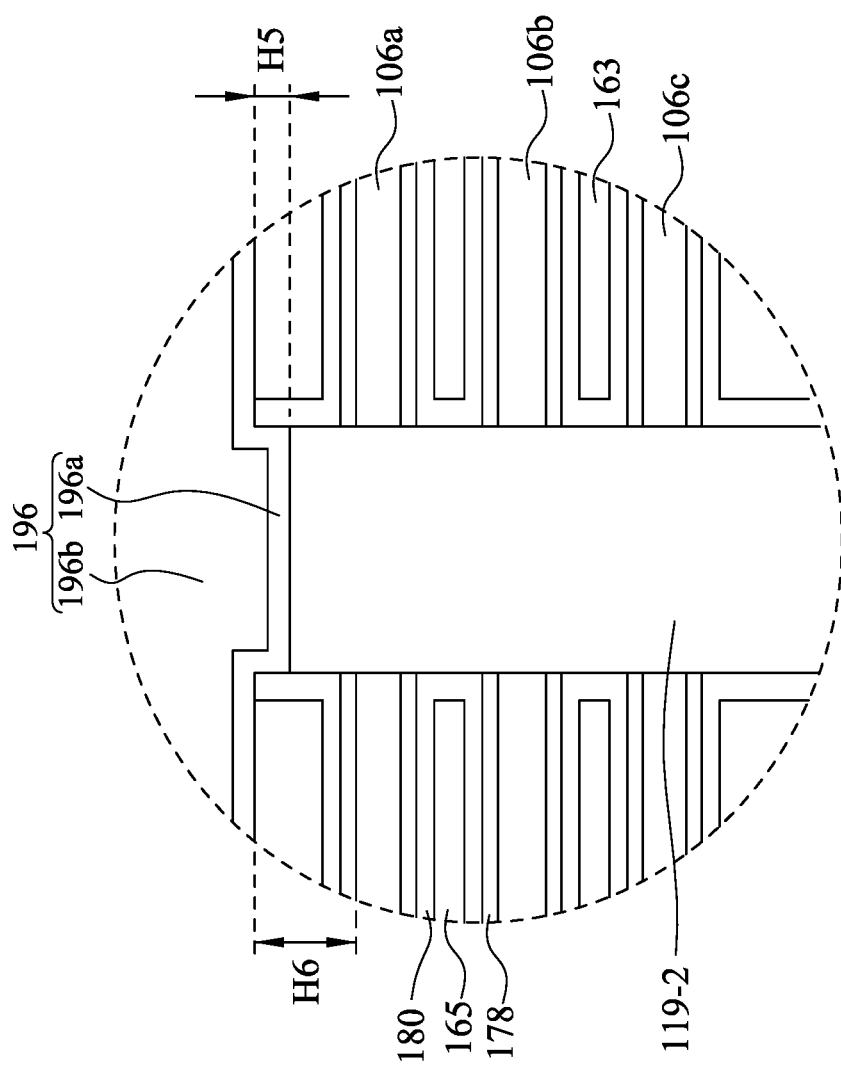


Fig. 26A-1

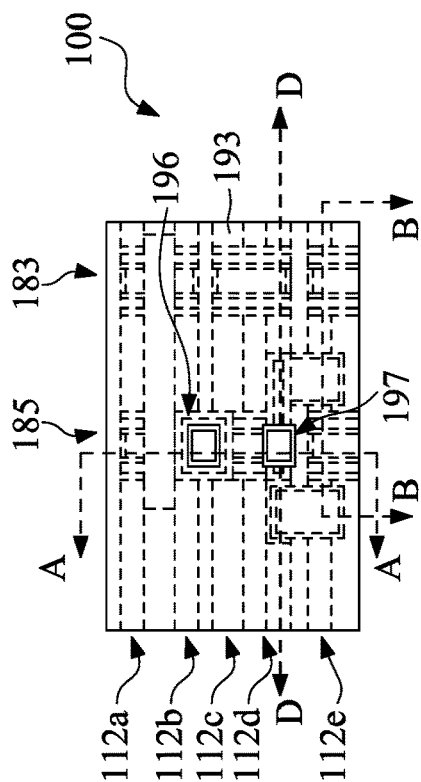


Fig. 26C

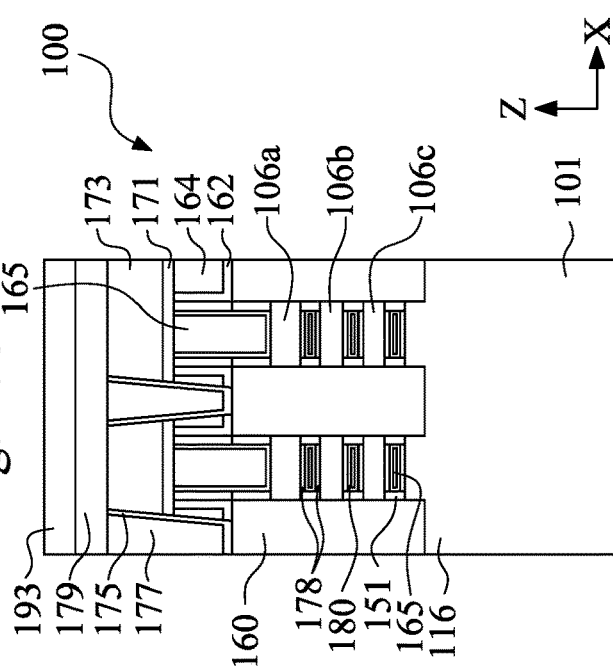


Fig. 26B

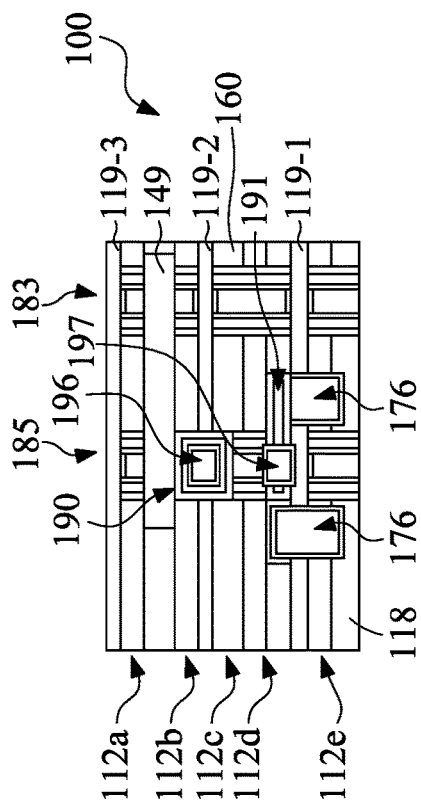


Fig. 26C-1

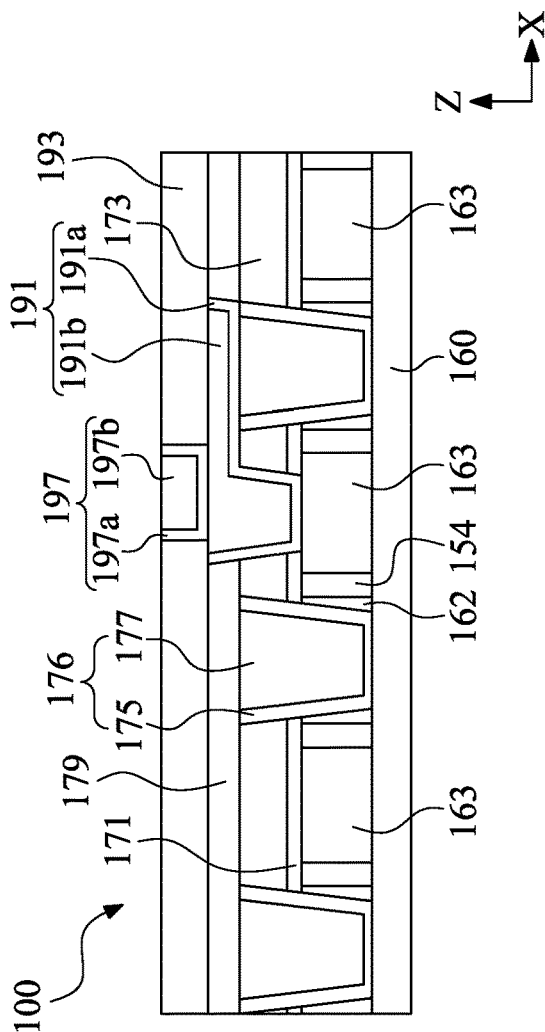
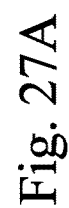
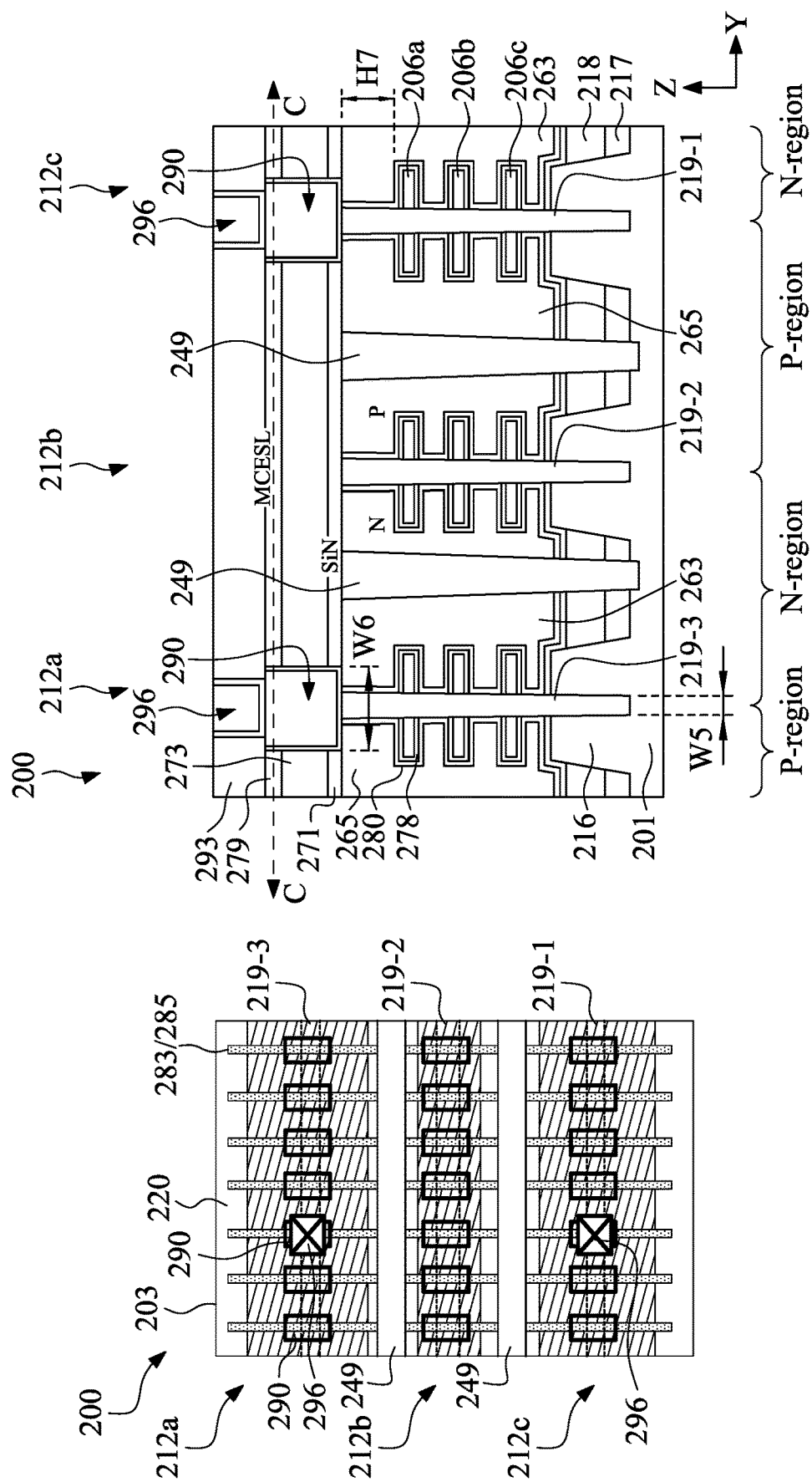


Fig. 26D



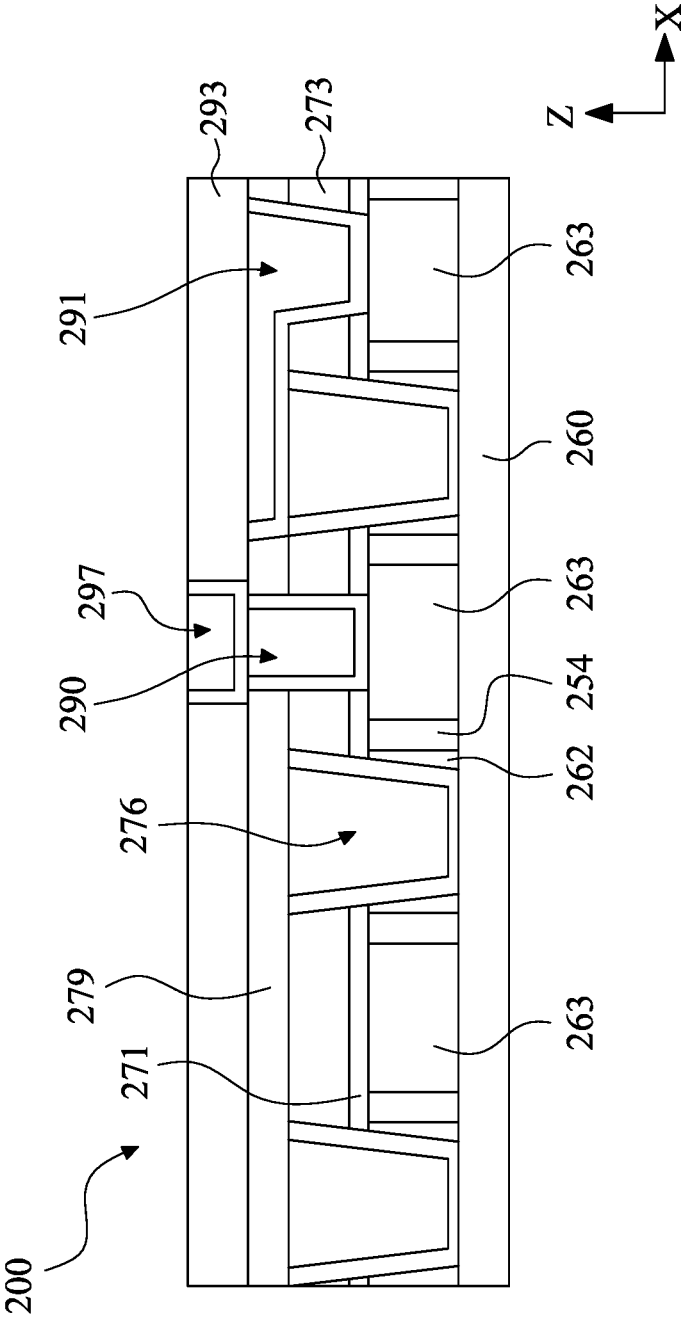


Fig. 27C

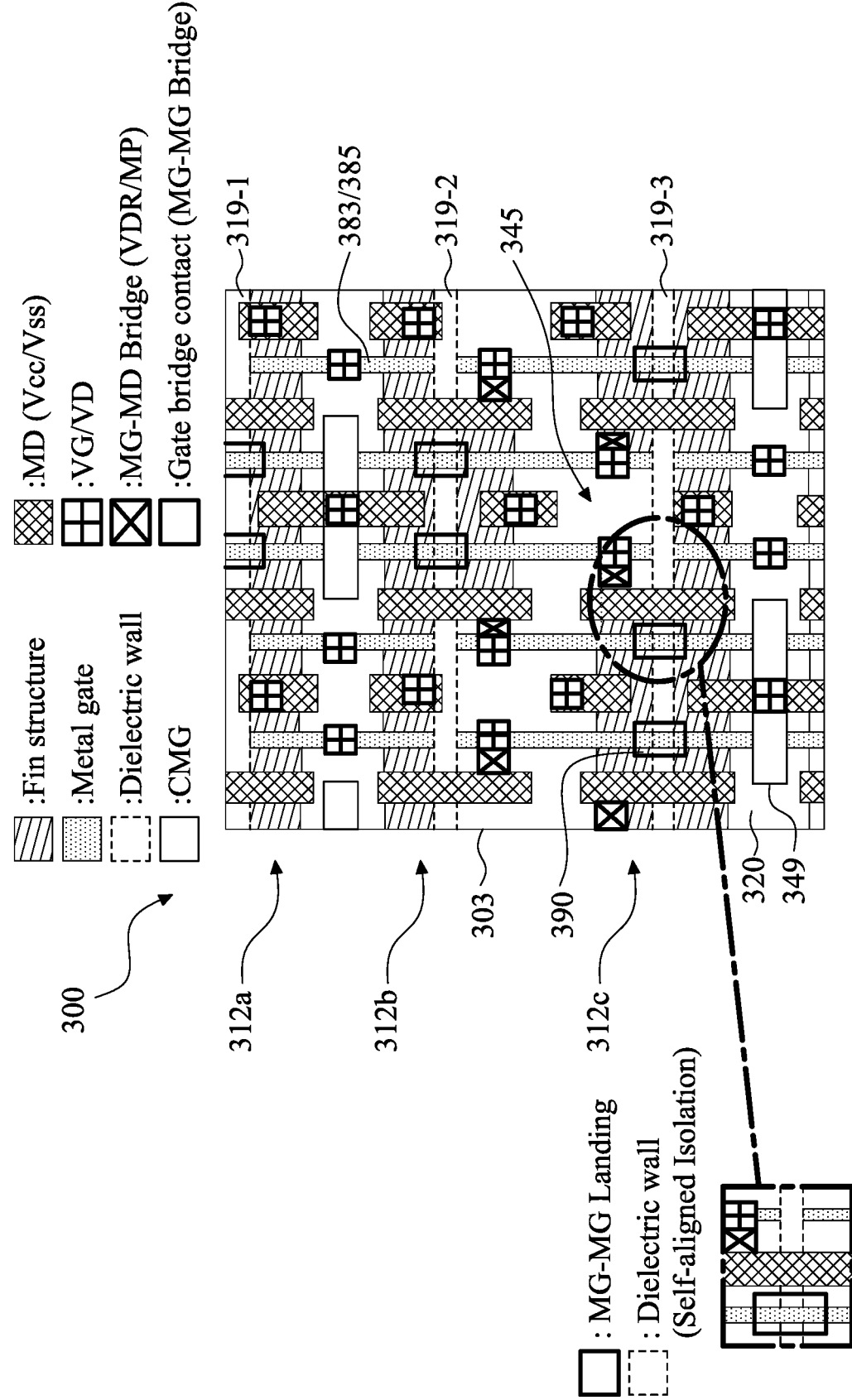


Fig. 28

SEMICONDUCTOR DEVICE STRUCTURE INCLUDING FORKSHEET TRANSISTORS AND METHODS OF FORMING THE SAME

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to U.S. Provisional Application Ser. No. 63/428,638 filed Nov. 29, 2022, which is incorporated by reference in its entirety.

BACKGROUND

[0002] The semiconductor integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. Such scaling down presents new challenge. For example, transistors using nanowire channels have been proposed to achieve increased device density, greater carrier mobility and drive current in a device. As device size reduces, there is a continuous need to improve processing and manufacturing ICs.

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIGS. 1-2 are perspective views of various stages of manufacturing a semiconductor device structure, in accordance with some embodiments.

[0005] FIGS. 3A-14A are cross-sectional side views of various stages of manufacturing the semiconductor device structure taken along line A-A of FIG. 3C, in accordance with some embodiments.

[0006] FIGS. 3B-14B are cross-sectional side views of various stages of manufacturing the semiconductor device structure taken along line B-B of FIG. 3C, in accordance with some embodiments.

[0007] FIGS. 3C-14C are top views of the semiconductor device structure of FIG. 2 in accordance with some embodiments.

[0008] FIGS. 15A-26A are cross-sectional views of various stages of manufacturing the semiconductor device structure taken along line A-A of FIG. 14C, in accordance with some embodiments.

[0009] FIGS. 15B-26B are cross-sectional side views of various stages of manufacturing the semiconductor device structure taken along line B-B of FIG. 14C, in accordance with some embodiments.

[0010] FIGS. 15C-26C are top views of various stages of manufacturing the semiconductor device structure of FIG. 14A.

[0011] FIG. 5A-1 is a cross-sectional view of one of various stages of manufacturing the semiconductor device structure, in accordance with some embodiments.

[0012] FIG. 8A-1 is a cross-sectional view of one of various stages of manufacturing the semiconductor device structure, in accordance with some embodiments.

[0013] FIGS. 20A-1 and 20A-2 illustrate cross-sectional views of one of various stages of manufacturing the semiconductor device structure, in accordance with some alternative embodiments.

[0014] FIGS. 21D-26D illustrate cross-sectional side views of various stages of manufacturing the semiconductor device structure taken along line D-D of FIG. 21C, in accordance with some embodiments.

[0015] FIG. 23A-1 illustrates an enlarged view of a portion of the semiconductor device structure of FIG. 23A, in accordance with some embodiments.

[0016] FIG. 26A-1 illustrates an enlarged view of a portion of the semiconductor device structure of FIG. 26A, in accordance with some embodiments.

[0017] FIG. 26C-1 is a top view of the semiconductor device structure of FIG. 26C showing locations of various devices/elements, in accordance with some embodiments.

[0018] FIG. 27A is a cross-sectional view of a semiconductor device structure, in accordance with some alternative embodiments.

[0019] FIG. 27B is a top view of the semiconductor device structure of FIG. 27A, in accordance with some embodiments.

[0020] FIG. 27C is a cross-sectional side view of the semiconductor device structure taken along line C-C of FIG. 27A, in accordance with some embodiments.

[0021] FIG. 28 is a top view of a semiconductor device structure, in accordance with some embodiments.

DETAILED DESCRIPTION

[0022] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations described.

[0023] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “over,” “on,” “top,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0024] While the embodiments of this disclosure are discussed with respect to nanosheet channel FETs, implementations of some aspects of the present disclosure may be used in other processes and/or in other devices, such as planar FETs, Fin-FETs, Horizontal Gate All Around (HGAA) FETs, Vertical Gate All Around (VGAA) FETs, and other suitable devices. A person having ordinary skill in the art will readily understand other modifications that may be made are contemplated within the scope of this disclosure. In cases where gate all around (GAA) transistor structures are adapted, the GAA transistor structures may be patterned by any suitable method. For example, the structures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the GAA structure.

[0025] FIGS. 1-28 show exemplary sequential processes for manufacturing a semiconductor device structure 100, in accordance with some embodiments. It is understood that additional operations can be provided before, during, and after processes shown by FIGS. 1-28, and some of the operations described below can be replaced or eliminated, for additional embodiments of the method. The order of the operations/processes may be interchangeable.

[0026] FIG. 1 is a perspective view of one of various stages of manufacturing the semiconductor device structure 100, in accordance with some embodiments. As shown in FIG. 1, a semiconductor device structure 100 includes a stack of semiconductor layers 104 formed over a front side of a substrate 101. The substrate 101 may be a semiconductor substrate. The substrate 101 may include a single crystalline semiconductor material such as, but not limited to silicon (Si), germanium (Ge), silicon germanium (SiGe), gallium arsenide (GaAs), indium antimonide (InSb), gallium phosphide (GaP), gallium antimonide (GaSb), indium aluminum arsenide (InAlAs), indium gallium arsenide (InGaAs), gallium antimony phosphide (GaSbP), gallium arsenic antimonide (GaAsSb) and indium phosphide (InP). In this embodiment, the substrate 101 is made of Si. In some embodiments, the substrate 101 is a silicon-on-insulator (SOI) substrate, which includes an insulating layer (not shown) disposed between two silicon layers. In one aspect, the insulating layer is an oxide.

[0027] The substrate 101 may include various regions that have been doped with impurities (e.g., dopants having p-type or n-type conductivity). Depending on circuit design, the dopants may be, for example boron for a p-type (or p-channel) field effect transistor (FET) and phosphorus for an n-type (or n-channel) FET.

[0028] The stack of semiconductor layers 104 includes alternating semiconductor layers made of different materials to facilitate formation of nanosheet channels in a multi-gate device, such as nanosheet channel FETs or forksheet FETs. In some embodiments, the stack of semiconductor layers 104 includes first semiconductor layers 106 (106a-106c) and

second semiconductor layers 108 (108a-108c). In some embodiments, the stack of semiconductor layers 104 includes alternating first and second semiconductor layers 106, 108. The first semiconductor layers 106 are aligned with the second semiconductor layers 108. The first semiconductor layers 106 and the second semiconductor layers 108 are made of semiconductor materials having different etch selectivity and/or oxidation rates. For example, the first semiconductor layers 106 may be made of Si and the second semiconductor layers 108 may be made of SiGe. In some examples, the first semiconductor layers 106 may be made of SiGe and the second semiconductor layers 108 may be made of Si. In some cases, the SiGe in the first or second semiconductor layers 106, 108 can have a germanium composition percentage between about 10% and about 80%. Alternatively, in some embodiments, either of the semiconductor layers 106, 108 may be or include other materials such as Ge, SiC, GeAs, GaP, InP, InAs, InSb, GaAsP, AlInAs, AlGaAs, InGaAs, GaInP, GaInAsP, or any combinations thereof.

[0029] The first semiconductor layers 106 or portions thereof may form nanosheet channels of the semiconductor device structure 100, which are to be constructed as forksheet FETs in later fabrication stages. As will be discussed in more details below, in forksheet FETs, both n-channel FETs and p-channel FETs are integrated in the same device structure. A dielectric wall separates the n-channel FETs and p-channel FETs. The term nanosheet is used herein to designate any material portion with nanoscale, or even microscale dimensions, and having an elongate shape, regardless of the cross-sectional shape of this portion. Thus, this term designates both circular and substantially circular cross-section elongate material portions, and beam or bar-shaped material portions including, for example, a cylindrical in shape or substantially rectangular cross-section. Portions of the nanosheet channel(s) of the semiconductor device structure 100 may be surrounded by a gate electrode. For example, the nanosheet channel(s) of a forksheet transistor may have at least three surfaces surrounded by the gate electrode. While the semiconductor device structure 100 is shown to include forksheet transistors, the semiconductor device structure 100 may include also include nanosheet transistor(s). The nanosheet transistors may be referred to as nanowire transistors, gate-all-around (GAA) transistors, multi-bridge channel (MBC) transistors, or any transistors having the gate electrode surrounding the channels.

[0030] It is noted that while three layers of the first semiconductor layers 106 and three layers of the second semiconductor layers 108 are alternately arranged as illustrated in FIG. 1, it can be appreciated that any number of first and second semiconductor layers 106, 108 can be formed in the stack of semiconductor layers 104, depending on the predetermined number of channels for the semiconductor device structure 100. In some embodiments, the number of first semiconductor layers 106, which is the number of channels, is between 2 and 8.

[0031] The first and second semiconductor layers 106, 108 are formed by any suitable deposition process, such as epitaxy. By way of example, epitaxial growth of the layers of the stack of semiconductor layers 104 may be performed by a molecular beam epitaxy (MBE) process, a metalorganic chemical vapor deposition (MOCVD) process, and/or other suitable epitaxial growth processes.

[00332] The substrate **101** may include a sacrificial layer **107** on the stack of semiconductor layers **104**. The sacrificial layer **107** protects the stack of semiconductor layers **104** during the subsequent processes and is removed along with a portion of a cladding layer **132** (FIG. 8A) prior to formation of the sacrificial gate stack (FIG. 9A). In cases where the first semiconductor layer **106** of the stack of semiconductor layers **104** is Si, the sacrificial layer **107** includes SiGe that can be epitaxially grown on the first semiconductor layer **106**.

[00333] Each first semiconductor layer **106** may have a thickness in a range between about 5 nm and about 30 nm. Each second semiconductor layer **108** may have a thickness that is equal, less, or greater than the thickness of the first semiconductor layer **106**. In some embodiments, each second semiconductor layer **108** has a thickness in a range between about 2 nm and about 50 nm. The second semiconductor layers **108** may eventually be removed and serve to define a vertical distance between adjacent channels for the semiconductor device structure **100**. The sacrificial layer **107** may have a thickness that is equal, less, or greater than the thickness of the first semiconductor layer **106**. The thickness of the sacrificial layer **107** may range from about 2 nm to 50 nm. The thickness of the first semiconductor layer **106**, the second semiconductor layer **108**, and the sacrificial layer **107** may vary depending on the application and/or device performance considerations.

[00334] A mask structure **110** is formed over the sacrificial layer **107**. The mask structure **110** may include an oxygen-containing layer and a nitrogen-containing layer. The oxygen-containing layer may be a pad oxide layer, such as a SiO₂ layer. The nitrogen-containing layer may be a pad nitride layer, such as Si₃N₄. The mask structure **110** may be formed by any suitable deposition process, such as chemical vapor deposition (CVD) process.

[00335] FIG. 2 is a perspective view of one of the various stages of manufacturing the semiconductor device structure **100**, in accordance with some embodiments. Fin structures **112** (**112a-112e**) are formed from the stack of semiconductor layers **104**. Each fin structure **112** has an upper portion including the semiconductor layers **106**, **108** and a lower portion which includes a well portion **116** formed from the substrate **101**. The fin structures **112** may be fabricated using multi-patterning operations including photo-lithography and etching processes. The etching process can include dry etching, wet etching, reactive ion etching (RIE), and/or other suitable processes. The etching process forms trenches **114** (e.g., **114a-114f**) in unprotected regions through the mask structure **110**, through the stack of semiconductor layers **104**, and into the substrate **101**, thereby leaving the plurality of extending fin structures **112** (e.g., **112a-e**). The trenches **114** extend along the X direction. The trenches **114** may be etched using a dry etch (e.g., RIE), a wet etch, and/or combination thereof.

[00336] As shown in FIG. 2, each fin structure **112a-e** may have a height H1 in a range of about 25 nm to about 300 nm. The fin structure **112a-e** may have a width W1. The width W1 may correspond to the device's channel width. In one embodiment, the width W1 is in a range between 5 nm to about 120 nm. The fins **112a-e** may be separated from each other by a distance. The distance between adjacent fin structures may be defined by the distance between a first sidewall of one fin structure and a second sidewall of the adjacent fin structure facing the first sidewall. For example,

the fin structure **112c** and the fin structure **112d** are separated by a distance D1. The fin structure **112d** and the fin structure **112e** are separated by a distance D2. The first and second distances D1, D2 may vary depending on the layouts of the fin structures in a SRAM cell and/or a logic cell. In one embodiment shown in FIG. 2, the distance D1 is greater than the distance D2. The distance D2 may be in a range from about 2 nm to about 40 nm, for example about 3 nm to about 30 nm. With the smaller distance D2 (i.e., reduced fin-to-fin spacing) between the fin structures **112c** and **112d**, layers of a dielectric wall **119** (FIG. 5A) subsequently formed in the trench **114d** may merge, while the trench **114c** between the fin structures **112c** and **112d** remains open after the deposition of layers of the dielectric wall due to the wider distance D1. The merged layers of the dielectric wall allow the nanosheet channels to attach to both sides of the dielectric feature wall and form forksheet transistors at a later stage. The reduced fin-to-fin spacing and fork-like nanosheet transistors enable greater device density (even with greater channel width) and superior area and performance scalability.

[00337] While not shown, the width of the fin structures **112a-112e** may vary depending on the channel width of the devices needed in the semiconductor device structure **100**. The devices with a wider channel may be more suitable for high-speed applications, such as a NAND device. The devices with a narrower channel may be more suitable for low-power and low-leakage applications, such as an inverter device. Trenches with wider width (e.g., trench **114c**) may be formed in regions where devices/transistors require higher voltage current and/or higher performance, while trenches with narrower width (e.g., trench **114b**) may be formed in regions where greater density of devices/transistors is desired.

[00338] Depending on the design layouts of the SRAM cell, the trenches **114e** and **114f** may have a width corresponding to the distance D1 or the distance D2. In one embodiment shown in FIG. 2, the trench **114e** has a width corresponding to the distance D1, and the trench **114f** has a width corresponding to the distance D2. In some embodiments, a fin structure (not shown) having a width greater than W1 may be disposed adjacent to and spaced apart the fin structure **112a** by the trench **114f**. Likewise, a fin structure (not shown) having a width greater than W2 may be disposed adjacent to and spaced apart the fin structure **112e** by the trench **114e**.

[00339] FIGS. 3C-14C are top views of the semiconductor device structure **100** of FIG. 2, which may represent a portion of the layout of active fin structures in a SRAM cell **103**. For example, a 6T SRAM cell may include two pull-up (PU) transistors, two pass-gate (PG) transistors, and two pull-down (PD) transistors. In one embodiment shown in FIGS. 3C-14C, the fin structures **112b** and **112c** can be used to form PD transistors or PG transistors and the fin structure **112a** can be used to form PU transistors in the 6T SRAM cell. FIGS. 3A-14A are cross-sectional side views of various stages of manufacturing the semiconductor device structure **100** taken along line A-A of FIG. 3C, in accordance with some embodiments. FIGS. 3B-14B are cross-sectional side views of various stages of manufacturing the semiconductor device structure **100** taken along line B-B of FIG. 3C, in accordance with some embodiments.

[0040] In FIGS. 4A-4C, a dielectric material **117** is formed in the trenches **114a-114f** and over the top surface of the

mask structure **110**. The dielectric material **117** is deposited so that the fins **112a-e** are embedded in the dielectric material **117**. The dielectric material **117** may include, but are not limited to, SiOx, SiN, SiON, SiCN, SiOCN, AlSi_xO_y, Al₂O₃, hafnium oxide (HfO₂), hafnium silicate (HfSiO), hafnium silicon oxynitride (HfSiON), hafnium aluminum oxide (HfAlO), hafnium lanthanum oxide (HfLaO), hafnium zirconium oxide (HfZrO), hafnium tantalum oxide (HfTaO), hafnium titanium oxide (HfTiO), lanthanum oxide (LaO), zirconium oxide (ZrO), zirconium silicon oxide (ZrSiO), titanium oxide (TiO), tantalum oxide (Ta₂O₅), yttrium oxide (Y₂O₃), any suitable low-k materials, any suitable high-k materials, or any combination thereof. The dielectric material **117** may be formed by any suitable process, such as a chemical vapor deposition (CVD), plasma enhanced CVD (PECVD), flowable CVD (FCVD), atomic layer deposition (ALD), or physical vapor deposition (PVD) process.

[0041] In FIGS. 5A-5C, a planarization operation, such as a chemical mechanical polishing (CMP) method, is performed such that the top of the sacrificial layers **107** is exposed. Next, a removal process is performed to remove the dielectric material **117** from the trenches **114a** and **114c**. The removal process may be any suitable etch process, such as dry etch, wet, etch, or a combination thereof. The removal process may be selective etch processes that remove portions of dielectric material **117** but not the mask structure **110**, the sacrificial layers **107**, the first semiconductor layers **106**, and the second semiconductor layers **108**. Because the trenches **114a** and **114c** have a larger dimension (i.e., distance D1, FIG. 2) in the Y direction compared to that of the trench **114b**, **114d**, **114e**, **114f** (FIG. 2), the etchant removes more of the dielectric material **117** in the trenches **114a** and **114c** than the dielectric material **117** in the trenches **114b**, **114d**, **114e**, **114f**. As a result, the dielectric material **117** in the trenches **114a**, **114c**, **114d** are etched at a faster rate than the etch rate of the dielectric material **117** in the trench **114b**. The removal process is performed until the dielectric material **117** in the trenches **114b**, **114d**, **114e**, **114f** are completely etched away. In some embodiments, bottom portions of the dielectric material **117** in the trenches **114b**, **114d**, **114e**, **114f** may remain, as shown in FIGS. 5A and 5C. In some embodiments, the removal process is continued until the dielectric material **117** in the trenches **114b**, **114d**, **114e**, **114f** are completely etched away. In some embodiments, the removal process is continued until the dielectric material **117** in the trenches **114a**, **114c**, **114e** are completely etched away, such as an alternative embodiment shown in FIG. 5A-1. In one exemplary embodiment, the dielectric material **117** remaining in the trenches **114b**, **114d**, **114f** becomes a dielectric wall **119**. The dielectric wall **119** extends all the way down to the well portions **116** of the substrate **101**. Each dielectric wall **119** isolates adjacent active fin structures (e.g., fin structures **112b**, **112c** and fin structures **112d**, **112e**), which are to be formed as a forksheet transistor in a SRAM cell. The dielectric walls **119** avoid electrical shorts at transistors, such as two adjacent pull-up transistors in the SRAM cell. While not shown, the top of the dielectric walls **119** may be slightly etched and has a concave profile due to etching effects from the removal process on the dielectric material **117**.

[0042] In FIGS. 6A-6C, a resist layer **141** is formed on the exposed surfaces of the semiconductor device structure **100**. The resist layer **141** may be any suitable masking material, such as a photoresist layer, a BARC (bottom anti-reflective

coating) layer, a SOG (spin-on-glass) layer, or a SOC (spin-on-carbon) layer, and may be deposited by spin coating or any suitable deposition technique. In cases where the dielectric material **117** in the trenches **114a**, **114c**, **114e** were completely removed, the resist layer **141** will fill up the trenches **114a**, **114c**, **114e** entirely. In either case, the resist layer **141** is patterned to expose certain regions of the semiconductor device structure **100**. Depending on the design layout, portions of the fin structures **112a-e** at certain regions may be removed. In one embodiment, the resist layer **141** is patterned so that the fin structure **112d** at one side of the dielectric wall **119** is removed. The removal of the portion of the fin structure **112d** forms an end cut **145** in the fin structure **112d**. Therefore, the remaining fin structure **112d** along the X-direction is segmented. The segmentation isolates some of the transistors from another. The end cut **145** may have any shape (e.g., a rectangular shape), depending on the application/design layout. In one embodiment, the end cut **145** extends along the X-direction and has a width similar to the width of the fin structure **112d**. The end cut **145** may be performed by protecting portions of the fin structures by the resist layer **141**. The exposed portions of the fin structure (e.g., fin structure **112d**) not covered by the resist layer **141** are removed in one or more etch processes. In one embodiment shown in FIG. 6A, the etch processes remove the fin structure **112d** (e.g., the stack of semiconductor layers **104** and the well portion **116**) to expose the sidewalls of the dielectric wall **119**. The etch processes are performed so that the top surface of the fin structure **112d** is at a level of, or slightly below an interface defined by the bottommost second semiconductor layer **108c** and the well portion **116**. Alternatively, the etch processes may be performed until the top portion of the substrate **101** is exposed. That is, the entire fin structure **112d** is removed. In one embodiment, the etch processes are performed so that the top surface of the fin structure **112d** is at an elevation below the interface defined by the bottommost second semiconductor layer **108c** and the well portion **116** by a height H2 of about 1 nm to about 3 nm.

[0043] In some embodiments, which can be combined with any other embodiments of the present disclosure, the end cut **145** may be omitted. That is, the fin structure **112d** is not removed, as shown in FIG. 8A-1. In such cases, the **112d** may serve as connect gate and use gate off-state voltage during the operation. In some embodiments, after the end cut **145** is formed (e.g., FIG. 6A), an insulating layer may be deposited in the end cut **145**. The insulating layer may have any shape depending on the profile of the end cut **145**.

[0044] In FIGS. 7A-7C, after formation of the end cut **145** in the fin structure **112d**, the resist layer **141** is removed using any suitable removal process, such as ashing, dry etch, wet etch, or a combination thereof. Then, an insulating material **118** is formed on the substrate **101**. The insulating material **118** fills the trenches **114a**, **114c**, **114e** (FIG. 5A), and the end cut **145** between neighboring fin structures **112** until the fin structures **112** are embedded in the insulating material **118**. The insulating material **118** may be made of silicon oxide, silicon nitride, silicon oxynitride (SiON), SiOCN, SiCN, fluorine-doped silicate glass (FSG), a low-k dielectric material, or any suitable dielectric material. In some embodiments, the insulating material **118** may be formed of a material different from the dielectric wall **118**. The insulating material **118** may be formed by any suitable

method, such as low-pressure chemical vapor deposition (LPCVD), plasma enhanced CVD (PECVD) or flowable CVD (FCVD).

[0045] Then, a planarization operation, such as a chemical mechanical polishing (CMP) method and/or an etch-back method, is performed to remove the insulating material **118** and the mask structures **110** until the top of the fin structures **112** (e.g., the sacrificial layer **107**) is exposed. Next, the insulating material **118** is recessed to form an isolation region (or shallow trench isolation (STI) region) **120**. The recess of the insulating material **118** exposes portions of the fin structures **112**, such as the stack of semiconductor layers **104**. A top surface of the insulating material **118** may be level with or slightly below an interface defined by the bottommost second semiconductor layer **108c** and the well portion **116**. In some embodiments, the insulating material **118** is recessed so that the top surface of the insulating material **118** and the top surface of the remaining fin structure **112d** are substantially co-planar.

[0046] In FIGS. 8A-8C, a cladding layer **132** is formed on the sidewalls of the stack of semiconductor layers **104**. In some embodiments, the cladding layer **132** is also formed on the exposed surfaces of the dielectric wall **119**. The cladding layer **132** may be formed on the exposed surfaces of the stack of semiconductor layers **104**, the sacrificial layer **107**, the dielectric wall **119**, and the isolation region **120**. The cladding layer **132** may be formed by a conformal process, such as an ALD process. Next, portions of the cladding layer **132** are removed by an anisotropic etch process so that the cladding layer **132** on horizontal surfaces of the fin structures **112a**, **112b**, **112c**, **112e** (e.g., top surfaces of the sacrificial layer **107**), the dielectric wall **119**, and the insulating material **118** is removed. The removal process does not remove the cladding layer **132** formed on the vertical surfaces (e.g., the sidewalls) of the fin structures **112a**, **112b**, **112c**, **112e** and the dielectric wall **119**. The cladding layer **132** may have a thickness ranging from about 2 nm to about 20 nm, for example about 5 nm to about 13 nm. In some embodiments, the cladding layer **132** includes a semiconductor material. In some embodiments, the cladding layer **132** and the second semiconductor layers **108** are made of the same material having the same etch selectivity. For example, the cladding layer **132** and the second semiconductor layers **108** may include SiGe. The cladding layer **132** and the second semiconductor layer **108** may be removed subsequently to create space for the gate electrode layer.

[0047] In FIGS. 9A-9C, one or more sacrificial gate stacks **142** are formed on the semiconductor device structure **100**. The sacrificial gate stacks **142** may each include a sacrificial gate dielectric layer **144**, a sacrificial gate electrode layer **146**, and a mask structure **148**. The sacrificial gate dielectric layer **144** may include one or more layers of dielectric material, such as SiO₂, SiN, a high-k dielectric material, and/or other suitable dielectric material. In some embodiments, the sacrificial gate dielectric layer **144** may be deposited by a CVD process, a sub-atmospheric CVD (SACVD) process, a FCVD process, an ALD process, a PVD process, or other suitable process. The sacrificial gate electrode layer **146** may include polycrystalline silicon (polysilicon). The mask structure **148** may include an oxygen-containing layer **150** and a nitrogen-containing layer **152**. The sacrificial gate electrode layer **146** and the mask structure **148** may be formed by various processes such as layer deposition, for example, CVD (including both LPCVD

and PECVD), PVD, ALD, thermal oxidation, e-beam evaporation, or other suitable deposition techniques, or combinations thereof.

[0048] The sacrificial gate stacks **142** may be formed by first depositing blanket layers of the sacrificial gate dielectric layer **144**, the sacrificial gate electrode layer **146**, and the mask structure **148**, followed by patterning and etching processes. By patterning the sacrificial gate stack **142**, the stacks of semiconductor layers **104** of the fins **112a-112e** are partially exposed on opposite sides of the sacrificial gate stack **142**. While two sacrificial gate stacks **142** are shown, the number of the sacrificial gate stacks **142** is not limited to two. More than two sacrificial gate stacks **142** may be arranged along the X-direction in some embodiments. Next, a spacer **154** is formed on sidewalls of the sacrificial gate stacks **142**, as shown in FIGS. 10B and 10C. The spacer **154** may be formed by first depositing a conformal layer (e.g., by an ALD process) that is subsequently etched back (e.g., by RIE) to form sidewall spacers **154**. During the anisotropic etch process, most of the spacer **154** is removed from horizontal surfaces, such as the tops of the fin structures **112a-112e**, the cladding layers **132**, the dielectric walls **119**, leaving the spacers **154** on the vertical surfaces, such as the sidewalls of sacrificial gate stacks **142**. The spacer **154** may be made of a dielectric material such as silicon oxide, silicon nitride, silicon carbide, silicon oxynitride, SiCN, silicon oxycarbide, SiOCN, and/or combinations thereof.

[0049] In FIGS. 10A-10C, exposed portions of the fin structures **112** and exposed portions of the cladding layers **132** not covered by the sacrificial gate stacks **142** and the spacers **154** are selectively recessed by using one or more suitable etch processes, such as dry etch, wet etch, or a combination thereof. In some embodiments, exposed portions of the stacks of semiconductor layers **104** of the fin structures **112** are removed, exposing portions of the well portions **116**, as shown in FIG. 10B. In some embodiments, the exposed portions of the fin structures **112** are recessed to a level at or slightly below the top surface of the insulating material **118**. During removal of the portions of the fin structures **112** and the cladding layers **132**, portions of the dielectric walls **119** not covered by the sacrificial gate stacks **142** and the spacers **154** may also be removed. Therefore, the etch processes should be performed so that the height of the dielectric walls **119** between the adjacent subsequent epitaxial S/D features (e.g., epitaxial S/D features **160** in FIGS. 13A-13C) is still enough to prevent the epitaxial S/D features of a NMOS device from merging with the epitaxial S/D features of a PMOS device.

[0050] In FIGS. 11A-11C, edge portions of each second semiconductor layer **108** (e.g., **108a**, **108b**) of the stack of semiconductor layers **104** are removed horizontally along the X direction. The removal of the edge portions of the second semiconductor layers **108** forms cavities. In some embodiments, the portions of the second semiconductor layers **108** are removed by a selective wet etching process. In cases where the second semiconductor layers **108** are made of SiGe and the first semiconductor layers **106** are made of silicon, the second semiconductor layer **108** can be selectively etched using a wet etchant such as, but not limited to, ammonium hydroxide (NH₄OH), tetramethylammonium hydroxide (TMAH), ethylenediamine pyrocatechol (EDP), or potassium hydroxide (KOH) solutions.

[0051] After removing edge portions of each second semiconductor layers **108**, a dielectric layer is deposited in the

cavities to form dielectric spacers **151**. The dielectric spacers **151** may be made of a low-K dielectric material, such as SiON, SiCN, SiOC, SiOCN, or SiN. The dielectric spacers **151** may be formed by first forming a conformal dielectric layer using a conformal deposition process, such as ALD, followed by an anisotropic etching to remove portions of the conformal dielectric layer other than the dielectric spacers **151**. The dielectric spacers **151** are protected by the first semiconductor layers **106** during the anisotropic etching process. The remaining second semiconductor layers **108** (e.g., **108a**, **108b**, **108c**) are capped between the dielectric spacers **151** along the X direction.

[0052] In FIGS. **12A-12C**, epitaxial S/D features **160** are formed on the well portions **116** of the fin structures **112a-112e**. The epitaxial S/D features **160** may be the S/D regions. For example, one of a pair of epitaxial S/D features **160** located on one side of the stack of semiconductor layers **104** can be a source region, and the other of the pair of epitaxial S/D features **160** located on the other side of the stack of semiconductor layers **104** can be a drain region. A pair of epitaxial S/D features **160** includes a source epitaxial feature **160** and a drain epitaxial feature **160** connected by the nanosheet channels (i.e., the first semiconductor layers **106**). Source/drain region(s) may refer to a source or a drain, individually or collectively dependent upon the context. In this disclosure, a source and a drain are interchangeably used, and the structures thereof are substantially the same.

[0053] For n-channel FETs, the epitaxial S/D features **160** may include one or more layers of Si, SiP, SiC, SiCP, or a group III-V material (InP, GaAs, AlAs, InAs, InAlAs, InGaAs). In some embodiments, the epitaxial S/D features **160** may be doped with n-type dopants, such as phosphorus (P), arsenic (As), etc., for n-type devices. For p-channel FETs, the epitaxial S/D features **160** may include one or more layers of Si, SiGe, SiGeB, Ge, or a group III-V material (InSb, GaSb, InGaSb). In some embodiments, the epitaxial S/D features **160** may be doped with p-type dopants, such as boron (B). The epitaxial S/D features **160** may grow both vertically and horizontally to form facets, which may correspond to crystalline planes of the material used for the substrate **101**. The epitaxial S/D features **160** may be formed by an epitaxial growth method using CVD, ALD or MBE.

[0054] In FIGS. **13A-13C**, a contact etch stop layer (CESL) **162** is formed on the epitaxial S/D features **160**, the tops of dielectric walls **119**, and the nitrogen-containing layer **152** of the mask structure **148**. The CESL **162** is also formed on a portion of the insulating material **118** and the top surface of the recessed fin structure **112d** adjacent the end cut **145** (FIG. **8A**). The CESL **162** may include an oxygen-containing material or a nitrogen-containing material, such as silicon nitride, silicon carbon nitride, silicon oxynitride, carbon nitride, silicon oxide, silicon carbon oxide, or the like, or a combination thereof. The CESL **162** may be formed by CVD, PECVD, ALD, or any suitable deposition technique. In some embodiments, the CESL **162** is a conformal layer formed by the ALD process. Next, a first interlayer dielectric (ILD or ILD0) layer **164** is formed on the CESL **162**. The materials for the first ILD layer **164** may include an oxide formed from tetraethylorthosilicate (TEOS), un-doped silicate glass, or doped silicon oxide such as borophosphosilicate glass (BPSG), fused silica glass (FSG), phosphosilicate glass (PSG), boron doped silicon glass (BSG), and/or other suitable dielectric materials. The

first ILD layer **164** may be deposited by a PECVD process or other suitable deposition technique. In some embodiments, after formation of the first ILD layer **164**, the semiconductor device structure **100** may be subject to a thermal process to anneal the first ILD layer **164**.

[0055] In FIGS. **14A-14C**, a planarization process, such as a CMP process, is performed until the tops of the sacrificial gate electrode layer **146** and the spacers **154** are exposed. The planarization process removes portions of the first ILD layer **164** and the CESL **162** disposed on the sacrificial gate stacks **142**. In some embodiments, the first ILD layer **164** may be recessed to a level at the top of the sacrificial gate electrode layer **146**. In such cases, a nitrogen-containing layer (not shown), such as a SiCN layer, may be formed on the recessed first ILD layer **164** to protect the first ILD layer **164** during subsequent etch processes.

[0056] FIGS. **15A-26A** are cross-sectional views of various stages of manufacturing the semiconductor device structure **100** taken along line A-A of FIG. **14C**, in accordance with some embodiments. FIGS. **15C-26C** are top views of various stages of manufacturing the semiconductor device structure **100** of FIG. **14A**. FIGS. **15B-26B** are cross-sectional side views of various stages of manufacturing the semiconductor device structure **100** taken along line B-B of FIG. **14C**, in accordance with some embodiments. In FIGS. **15A-15C**, the sacrificial gate electrode layer **146** (FIG. **14A**) and the sacrificial gate dielectric layer **144** (FIG. **14A**) are removed, exposing the top surfaces of the cladding layers **132** and the stacks of semiconductor layers **104** (e.g., topmost first semiconductor layer **106a**). The sacrificial gate electrode layer **146** may be first removed by any suitable process, such as dry etch, wet etch, or a combination thereof, followed by the removal of the sacrificial gate dielectric layer **144**, which may be performed by any suitable process, such as dry etch, wet etch, or a combination thereof. In some embodiments, a wet etchant such as a tetramethylammonium hydroxide (TMAH) solution can be used to selectively remove the sacrificial gate electrode layer **146** but not the spacers **154**, the dielectric walls **119**, the CESL **162**, and the first ILD layer **164**.

[0057] Next, the sacrificial layers **107**, cladding layers **132** and the second semiconductor layers **108** are removed. The removal process exposes portions of the dielectric walls **119**, the first semiconductor layers **106**, and a portion of the insulating material **118**. The removal process may be any suitable processes, such as dry etch, wet etch, or a combination thereof. The removal process may be a selective etch process that removes the sacrificial layers **107**, the cladding layers **132** and the second semiconductor layers **108** but not the first semiconductor layers **106**, the spacers **154**, the dielectric walls **119**, and the CESL **162**. In cases where the sacrificial layers **107**, the cladding layers **132**, and the second semiconductor layers **108** are made of SiGe, and the first semiconductor layers **106** are made of silicon, a selective wet etching including an ammonia and hydrogen peroxide mixtures (APM) may be used. As a result of the etch process, openings **166** are formed, leaving the first semiconductor layers **106** (e.g., first semiconductor layers **106a**, **106b**, **106c**) protruded from opposing sides of the first dielectric feature **130**. Specifically, each of the first semiconductor layers **106a**, **106b**, **106c** has a first end in contact with the dielectric wall **119** and a second end (i.e., distal end) extending away from the first end, as shown in FIG. **15A**. Having the first end of the first semiconductor layers **106a**,

106b, **106c** directly connected to a portion of the first dielectric feature **130** saves the space for subsequent metal gate and increases the overall pattern density. The portions of the first semiconductor layers **106** not covered by the dielectric spacers **151** are exposed in the openings **166**. The first semiconductor layers **106a-c** from the fins **112a**, **112b**, **112c**, **112e** serve as nanosheet channel regions for forksheet transistors to be formed at regions **153**, **155**, **157**.

[0058] In some embodiments, the first semiconductor layers **106a-c** at the region **153** on one side of the dielectric walls **119** (e.g., dielectric walls **119-1** and **119-3**) may be designated as a N-type FET or P-type FET of the forksheet transistor, the first semiconductor layers **106a-c** at the region **155** between the dielectric wall **119-2** and the dielectric wall **119-1** may be designated as a P-type FET or N-type FET of the forksheet transistor, and the first semiconductor layers **106a-c** at the region **157** between the dielectric wall **119-2** and the dielectric wall **119-3** may be designated as a N-type FET or P-type FET of the forksheet transistor. The region **155** may be disposed along the Y-direction between the region **153** and the region **157**. In one exemplary embodiment shown in FIG. 15A, the first semiconductor layers **106a-c** at the region **153** on the right-hand side of the dielectric wall **119-1** is designated as a N-type FET of the forksheet transistor, the first semiconductor layers **106a-c** at the region **155** on the right-hand side of the dielectric wall **119-2** is designated as a P-type FET of the forksheet transistor, and the first semiconductor layers **106a-c** at the region **157** is designated as a N-type FET of the forksheet transistor. Alternatively, the first semiconductor layers **106a-c** at the region **153** on the right-hand side of the dielectric wall **119-1** may be designated as a P-type FET of the forksheet transistor, the first semiconductor layers **106a-c** at the region **155** on the right-hand side of the dielectric wall **119-2** may be designated as a N-type FET of the forksheet transistor, and the first semiconductor layers **106a-c** at the region **157** may be designated as a P-type FET of the forksheet transistor. It should be noted that while the region **155** is shown as immediately adjacent to the regions **153**, **157**, the regions **153**, **155** and **157** may be separated from each other by other devices, transistors, or features.

[0059] In some embodiments, the top surface of the dielectric walls **119** (e.g., dielectric wall **119-2**) is higher than the top surface of the topmost first semiconductor layer **106a** by a height **H3**. The height **H3** may be in a range of about 25 nm to about 60 nm. The height **H3** may be adjusted in accordance with the application and may be done during formation of the fin structures **112a-e** as discussed above with respect to FIG. 3A and/or during removal of the mask structure **110** as discussed above with respect to FIG. 7A.

[0060] In FIGS. 16A-16C, an interfacial layer (IL) **178** is formed to surround at least three surfaces (except for the surfaces being in contact with the dielectric wall **119**) of the first semiconductor layers **106** (e.g., first semiconductor layers **106a**, **106b**, **106c**). The IL **178** may also form on the exposed surfaces of the well portion **116** of the substrate **101** and the insulating material **118**. In some embodiments, the IL **178** may form on the first semiconductor layers **106** but not on the exposed dielectric walls **119**. The IL **178** may include or be made of an oxygen-containing material or a silicon-containing material, such as silicon oxide, silicon oxynitride, oxynitride, hafnium silicate, etc. The IL **178** may be formed by CVD, ALD or any suitable conformal deposition technique. In one embodiment, the IL **178** is formed

using ALD. The thickness of the IL **178** is chosen based on device performance considerations. In some embodiments, the IL **178** has a thickness ranging from about 0.5 nm to about 2 nm.

[0061] Next, a high-k (HK) dielectric layer **180** is formed on the exposed surfaces of the semiconductor device structure **100**. In some embodiments, the HK dielectric layer **180** is formed on the IL **178**, a portion of the insulating material **118**, and on the exposed surfaces of the dielectric walls **119**, as shown in FIG. 16A. Suitable materials for the HK dielectric layer **180** may include, but are not limited to, SiN, SiON, SiCN, SiOCN, AlSi₃O₅, Al₂O₃, or the like. Other suitable high-k materials, such as hafnium oxide (HfO₂), hafnium silicate (HfSiO), hafnium silicon oxynitride (Hf-SiON), hafnium aluminum oxide (HfAlO), hafnium lanthanum oxide (HfLaO), hafnium zirconium oxide (HfZrO), hafnium tantalum oxide (HfTaO), hafnium titanium oxide (HfTiO), lanthanum oxide (LaO), zirconium oxide (ZrO), zirconium silicon oxide (ZrSiO), titanium oxide (TiO), tantalum oxide (Ta₂O₅), yttrium oxide (Y₂O₃), or any suitable material having a k value greater than 7, may also be used. The HK dielectric layer **180** may be a conformal layer formed by a conformal process, such as an ALD process or a CVD process. The HK dielectric layer **180** may have a thickness of about 0.5 nm to about 3 nm, which may vary depending on the application.

[0062] In FIGS. 17A-17C, after formation of the IL **178** and the HK dielectric layer **180**, a first gate electrode layer **165** and a second gate electrode layer **163** are formed in the openings **166** (FIG. 16A). In some embodiments, the first gate electrode layer **165** may be formed in the openings **166** at the regions **153** and **157**, and the second gate electrode layer **163** may be formed in the openings **166** at the region **155**. The first gate electrode layer **165** is formed on the HK dielectric layer **180** to surround a portion of each first semiconductor layer **106a**, **106b**, **106c** and over the insulating material **118**. The first gate electrode layer **165** may include one or more layers of conductive material, such as polysilicon, aluminum, copper, titanium, tantalum, tungsten, cobalt, molybdenum, tantalum nitride, nickel silicide, cobalt silicide, TiN, WN, TiAl, TiAlN, TaCN, TaC, TaSiN, metal alloys, other suitable materials, and/or combinations thereof.

[0063] The first and second gate electrode layers **165**, **163** may be formed by first forming the first gate electrode layer **165** in the openings **166** at regions **153**, **155**, **157**. The first gate electrode layer **165** may be deposited so that at least the forksheet transistors to be formed at the regions **153**, **155**, **157** are submerged in the first gate electrode layer **165**. The first gate electrode layer **165** may be formed to a predetermined height above the dielectric wall **119**. In some embodiments, the first gate electrode layer **165** is deposited to a height over a top surface of the HK dielectric layer **180** over the dielectric wall **119**. In some embodiments, the first gate electrode layer **165** may be formed using multiple layers, each layer deposited sequentially adjacent to each other using a highly conformal deposition process such as ALD. Other deposition technique such as PVD, CVD, or electroplating may also be used. Thereafter, a patterned resist layer (not shown) is formed to cover N-type FETs, such as N-type FETs of the forksheet transistors at the regions **153** and **157**, while the P-type FETs, such as P-type FETs of the forksheet transistors at the region **155**, are left uncovered. The patterned resist layer may be formed by first forming a blanket layer on the semiconductor device structure **100**, followed

by patterning and etching processes to remove portions of the blanket layer at selected regions to form the patterned resist layer. The patterned resist layer may be any suitable masking material, such as a photoresist layer, a BARC (bottom anti-reflective coating) layer, a SOG (spin-on-glass) layer, or a SOC (spin-on-carbon) layer, and may be deposited by spin coating or any suitable deposition technique.

[0064] While not shown, the first gate electrode layer **165** may include a capping layer, a barrier layer, an n-metal work function layer, a p-metal work function layer, and a fill material. The capping layer and the barrier layer may be metal oxides, metal nitrides, metal silicates, transition metal-oxides, transition metal-nitrides, transition metal-silicates, oxynitrides of metals, metal aluminates, zirconium silicate, zirconium aluminate, combinations of these, or the like. The barrier layer may be formed of a material different from the capping layer. The n-metal work function layer may be formed from a metallic material such as W, Cu, AlCu, TiAlC, TiAlN, Ti, TiN, Ta, TaN, Co, Ni, Ag, Al, TaAl, TaAlC, TaC, TaCN, TaSiN, Mn, Zr, other suitable n-type work function materials, or combinations thereof. The p-metal work function layer may be formed from a metallic material such as W, Al, Cu, TiN, Ti, TiAlN, Ta, TaN, Co, Ni, TaC, TaCN, TaSiN, TaSi₂, NiSi₂, Mn, Zr, ZrSi₂, TaN, Ru, AlCu, Mo, MoSi₂, WN, other metal oxides, metal nitrides, metal silicates, transition metal-oxides, transition metal-nitrides, transition metal-silicates, oxynitrides of metals, metal aluminates, zirconium silicate, zirconium aluminate, combinations of these, or the like.

[0065] Next, portions of the first gate electrode layer **165** at the region **155** not covered by the patterned resist layer are removed and a second gate electrode layer **163** is formed in the region where the first gate electrode layer **165** was removed. The patterned resist layer **141** protects portions of first gate electrode layer **165** at the regions **153** and **157** so that the first gate electrode layer **165** at the region **155** are removed. The first gate electrode layer **165** may be removed using any suitable processes, such as dry etch, wet etch, or a combination thereof. The removal process may be a selective etch process that removes the first gate electrode layer **165** but not the HK dielectric layer **180**. Thereafter, the second gate electrode layer **163** is formed on the exposed HK dielectric layer **180** and in the region where the first gate electrode layer **165** was removed. The second gate electrode layer **163** may be deposited so that at least the forksheet transistors at the regions **155** are submerged in the second gate electrode layer **163**. The second gate electrode layer **163** surrounds a portion of each first semiconductor layer **106** at the region **155**. In some embodiments, the second gate electrode layer **163** is deposited so that the top surface of the second gate electrode layer **163** is substantially co-planar with the first gate electrode layer **165**. In some embodiments, the second gate electrode layer **163** is chemically different than the first gate electrode layer **165**. Alternatively, the second gate electrode layer **163** may include the same material as the first gate electrode layer **165**. Likewise, the second gate electrode layer **163** may be formed using multiple layers, each layer deposited sequentially adjacent to each other using a highly conformal deposition process such as ALD. The second gate electrode layer **163** may also include a capping layer, a barrier layer, an n-metal work function layer, a p-metal work function layer, such as those used for the first gate electrode layer **165**. Each layer in the first and second gate electrode layers **165**, **163** may be

chosen depending on the threshold voltage and application of the NMOS or PMOS devices needed for regions **153**, **155**, **157**.

[0066] After the formation of the first and second gate electrode layers **165**, **163**, the patterned resist layer is removed using any suitable removal process, such as ashing, dry etch, wet etch, or a combination thereof. As a result of the formation of the first and second gate electrode layers **165**, **163**, the metal gate structures **183**, **185** are formed and extended along the Y-direction. The forksheet transistors **182**, **184**, **186** are therefore formed at the regions **153**, **155** and **157**, respectively. In one exemplary embodiment, the forksheet transistor **182** includes a NMOS device **182-1** extending outwardly from one side of the dielectric wall **119-1**, a PMOS device **184-1** extending outwardly from one side of the dielectric wall **119-2**, a NMOS device **184-2** extending outwardly from another side of the dielectric wall **119-2**, and a NMOS device **186-1** extending outwardly from one side of the dielectric wall **119-3**.

[0067] In FIGS. **18A-18C**, a planarization operation, such as a CMP method, is performed. The planarization operation may be performed until the top surface of the dielectric walls **119** is exposed. In some embodiments, the CMP method may use a NH₄OH or KOH based slurry for polishing. At this stage, the top surface of the dielectric walls **119**, the HK dielectric layer **180**, and the first and second gate electrode layers **165**, **163**, and the first ILD layer **164** are substantially co-planar. As a result of the planarization operation, the N-type FETs of the forksheet transistors at the regions **153**, **155**, **157** are separated from each other by the dielectric walls **119**. In some embodiments, the planarization operation is performed so that the height between the top surface of the dielectric walls **119** (e.g., dielectric wall **119-2**) and the top surface of the topmost first semiconductor layer **106a** is reduced from the height H3 (FIG. **15A**) to the height H4. The height H4 may be referred to as a gate height. In various embodiments, the height H4 is about 15 nm or less, for example about 3 nm to about 10 nm. When the gate height is kept at about 15 nm or below, the overlap distance between the subsequent metal gate contact and the subsequent source/drain metal contact is reduced, which in turn reduces overall capacitance for the device. However, if the height H4 is less than about 3 nm, the subsequent gate contact vias (e.g., gate contact via **190**) may accidentally punch into/through the dielectric walls **119** and result in an open circuit failure. On the other hand, if the height H4 is over 15 nm, the benefit associated with having a reduced gate height is diminished.

[0068] Next, a mask layer **143** is formed on the semiconductor device structure **100**. The mask layer **143** may be any suitable masking material. In some embodiments, the mask layer **143** is formed of a nitrogen-containing material, such as silicon nitride (SiN). The mask layer **143** may be a single layer photoresist or a tri-layer photoresist. For example, the mask layer **143** may use the same material as the resist layer **141** and may be deposited using the same fashion as the resist layer **141**. As will be discussed below with respect to FIGS. **19A-19C**, the mask layer **143** is to be patterned to define regions where cut metal gate (CMG) openings are needed in accordance with the design layout.

[0069] In FIGS. **19A-19C**, the mask layer **143** is patterned to define regions where CMG openings **147** are to be formed. As can be seen in FIG. **19B**, the CMG opening **147** separates the metal gates **183**, **185** into two metal gate

stacks, respectively. A gate-cut process is then performed using the patterned mask layer 143 as an etching mask. The gate-cut process may be a dry etch, wet etch, reactive ion etch (RIE), and/or other suitable processes. The gate-cut process removes portions of the first gate electrode layer 165, the HK dielectric layer 180, the insulating material 118, and the dielectric material 117 to expose a top surface of the substrate 101. In some embodiments, the gate-cut process may be an over-etch process that removes a portion of the substrate 101. In such cases, the CMG openings 147 may extend into a portion of the exposed substrate 101. The etch process may use one or more etchant gases such as BCl_3 , Cl_2 , SiCl_4 , or any suitable etchant chemicals. The CMG openings 147 are to be filled a dielectric material and form isolation structures. The isolation structures ensure the two metal gate stacks obtained from cutting the metal gate structures 183, 185 are not in electrical contact with each other. Short circuiting therebetween is therefore prevented.

[0070] In FIGS. 20A-20C, the CMG openings 147 (FIG. 19A) are filled with a dielectric material. The CMG openings filling with the dielectric material form an isolation structure or so-called cut metal gate (CMG) structure 149. The dielectric material may be made of an oxygen-containing material, such as silicon oxide; a nitrogen-containing material, such as SiN, silicon oxynitride (SiON), SiOCN, SiCN; a low-K dielectric material; or any suitable dielectric material. The dielectric material may include the same or different material than the insulating material 118, and may be formed by any suitable process, such as a CVD, PECVD, FCVD, or ALD process. Once the CMG structures 149 are formed, a planarization process, such as a CMP process, may be performed to remove portions of the dielectric material formed over the mask layer 143. The planarization process is performed until a portion of the dielectric walls 119 is exposed. In one embodiment, the CMG structure 149 extends through the first gate electrode layer 165, the HK dielectric layer 180, the insulating material 118, the dielectric material 117, and into the substrate 101. The CMG structure 149 within the insulating material 118 may have a depth of about 0 nm to about 60 nm. After the planarization process, the top surfaces of the CMG structure 149, the first gate electrode layer 165, the second electrode layer 163, the HK dielectric layer 180, and the dielectric walls 119 are substantially co-planar.

[0071] Having the CMG structure 149 formed at the regions between adjacent dielectric walls 119 (e.g., dielectric walls 119-2 and 119-3) is advantageous because the CMG structure 149 is less likely to land on (and damage) the dielectric walls 119 (e.g., dielectric wall 119-1) due to larger spacing between the adjacent dielectric walls 119. Even if the CMG structure 149 is slightly misaligned, the CMG structure 149 can still properly separate the gate structures as intended without the risk of damaging the dielectric walls 119.

[0072] FIGS. 20A-1 and 20A-2 illustrate cross-sectional views of one of various stages of manufacturing the semiconductor device structure 100 in accordance with some alternative embodiments. The embodiment in FIG. 20A-1 is substantially identical to that shown in FIG. 20A except that the CMG structure 149 does not extend into a portion of the exposed substrate 101. Instead, the CMG structure 149 extends into a portion of the insulating material 118. In some embodiments, the CMG structure 149 extends into the insulating material 118 without touching the dielectric mate-

rial 117. Likewise, the embodiment in FIG. 20A-2 is substantially identical to that shown in FIG. 20A except that the dielectric material 117 was previously removed from the trenches 114a, 114c, 114e during the removal process, as the alternative embodiment shown in FIG. 5A-1. In this embodiment, the CMG structure 149 extends into a portion of the insulating material 118 without touching the exposed substrate 101.

[0073] In FIGS. 21A-21C, a first etch stop layer 171 is formed on the top surfaces of the CMG structure 149, the first gate electrode layer 165, the second electrode layer 163, the HK dielectric layer 180, and the dielectric walls 119. The first etch stop layer 171 may include a nitrogen-containing material, such as silicon nitride, silicon carbon nitride, silicon oxynitride, carbon nitride, or the like, or a combination thereof. In some embodiments, the first etch stop layer 171 includes a carbide and a nitride. In some embodiments, the first etch stop layer 171 includes an oxide and a nitride. The first etch stop layer 171 may be formed by CVD, PECVD, ALD, or any suitable deposition technique. In some embodiments, the first etch stop layer 171 includes SiN. Thereafter, a second interlayer dielectric (ILD or ILD1) layer 173 is formed on the first etch stop layer 171. The materials for the second ILD layer 173 may include the same material as the first ILD layer 164, and may be deposited by a PECVD process or other suitable deposition technique.

[0074] After the formation of the second ILD layer 173, S/D contact openings are formed through the second ILD layer 173, the first etch stop layer 171, the first ILD layer 164, and the CESL 162 to expose a top surface of the epitaxial S/D features 160. A silicide layer (not shown) is then formed on the exposed epitaxial S/D features 160, and a S/D contact 176 is formed in the S/D contact openings on the silicide layer. The S/D contact 176 may include a barrier layer 175 and an electrically conductive material layer 177 formed on the barrier layer 175. The S/D contact 176 may be electrically connected to the epitaxial S/D features 160. The barrier layer 175 may include TiN, TaN, or the like, and the electrically conductive material layer 177 may include Ru, Mo, Co, Ni, W, Ti, Ta, Cu, Al, TiN, TaN, or the like. FIGS. 21D-26D illustrate cross-sectional side views of various stages of manufacturing the semiconductor device structure 100 taken along line D-D of FIG. 21C, in accordance with some embodiments. It should be noted that additional S/D contact 176 and gate electrode layers 163 are also shown in FIG. 26D for ease of discussion.

[0075] The S/D contact 176 may be formed by first depositing the barrier layer 175 on the second ILD layer 173 and within the S/D contact openings by a conformal deposition process, such as an ALD process. The electrically conductive material layer 177 is then formed on the barrier layer 175. The electrically conductive material layer 177 fills up the S/D contact openings and over the second ILD layer 173. A planarization operation, such as a CMP method, is then performed until the second ILD layer 173 is exposed. After the planarization operation, the top surfaces of the second ILD layer 173, the barrier layer 175, and the electrically conductive material layer 177 are substantially co-planar.

[0076] In FIGS. 22A-22D, a second etch stop layer 179 is formed on the second ILD layer 173. The second etch stop layer 179 may include the same material as the first etch stop layer 171, and may be deposited using the same fashion as the first etch stop layer 171. In some embodiments, the first

and second etch stop layers **171**, **179** may include the same material but with different ratio of the composition in order to have different etch selectivity. The first and second etch stop layers **171**, **179** may be a single layer or a multi-layer structure. The first etch stop layer **171**, the second ILD layer **173**, and the second etch stop layer **179** may have a combined thickness of about 20 nm to about 60 nm, for example about 25 nm to about 30 nm.

[0077] In FIGS. 23A-23D, gate contact openings **181** and bridge contact openings **187** are formed. The gate contact openings **181** and bridge contact openings **187** are intended to be filled with a conductive material to form conductive features therein. The gate contact openings **181** extend through the second etch stop layer **179**, the second ILD layer **173**, and the first etch stop layer **171** to expose the gate electrode layer (e.g., gate electrode layers **163**, **165**). In various embodiments, the gate contact openings **181** are formed at regions where the dielectric walls **119** are located. In some embodiments, the gate contact openings **181** may be formed through a first patterned resist layer (not shown) at selected regions where the dielectric wall **119** separates gate electrode layers **163**, **165** and each gate electrode layers **163**, **165** surrounds an effective nanosheet channel region (e.g., first semiconductor layers **106a-c**). The bridge contact openings **187** may be formed through the second etch stop layer **179** at selected regions where the gate electrode layers **163/165** and S/D contacts **176** are located. The bridge contact openings **187** therefore overlap with the gate contact opening **181**. The gate contact openings **181** and bridge contact openings **187** may be formed by any suitable process, such as one or more etch processes. In some embodiments, the gate contact openings **181** and the bridge contact openings **187** are a result of a dual-damascene process. The gate contact opening **181** may be a via opening formed through the second etch stop layer **179**, the second ILD layer **173**, and the first etch stop layer **171** to expose a portion of the gate electrode layers **163**, **165**. The bridge contact opening **187** may be a trench opening formed in the second etch stop layer **179**. In some embodiments, the bridge contact opening **187** is formed so that it is disposed directly above the gate contact opening **181** and extended over the S/D contact **176** and the gate electrode layer **163**, **165**. Depending on the design layout, at least one gate contact opening **181** may be disposed at the region where the fin structure **112d** was partially removed, as one exemplary embodiment shown in FIGS. 23A and 23C. More or less gate contact openings **181** and bridge contact openings **187** are also contemplated.

[0078] FIG. 23A-1 illustrates an enlarged view of a portion of the semiconductor device structure **100** of FIG. 23A, in accordance with some embodiments. As shown in FIG. 23A-1, the one or more etch processes may be performed so that the gate contact openings **181** further extends into a portion of the dielectric walls **119**. In such cases, the dielectric wall **119** (e.g., dielectric wall **119-2**) may be recessed so that the top surface of the dielectric wall **119** is at a level below the top surface of the gate electrode layers **163**, **165** by a height **H5**. In some embodiments, the height **H5** is in a range of about 1 nm to about 3 nm. The recess of the dielectric walls **119** allows more contact surface area of the subsequent metal gate bridge (**190**, FIG. 24A) with the gate electrode layers **163**, **165**, thereby improving gate control of the devices.

[0079] In FIGS. 24A-24D, gate bridge contact **190** and metal on gate bridge contact **191** are formed in the gate contact openings **181** and the bridge contact openings **187**, respectively. The gate bridge contact **190** may include a barrier layer **190a** and an electrically conductive material layer **190b** formed on the barrier layer **190a**. Likewise, the metal on gate bridge contact **191** may include a barrier layer **191a** and an electrically conductive material layer **191b** formed on the barrier layer **191a**. The barrier layer **190a**, **191a** may include the same material as the barrier layer **175**, and the electrically conductive material layer **190b**, **191b** may include the same material as the electrically conductive material layer **177**. A planarization operation, such as a CMP method, is then performed until the second etch stop layer **179** is exposed. After the planarization operation, the top surfaces of the second etch stop layer **179**, the barrier layer **190a**, **191a**, and the electrically conductive material layer **191a**, **191b** are substantially co-planar. The metal on gate bridge contact **191** may be considered as an integrated body having a lower portion (i.e., gate bridge contact **190**) in contact with the gate electrode layer (e.g., gate electrode layer **163**) and an upper portion extending from over the gate electrode layer to the S/D contact **176**.

[0080] In various embodiments, the dielectric wall **119** (e.g., dielectric wall **119-2**) along the Y-direction may have a width **W3** and the gate bridge contact **190** along the Y-direction may have a width **W4** that is greater than the width **W3**. In some embodiments, the width **W3** and the width **W4** are at a ratio (**W3:W4**) of about 1:1.5 to about 1:5, for example about 1:2 to about 1:3. Having the gate bridge contact **190** formed with a larger size than that of the dielectric wall **119** ensures that the gate bridge contact **190** is in contact with both the gate electrode layers **163** and gate electrode layer **165**, and that the subsequent gate via contact **196** is in direct contact with the gate bridge contact **190**, which minimizes the contact failure due to misalignment of the gate via contact **196** with the gate electrode layers **163**, **165**. The formation of the gate bridge contact **190** on the dielectric walls **119** (e.g., dielectric wall **119-2**) allows the subsequent gate via contact **196** (FIG. 26A) to land directly on the gate bridge contact **190** without accidentally punching into/through the dielectric walls **119**, which may otherwise result in an open circuit failure if the gate via contact **196** were not in contact with the gate electrode layers **163**, **165**.

[0081] In FIGS. 25A-25D, a patterned third ILD layer (ILD2) **193** is formed on the second etch stop layer **179**. The third ILD layer **193** may include the same material as the second ILD layer **173** and may be deposited using the same deposition technique as the second ILD layer **173**. The patterned third ILD layer **193** defines via contact openings **195a**, **195b** at locations where the gate bridge contact **190** (FIG. 24C) and metal on gate bridge contact **191** (FIG. 24C) are located, respectively.

[0082] In FIGS. 26A-26D, gate via contact **196** and via bridge contact **197** are formed in the via contact openings **195a**, **195b**, respectively. The gate via contact **196** may include a barrier layer **196a** and an electrically conductive material layer **196b** formed on the barrier layer **196a**. Likewise, the via bridge contact **197** may include a barrier layer **197a** and an electrically conductive material layer **197b** formed on the barrier layer **197a**. The barrier layer **196a**, **197a** may include the same material as the barrier layer **175**, and the electrically conductive material layer **196b**, **197b** may include the same material as the electrically

conductive material layer 177. A planarization operation, such as a CMP method, is then performed until the third ILD layer 193 is exposed. After the planarization operation, the top surfaces of the third ILD layer 193, the barrier layer 196a, 197a, and the electrically conductive material layer 196a, 197b are substantially co-planar.

[0083] FIG. 26A-1 illustrates a portion of a bottom of the gate via contact 196 is further extended into a top portion of the dielectric wall 119 (e.g., dielectric wall 119-2). Specifically, portions of the barrier layer 196a and the electrically conductive material layer 196b are extended into the top portion of the dielectric wall 119-2. Therefore, the effective contact area between the gate via contact 196 and the gate electrode layers 163, 165 is increased. FIG. 26C-1 is a top view of the semiconductor device structure 100 of FIG. 26C showing locations of various devices/elements, in accordance with some embodiments. For the sake of illustration purpose, certain layers such as the ILD layers 173, 193, the etch stop layers 171, 179, etc., have been omitted. In addition, various features, such as the S/D contact 176, the gate bridge contact 190, the metal on gate bridge contact 191, the gate via contact 196, the via bridge contact 197, etc., are not drawn to scale. Instead, the dimensions of the various features may be arbitrarily increased or reduced for clarity of illustration. In some embodiments, the height H6 between the top surface of the gate electrode layer (e.g., gate electrode layer 165) and the top surface of the topmost first semiconductor layer 106a may be in a range of about 15 nm or less, for example about 3 nm to about 10 nm. The height H6 may correspond to the gate height of the gate electrode layers 163, 165.

[0084] FIG. 27A is a cross-sectional view of a semiconductor device structure 200 in accordance with some alternative embodiments. FIG. 27B is a top view of the semiconductor device structure 200 of FIG. 27A, which may represent a portion of design layout of active fin structures in a logic cell 203. FIG. 27C is a cross-sectional side view of the semiconductor device structure 200 taken along line C-C of FIG. 27A, in accordance with some embodiments. The embodiment in FIG. 27A is similar to the embodiment of FIG. 26A except that certain fin structures, such as a fin structure 212c, are not partially removed. Since the semiconductor device structure 200 has features similar to that of the semiconductor device structure 100 of FIG. 26A, the features of the semiconductor device structure 200 will be assigned with reference numerals in the form of “2xx” that are identical or substantially similar to features of the semiconductor device structure 100 having reference numerals in the form “1xx” for the purpose of simplicity and clarity. Most of the features in FIG. 27A-27C are similar to the features shown in FIG. 26A and therefore will not be repeated for the sake of brevity.

[0085] As can be seen in FIGS. 27A-27C, the fin structures 212a, 212b, 212c are extended along the Y-direction and a plurality of metal gate structures 283/285 are extended over the fin structures 212a, 212b, 212c along the X-direction. The isolation structure or so-called cut metal gate (CMG) structures 349 are disposed across the metal gate structures 283/285. Gate bridge contacts 290 are formed directly on the dielectric walls 219 and in contact with both the gate electrode layers 263, 265. The gate via contact 296 is formed on the gate bridge contact 290, which is in contact with the gate electrode layer (e.g., gate electrode layer 263). The gate bridge contact 290 extend through the second etch

stop layer 279, the second ILD layer 273, and the first etch stop layer 271, and the gate via contact 296 extend through the third ILD layer 293 to contact with the gate bridge contact 290. Likewise, the dielectric wall 219 along the Y-direction may have a width W5 and the gate bridge contact 290 along the Y-direction may have a width W6 that is greater than the width W5. In some embodiments, the width W5 and the width W6 are at a ratio (W5:W6) of about 1:1.5 to about 1:5, for example about 1:2 to about 1:3. Having the gate bridge contact 290 formed with a larger size than that of the dielectric wall 219 ensures that the gate bridge contact 290 is in contact with both the gate electrode layers 263 and gate electrode layer 265, and that the gate via contact 296 is in direct contact with the gate bridge contact 290, which minimizes the contact failure due to misalignment of the gate via contact 296 with the gate electrode layers 263, 265. The formation of the gate bridge contact 290 on the dielectric walls 219 (e.g., dielectric wall 219-3) allows the gate via contact 296 to land directly on the gate bridge contact 290 without accidentally punching into/through the dielectric walls 219, which may otherwise result in an open circuit failure if the gate via contact 296 were not in contact with the gate electrode layers 263, 265.

[0086] In some embodiments, the height H7 between the top surface of the gate electrode layer (e.g., gate electrode layer 263) and the top surface of the topmost first semiconductor layer 206a may be in a range of about 15 nm or less, for example about 3 nm to about 10 nm. The height H7 may correspond to the gate height of the gate electrode layers 163, 165. When the gate height is kept at about 15 nm or below, the overlap distance between the metal on gate bridge contact 291 and the source/drain contact 276 is decreased, which in turn reduces overall capacitance for the device.

[0087] FIG. 27C further illustrates the semiconductor device structure 200 includes a metal on gate bridge contact 291 and a via bridge contact 297. The metal on gate bridge contact 291 may include a lower portion in contact with the gate electrode layer (e.g., gate electrode layer 263) and an upper portion extending from over the gate electrode layer 263 to over the S/D contact 276. The via bridge contact 297 extends through the third ILD layer 293 and in direct contact with the gate bridge contact 290. The via bridge contact 297 is electrically connected to the gate electrode layer 263 via the gate bridge contact 290.

[0088] FIG. 28 is a top view of a semiconductor device structure 300, which may represent a portion of design layout of active fin structures in a SRAM cell 303, in accordance with some embodiments. 103. The features of the semiconductor device structure 300 will be assigned with reference numerals in the form of “3xx” that are identical or substantially similar to features of the semiconductor device structure 100 having reference numerals in the form “1xx” for the purpose of simplicity and clarity. Most features in FIG. 28 are similar to the features shown in FIG. 27B and therefore will not be repeated for the sake of brevity. In one embodiment, the semiconductor device structure 300 is a forksheet 6T SRAM cell including pull-up (PU) transistors, pass-gate (PG) transistors, and pull-down (PD) transistors. Likewise, gate bridge contacts 390 are formed directly on the dielectric walls 319 (319-1, 319-2, 319-3) and in contact with both P-type and N-type gate electrode layers (in the metal gate structure 383/385). The isolation structure or cut metal gate (CMG) structures 349 are disposed across the metal gate structures 383/385 at selected

regions. The formation of the gate bridge contact **390** on the dielectric walls **319** (e.g., dielectric wall **319-3**) allows gate via contact (not shown, such as gate via contact **296** in FIGS. **27A** and **27B**) to land directly on the gate bridge contact **390** without accidentally punching into/through the dielectric walls **319**. In addition, the source/drain contact (MD) and the metal gate (MG) located at the end-cut region **345** can be electrically connected through MG-MD bridge, which allows CMG structures **349** to cut on the STI **320** instead of the dielectric walls **319**. As a result, the CMG etch-through damage to the dielectric walls **319** and VG high-R and/or open risk (due to VG landing on dielectric wall) are prevented, which is advantageous for scaling and design feasibility.

[0089] It is understood that the semiconductor device structures **100**, **200** may undergo further complementary metal oxide semiconductor (CMOS) and/or back-end-of-line (BEOL) processes to form various features such as transistors, contacts/vias, interconnect metal layers, dielectric layers, passivation layers, etc. For example, the semiconductor device structure **100** may also include backside contacts (not shown) on the backside of the substrate **101** by flipping over the semiconductor device structure **100**, removing the substrate **101**, and selectively connecting source or drain feature/terminal of the epitaxial S/D features **160** to a backside power rail (e.g., positive voltage VDD or negative voltage VSS) through the backside contacts. Depending on the application, the source or drain feature/terminal of the epitaxial S/D features **160** and the gate electrode layers **163**, **165** may be connected to a frontside power source.

[0090] Various embodiments described herein offer multiple advantages over the state-of-art technology. According to embodiments of the present disclosure, a gate bridge contact is formed between and in contact with a gate via contact and a dielectric wall. The gate bridge contact is larger in size (e.g., width) than that of the gate via contact and the dielectric wall to ensure electrical contact with gate electrode layers at N-type region and P-type region. The larger size of the gate bridge contact also allows a gate via contact to land directly on the gate bridge contact without accidentally punching into/through the dielectric walls, which may otherwise result in an open circuit failure if the gate via contact were not in contact with the gate electrode layer. The use of the gate bridge contact avoid possible damage to the dielectric walls and the gate electrode layers (and thus contact failure) due to misalignment of the gate via contact with the gate electrode layers. In addition, after formation of the gate electrode layers, a planarization process may be performed on the gate electrode layers until gate electrode layers at N-type region and gate electrode layers at P-type region are fully blocked by the dielectric wall. This keeps the gate height at a minimum so that the overlap distance between the metal gate contact and source/drain contact can be reduced for a lower capacitance of the device. Lastly, the use of the gate bridge contact allows cut metal gate (CMG) structures to be formed at regions between adjacent dielectric walls instead of on the dielectric walls. As a result, the CMG structures are less likely to land on (and damage) the dielectric walls due to larger spacing between the adjacent dielectric walls. Even if the CMG structure is slightly misaligned, the CMG structure can still properly separate the gate structures as intended without the risk of damaging the dielectric walls.

[0091] An embodiment is a semiconductor device structure. The structure includes a first dielectric wall, a plurality of first semiconductor layers vertically stacked and extending outwardly from a first side of the first dielectric wall, a plurality of second semiconductor layers vertically stacked and extending outwardly from a second side of the first dielectric wall. The structure also includes a first gate electrode layer surrounding at least three surfaces of each of the first semiconductor layers, the first gate electrode layer having a first conductivity type, and a second gate electrode layer surrounding at least three surfaces of each of the second semiconductor layers, the second gate electrode layer having a second conductivity type opposite the first conductivity type. The structure further includes a gate bridge contact disposed on the first dielectric wall, and a gate via contact disposed on the gate bridge contact.

[0092] Another embodiment is a semiconductor device structure. The structure includes a first dielectric wall, a plurality of first semiconductor layers vertically stacked and extending outwardly from a first side of the first dielectric wall, a second dielectric wall disposed in parallel with the first dielectric wall, a plurality of second semiconductor layers vertically stacked and extending outwardly from both sides of the second dielectric wall, a third dielectric wall disposed in parallel with the first dielectric wall, a plurality of third semiconductor layers vertically stacked and extending outwardly from a first side of the third dielectric wall, a gate structure disposed over a portion of the first, second, and third dielectric walls, a source/drain contact disposed over a portion of the first dielectric wall and in contact with an epitaxial source/drain feature, a metal on gate bridge contact disposed over the gate structure and in contact with a portion of the gate structure and the source/drain contact, and a gate bridge contact disposed over a portion of the gate structure and in contact with the second dielectric wall.

[0093] A further embodiment is a method. The method includes forming first, second, third, and fourth fin structures from a substrate, wherein the first fin structure includes a first plurality of semiconductor layers, the second fin structure includes a second plurality of semiconductor layers, the third fin structure includes a third plurality of semiconductor layers, and the fourth fin structure includes a fourth plurality of semiconductor layers, and wherein each of the first, second, third, and fourth plurality of semiconductor layers comprises first semiconductor layers and second semiconductor layers. The method also includes forming a first dielectric wall between the first fin structure and the second fin structure, forming a second dielectric wall between the third fin structure and the fourth fin structure, forming a sacrificial gate stack over a portion of the first, second, third, and fourth fin structures, and over a portion of the first and second dielectric walls, removing a portion of the first, second, third, and fourth fin structures not covered by the sacrificial gate stack, removing the sacrificial gate stack to expose portions of the first, second, third, and fourth fin structures, selectively removing the second semiconductor layers of the first, second, third, and fourth plurality of semiconductor layers. The method also includes surrounding a first gate electrode layer over at least three surfaces of each of the first semiconductor layer of the third plurality of semiconductor layers, wherein the first gate electrode layer has a first conductivity type. The method further includes surrounding a second gate electrode layer over at least three surfaces of each of the first semiconductor layer of the fourth

plurality of semiconductor layers, wherein the second gate electrode layer has a second conductivity type opposite the first conductivity type. The method further includes performing a planarization operation so that top surfaces of the first and second dielectric walls and top surfaces of the first and second gate electrode layers are substantially co-planar, forming a gate bridge contact on the second dielectric wall, and forming a gate via contact on the gate bridge contact.

[0094] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

1. A semiconductor device structure, comprising:
 - a first dielectric wall;
 - a plurality of first semiconductor layers vertically stacked and extending outwardly from a first side of the first dielectric wall;
 - a plurality of second semiconductor layers vertically stacked and extending outwardly from a second side of the first dielectric wall;
 - a first gate electrode layer surrounding at least three surfaces of each of the first semiconductor layers, the first gate electrode layer having a first conductivity type;
 - a second gate electrode layer surrounding at least three surfaces of each of the second semiconductor layers, the second gate electrode layer having a second conductivity type opposite the first conductivity type;
 - a gate bridge contact disposed on the first dielectric wall; and
 - a gate via contact disposed on the gate bridge contact.
2. The semiconductor device structure of claim 1, further comprising:
 - a second dielectric wall; and
 - a plurality of third semiconductor layers vertically stacked and extending outwardly from a first side of the second dielectric wall.
3. The semiconductor device structure of claim 2, wherein the second gate electrode layer surrounds at least three surfaces of each of the third semiconductor layers.
4. The semiconductor device structure of claim 3, further comprising:
 - a cut metal gate (CMG) structure disposed between the first dielectric wall and the second dielectric wall.
5. The semiconductor device structure of claim 4, wherein the CMG structure is in contact with the second gate electrode layer.
6. The semiconductor device structure of claim 1, wherein the gate bridge contact is further in contact with the first gate electrode layer and the second electrode layer.
7. The semiconductor device structure of claim 6, wherein the gate bridge contact has a first width and the first dielectric wall has a second width less than the first width.

8. The semiconductor device structure of claim 6, wherein a portion of a bottom of the gate bridge contact is extended into a top portion of the first dielectric wall.

9. The semiconductor device structure of claim 4, further comprising:

- an epitaxial source/drain contact disposed in contact with the first dielectric wall;
- a metal on gate bridge contact, comprising:
 - a lower portion in contact with the first gate electrode layer; and
 - an upper portion extending from over the first gate electrode layer to the epitaxial source/drain contact.

10. The semiconductor device structure of claim 9, further comprising:

- a via bridge contact in contact with the metal on gate bridge contact.

11. The semiconductor device structure of claim 1, further comprising:

- a third dielectric wall; and
- a plurality of fourth semiconductor layers vertically stacked and extending outwardly from a first side of the third dielectric wall.

12. The semiconductor device structure of claim 11, further comprising:

- an interfacial layer surrounding at least three surfaces of the first, second, third, and fourth semiconductor layers; and
- a high-k dielectric layer formed on the interfacial layer.

13. The semiconductor device structure of claim 12, wherein the high-k dielectric layer is in contact with exposed second side of the third dielectric wall.

14. A semiconductor device structure, comprising:

- a first dielectric wall;
- a plurality of first semiconductor layers vertically stacked and extending outwardly from a first side of the first dielectric wall;
- a second dielectric wall disposed in parallel with the first dielectric wall;
- a plurality of second semiconductor layers vertically stacked and extending outwardly from both sides of the second dielectric wall;
- a third dielectric wall disposed in parallel with the first dielectric wall;
- a plurality of third semiconductor layers vertically stacked and extending outwardly from a first side of the third dielectric wall;
- a gate structure disposed over a portion of the first, second, and third dielectric walls;
- a source/drain contact disposed over a portion of the first dielectric wall and in contact with an epitaxial source/drain feature;
- a metal on gate bridge contact disposed over the gate structure and in contact with a portion of the gate structure and the source/drain contact; and
- a gate bridge contact disposed over a portion of the gate structure and in contact with the second dielectric wall.

15. The semiconductor device structure of claim 14, wherein the gate structure further comprises:

- a first gate electrode layer surrounding at least three surfaces of each of the second semiconductor layers located on a first side of the second dielectric wall, the first gate electrode layer having a first conductivity type; and

a second gate electrode layer surrounding at least three surfaces of each of the second semiconductor layers located on a second side of the second dielectric wall, the second gate electrode layer having a second conductivity type opposite the first conductivity type.

16. The semiconductor device structure of claim **15**, wherein the gate bridge contact has a bottom in contact with the second dielectric wall, the first gate electrode layer, and the second electrode layer.

17. The semiconductor device structure of claim **15**, further comprising:

a gate via contact disposed above and in contact with the gate bridge contact.

18. The semiconductor device structure of claim **17**, further comprising:

a cut metal gate (CMG) structure disposed between the second dielectric wall and the third dielectric wall, the CMG structure extending through an entire thickness of the second gate electrode layer.

19. A method for forming a semiconductor device structure, comprising:

forming first, second, third, and fourth fin structures from a substrate, wherein the first fin structure includes a first plurality of semiconductor layers, the second fin structure includes a second plurality of semiconductor layers, the third fin structure includes a third plurality of semiconductor layers, and the fourth fin structure includes a fourth plurality of semiconductor layers, and wherein each of the first, second, third, and fourth plurality of semiconductor layers comprises first semiconductor layers and second semiconductor layers;

forming a first dielectric wall between the first fin structure and the second fin structure;

forming a second dielectric wall between the third fin structure and the fourth fin structure;

forming a sacrificial gate stack over a portion of the first, second, third, and fourth fin structures, and over a portion of the first and second dielectric walls;

removing a portion of the first, second, third, and fourth fin structures not covered by the sacrificial gate stack; removing the sacrificial gate stack to expose portions of the first, second, third, and fourth fin structures;

selectively removing the second semiconductor layers of the first, second, third, and fourth plurality of semiconductor layers;

surrounding a first gate electrode layer over at least three surfaces of each of the first semiconductor layer of the third plurality of semiconductor layers, wherein the first gate electrode layer has a first conductivity type;

surrounding a second gate electrode layer over at least three surfaces of each of the first semiconductor layer of the fourth plurality of semiconductor layers, wherein the second gate electrode layer has a second conductivity type opposite the first conductivity type;

performing a planarization operation so that top surfaces of the first and second dielectric walls and top surfaces of the first and second gate electrode layers are substantially co-planar;

forming a gate bridge contact on the second dielectric wall; and

forming a gate via contact on the gate bridge contact.

20. The method of claim **19**, further comprising: after performing the planarization operation, forming a cut metal gate (CMG) structure, wherein the CMG structure extends through an entire thickness of at least the second gate electrode layer.

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